

CE-QUAL-W2: A Two-Dimensional, Laterally Averaged, Hydrodynamic and Water Quality Model, Version 4.5.5

User Manual: Part 5 – Model Utilities

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Preface

This manual documents the two-dimensional, laterally averaged, hydrodynamic and water quality model CE-QUAL-W2. As in all complex models, there have been many contributors. This re-write of the User Manual was based on prior User Manuals: Environmental and Hydraulic Laboratories (1986), Cole and Buchak (1995) Version 2, and Cole and Wells (2000) Version 3.0 through Cole and Wells (2019) Version 4.1. Hence, one can think of the primary author as merely an editor of past documents, rather than reflecting one person's sole authorship. This updated User Manual contains numerous corrections, new figures, new sections, additional documentation, and improvements in organization and presentation of information compared to Cole and Wells (2019).

This section of the User Manual Part 5 documents the model utilities used in supporting the model, the model release notes and a detailed list of bug fixes.

The other sections of the User Manual are divided into multiple sections for ease of updating and editing:

- User Manual Part 1: Introduction to CE-QUAL-W2, Model download package, how to run the model, model versions, changes between model versions
- User Manual Part 2: Theoretical basis for CE-QUAL-W2: hydrodynamics and water quality, particle transport and numerical scheme
- User Manual Part 3: Model input and output file descriptions and input/output file examples
- User Manual Part 4: Model examples
- User Manual Part 5: Release notes, bug fixes, differences in model versions, history of bug fixes, and other user manuals such as for the GUI interface, the water balance algorithm, and other external codes.

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Or for a specific section:

Cole, Tom (2026) "Water Balance Utility," in "CE-QUAL-W2: A two-dimensional, laterally averaged, hydrodynamic and water quality model, version 4.5.5, user manual part 5, model utilities and release notes," ed. S. Wells, Department of Civil and Environmental Engineering, Portland State University, Portland, OR.

1. Introduction

Several useful CE-QUAL-W2 model utilities are available for editing the control file, post-processing model results, and setting up model input files. A description of these utilities is included in this part of the User Manual.

Also, this document shows model changes and bug fixes, model changes between versions, and a history of bug fixes in earlier versions.

Hence, this section of the User Manual includes the following sections:

- Model utilities
 - Water balance utility
 - Converter utility from w2_con.npt control file to Excel version
 - GUI control file editor and bathymetry viewer for Versions 3.7-4.22
 - Post-processor provided by DSI, Inc. including a bathymetry viewer
 - Excel macro utility for writing out input files for CE-QUAL-W2
- Model release notes
 - Current Bug fixes
 - Model Known Limitations
 - Model changes between versions
 - Model bug fixes between versions

2. Water Balance Utility

Introduction: Scott Wells

The water balance utility computes the residual water required to match field data of water surface elevation given a model prediction of water level. This often works well for reservoirs or lakes in understanding the water inflows and outflows required to match the water level. This is not generally used for rivers where usually a flow balance occurs at gaging stations. In some cases though it does not work well. Why? Below is a list of potential issues that may prevent this utility from matching water levels:

- Since this utility assumes a flat-water surface, wind-wave action and large seiche motions may cause the utility to over-estimate and under-estimate the inflow/outflow required.

- Since this utility uses the model volume-elevation curve, if the model volume-elevation curve is different from the actual volume-elevation curve errors may occur in flow required to match water levels.
- In some reservoirs that have large water surface variations over a short time frame (like reregulating reservoirs), this utility may not be able to reproduce the water surface variation since these reservoirs rarely have flat water surfaces. In this case, the user may end up using a variable elevation gate that mimics the water surface.

How to use the water balance utility GUI application

Primary author: Tom Cole

Running the water balance program

When the executable is run, a window appears that allows the following inputs (note that the executable runs under the Windows operating system only) as shown in Figure 1.

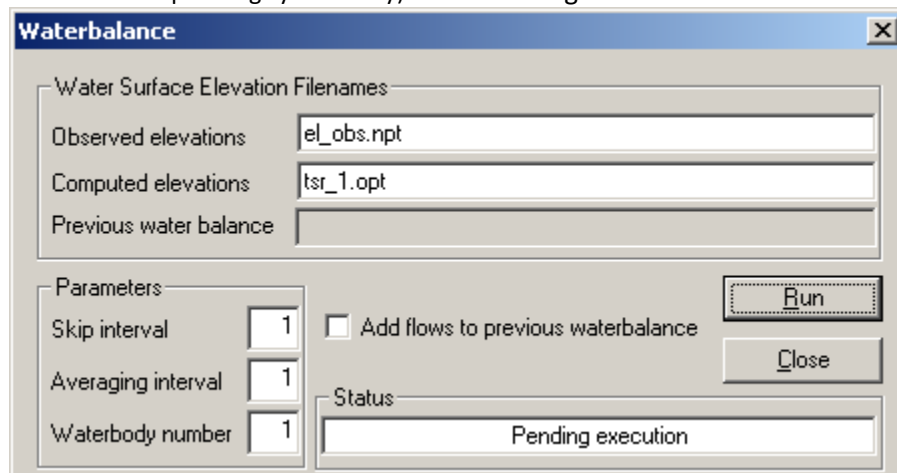


Figure 1. Dialog box for water balance utility.

:

When the dialog box first appears, default values populate the edit boxes. The user can then edit each one if the default values are not correct. Selecting Run will run the waterbalance utility to completion as shown in the following dialog box (Figure 2).

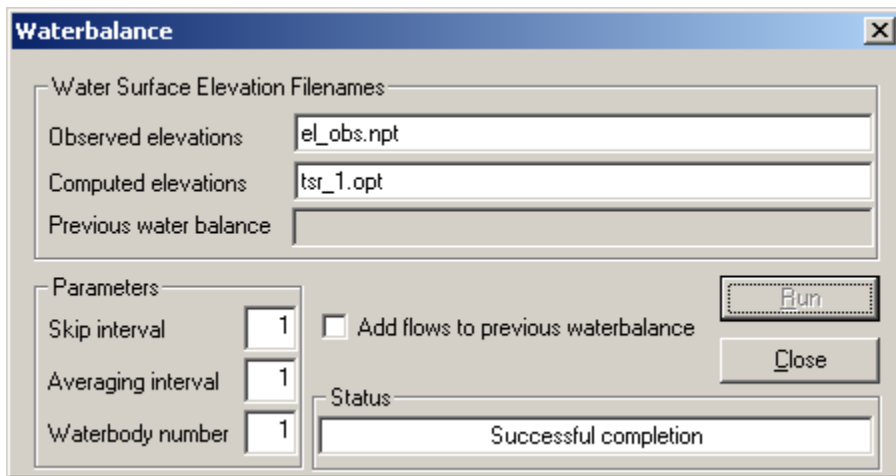


Figure 2. Dialog box for water balance utility if successful completion.

Observed elevations filename. This file consists of a Julian day and observed elevation as in the following example:

2000-2001 Oologah Reservoir observed water surface elevations

JDAY	ELO
90.792	195.453
90.833	195.456
90.875	195.459
90.917	195.441
90.958	195.444
91.000	195.450
91.042	195.441
91.083	195.441
91.125	195.447
91.167	195.441
91.208	195.444
91.250	195.438
91.292	195.432

This example skips the first 3 lines and is of a similar format to all other CE-QUAL-W2 time-varying inputs with a fixed format with eight columns each for the JDAY and ELO values. However, the utility will read in values using variable field lengths so long as the JDAY and ELO values are separated by a space. Data need not be at regular intervals that might cause a repeat of the same values. Sometimes better results have been obtained if the same values that repeat over a time interval are not included (such as the water surface elevation at day 91.083 above).

Also note that the degree of accuracy in the observed elevations can have an impact on the computed flows. It is up to the user to decide the necessary precision used in the observed water surface elevations.

Computed elevations filename. The following shows an example output file from a time series file (the columns to the right have not been shown):

JDAY,	DLT,	ELWS,	
92.000,	456.39,	195.37,	
93.000,	952.28,	195.28,	
94.000,	494.28,	195.27,	
95.000,	34.43,	195.17,	
96.000,	170.88,	195.08,	

97.000,	211.93,	194.99,
98.000,	255.11,	194.92,
99.000,	1459.08,	194.78,

The water balance utility reads in the [JDAY] and [ELWS] values and uses these in the water balance computations. The user must turn on time series output in the model control file and specify the segment at which the water surface elevation values are output (typically the segment next to the dam for reservoirs). Information on how to accomplish this is given in the User's Manual under the Time Series output file discussion.

Add to previous water balance . For various reasons, the water balance utility may not perfectly close the water balance the first time through the computations. Depending upon the discrepancies between computed and observed elevations, the utility may need to be used iteratively by rerunning the model using output from the first run of the water balance utility and then rerunning the water balance utility on the water surface elevations output in the new time series file. For a system with multiple branches, each iteration of the utility and the resulting output file can be saved as a separate file that is then incorporated as a distributed tributary for branch 2, then branch 3, etc. In the case of a system with only one branch, this approach cannot be used. Rather, the new flows generated at the second iteration need to be added to the previously computed flows and incorporated as an "improved" distributed tributary inflow file. This option allows the user to continue adding flows to the same inflow file.

The computed flows are contained in the "qwb.opt" file. For most simulations, these flows will generate water surface elevations sufficiently close to the observed elevations such that further refinement is unnecessary. However, as mentioned above, the solution may need to be iterated. Rarely, manual adjustment of the generated flows may be required. This is usually only needed when observed water surface elevations change significantly over a short time period.

Previous water balance filename. If the "Add to previous water balance" option is used, you must specify the existing water balance output file for the computed flows to be added to.

Skip interval. Some reservoirs have a lot of noise in the observed water surface elevation data, such as in peaking hydropower operations, and this option allows the user to specify how many observed elevations are ignored when computing the flows between observed elevations. For example, if water surface elevations are available on an hourly interval, the resulting flows generated by the water balance utility can have large + and – flows that are completely unrealistic as opposed to using observed elevations on a daily basis taken during periods of no hydropower generation. In order to smooth out the computed flows, a skip factor of 24 would result in computed flows being output on a daily basis with all of the "noise" generated by hydropower operations ignored over the 24-hour period.

Averaging interval. This option computes a running average of the water surface elevation based on the input value. This is an additional aid to smooth out water surface elevation "noise". For example, consider the case in which there is no inflow/outflow to the system, but there is considerable wind seiching. The water balance utility would compute alternating inflows and outflows from the system that, depending on the amount of seiching, could be very large when in reality there should not be any flows added to or subtracted from the system. Using a running average alone or in combination with skipping over a number of observed elevations specified in (4) can help alleviate many of the problems caused by an automated water balance computation.

Waterbody number. In the case of multiple waterbodies each of which has a separate bathymetry input file, the user must specify which waterbody (and thus which bathymetry file) the water balance is being computed for. This capability is necessary for modeling systems with multiple reservoirs.

Incorporating the computed flows into the simulation

The water balance utility can be used for lakes and reservoirs in which water surface elevations are a function of inflows and controlled outflows from the system. The utility computes the flows necessary to match observed water surface elevations (typically taken at the dam) and outputs them to the “qwb.opt” file. This file is composed of a Julian date and an inflow ($\text{m}^3 \text{sec}^{-1}$). The flows can be either positive or negative. Temperatures and/or constituent concentrations must also be provided in the corresponding temperature and constituent concentration input files if the computed flows are incorporated as inflows to the system. ***The water balance utility does not provide this information***, but this information needs to be provided by the user depending upon how the computed flows are incorporated into the simulation. Considerable thought should go into how best to incorporate temperature and constituent concentrations and is discussed in more detail below.

Note that negative flows use temperatures/concentrations in the waterbody when calculating the impact on the system of these flows rather than the temperatures/concentrations in the corresponding inflow temperature and constituent concentration files. This ensures that negative flows generate no change in temperature or constituent concentrations. However, positive flows can impact simulation results and care must be taken as to how the flows are incorporated into the simulation.

The flows required to complete the water balance are computed as a step function. If they are incorporated into the model as an additional inflow or outflow whose current values are being linearly interpolated, such as a branch inflow, then the resulting water balance will not be correct. Typically, the flows in the qwb.opt file are first included as a distributed tributary inflow assigned to the mainstem branch and interpolation [DTRIC] is turned “OFF”. The corresponding distributed tributary inflow temperatures are usually set to air temperatures in the qdt_br1.npt file. When running water quality, care must be taken as to what constituents should be included in the corresponding inflow constituent concentration file. Typically, only DO values are included if the distributed tributary option of incorporation is used, and they are set to saturated values corresponding to the observed air temperatures. Keep in mind that if the water balance flows are incorporated as branch inflows, then the mass loading of organic matter and nutrients will be increased as well.

The branch corresponding to the distributed tributary inflow is usually assigned to the mainstem branch of a reservoir. Using a distributed tributary minimizes the impact of the flow, temperature, and/or water quality associated with the distributed tributary by distributing the flow throughout all segments in a branch weighted by surface area. Be aware that large flows as a result of large errors in inflow/outflow measurements can and have had a significant impact on temperature and water quality calibration in the surface layers. Usually, this is not a problem, but sensitivity analyses should be conducted to see if the flow and associated temperature/constituent concentrations have an impact on the simulation results. If so, then the following discussion is of particular relevance.

As emphasized previously, a great deal of thought should go into how the flows generated from the water balance utility are incorporated into the simulation. As discussed previously, these are typically incorporated as distributed tributary inflows so as to minimize the impact of the flows on the simulation. However, this may not always be the best, most accurate, or most realistic method. For example, suppose that the water balance flows are consistently negative. This would indicate that either inflows are consistently overestimated or outflows are consistently underestimated. Obviously, incorporating the flows as a

positive increase in the outflows as opposed to subtracting them from the inflows can potentially have a very significant impact on simulation results. In this case, sensitivity analyses should be conducted to determine which method improves the simulation results. If, say, hypolimnetic temperatures are consistently being underestimated, then incorporating the flows into a hypolimnetic outflow could improve the simulation results. Conversely, if hypolimnetic temperatures were being overpredicted, then the inflows should probably be reduced. The key point to keep in mind is that there are a number of different ways to incorporate the computed flows, and they generally should all be tested to determine the best way to incorporate the computed flows into the simulation.

As another example, consider the case in which the generated flows are consistently positive and a branch in which sometimes significant inflows are ungauged. In this case, a sensitivity analyses should be performed to determine if incorporating the flows or a portion of the flows into the ungauged branch inflow improves model results. Oftentimes, the model can be used as a guide as to how best to incorporate the computed flows into the simulation.

Example from Tom Cole

Walter F. George is a U.S. Army Corps of Engineer reservoir located on the Chattahoochee River in Alabama. The reservoir is operated as a peaking hydropower facility. During calibration, the model consistently underpredicted hypolimnetic temperatures by 0.5-1°C. Wind sheltering could be adjusted to increase hypolimnetic temperatures, but this adjustment always adversely impacted thermocline depth. After considerable thought, it was concluded that including possible seepage at the dam might improve hypolimnetic temperature predictions. A portion of the distributed tributary flows were incorporated as an additional outflow at the bottom of the dam. The final value used was $5 \text{ m}^3 \text{ sec}^{-1}$, which was less than 1% of the average outflows and brought hypolimnetic temperatures into almost exact agreement with observed temperatures. Further investigation of the outflows revealed that during times of no power generation, an additional flow of $5.1 \text{ m}^3 \text{ sec}^{-1}$ was specified in a file that was not originally sent as part of the outflow data. Thus, the model pointed the way as to how best to incorporate the computed flows and was a surprisingly accurate indicator of what was occurring in the prototype.

How to use the water balance utility console application

A console application that reads an input file is also available to perform the waterbalance. This new console version of the water balance code provides the following updates:

1. This is not a windows dialog box driven code so it can easily be used in batch files.
2. There is now an input file for model parameters for the water balance utility and an output file for model errors.
3. There are several enhancements: one has more flexibility over file naming, the number of header lines to skip in input files, and the number of waterbodies to use in the analysis.

Input File: "WatBal.npt"

The new input file, "WatBal.npt" has the following format:

```
Water Balance input file for Console Application
"el_obs.npt"      , file for observations, time and water level time series in 2 columns
"tsr_8_seg36.opt" , tsr file for model predictions-assuming only one line skipped in
file header
"qwb1.npt"        , Output file name
1                 , NSKIPS - number of skips of data
1                 , NAV - averaging interval, number of data points to average
```

```

1,1      , waterbody to perform water balance: JW1:STARTWB,JW2:ENDWB:1,2=WB1&2;1,1==WB1
ONLY
3        , number of lines to skip in the header for the water level data file
0        , past water balance PWB file: Yes==1, No==0
"qwb.npt" , Previous Water Balance file name

```

The explanations of each line are shown in the example file above. If performing over multiple waterbodies, keep in mind that all waterbodies must have the same grid (i.e., ELBOT and vertical spacing must be the same). Currently all the flow correction is given to only 1 water balance file. A later option will be for multiple WB files so that one can distribute them across several waterbodies.

Output File: "WatBal_Errors.opt"

If there are errors, any errors will be shown in this text file. If no errors, then this file will not be written to the disk.

An example of the error file output is shown below:

```
Could not open simulated elevations file tsr_8_seg39.opt
```

Output File: "el_stats.opt"

This output file displays model error statistics of the water level and average flow rate in the qwb output file. Typical output from this program are shown below:

N	Mean Error	Absolute ME	RMS Error
238	0.00	0.00	0.00

Average water balance flow correction = .00 m³/s for period covering Julian day 1.04 to 239.94

Output File from water balance

The main output file from the water balance utility is a time series of flows necessary to match the water level.

A typical output file is shown below:

```

Computed flow to complete water balance
      1      1
      JDAY      QWB
1.040      0.00
2.000     24.34
3.000      2.33
4.000      2.34
5.000      3.72
6.000      2.33
7.000      4.67
8.000      2.34
9.000      3.71
10.000     3.29
11.000     4.66
12.000      2.34
13.000     3.69
14.000      2.34
15.000     3.32
16.000     3.31
17.000     3.49

```

How to Run the Console Water Balance Utility

The water balance utility is run by executing the exe file for 32 (**WBconsole32.exe**) or 64 bit (**WBconsole64.exe**) Windows operating system. One also needs the input file **WatBal.npt** in the directory of the executable.

3. Control File Converter for CE-QUAL-W2

This section outlines the steps required to convert a legacy application control file, **w2_con.npt**, and its companion **graph.npt** file into the Excel version of the control file or to convert a **w2_con.csv** file from 4.1 or 4.22 to 4.5.

This is a relatively simple process that involves these steps:

1. Copy **ConverterControlFile.exe** into the directory of a legacy application or copy the three files read by the program (**w2_con.npt**, **graph.npt**, and the bathymetry file) into a directory with the executable. If you are just converting a 4.1 or 4.22 **w2_con.csv** file to the new format, just copy **w2_con.csv** and the bathymetry file into a new directory. After execution (double click the converter executable), the following files are written to the directory:
 - a. **w2_con45.csv** – this is a csv file format of the control file. This will then need to be copied into a Version 4.5 example **w2_con.xlsm** template.
 - b. A csv form of the bathymetry file (if the bathymetry file was not already in csv format). The output bathymetry file is named **bthX.csv**, where X is the waterbody number. This new format is much easier for editing and analysis than the older file format. The file name in the **w2_con.csv** is also changed to **bthX.csv**.
2. Copy **w2_con.csv** into an existing Version 4.5 example file ***.xlsm** from another application (you can use one from the W2 Model Examples) by following these steps:
 - a. Open the file **w2_con.xlsm** (or it may have a different descriptive name, such as **w2_con_DeGray.xlsm**) from an existing example problem supplied in the CE-QUAL-W2 example problems.

Row	Column A	Column B	Column C	Column D	Column E	Column F	Column G	Column H	Column I	Column J	Column K	Column L	Column M
1	Note COL A and B are not written out to w2_con.csv	w2_con.csv file format	CE-QUAL-W2 Version	4.2.1									
2			Control File version	4.2.1	w2_con.csv								
3	Fixed length of file except when more than 5 algae, 5 zooplankton, 5 macrophytes, 5 structures, 5 periphyton groups.	TITLE C	Title comments: next 10 lines										
4			"Version 3.7 Long Lake Model"										
5	The # of rows though changes with the # of active water quality constituents.	Any comment - this is written only to the SNP file	"Workshop water balance problem"										
6	Do not change the file tab name for this sheet since the output file name is tied to the name of the tab												
7	NWB: # of waterbodies		NWB	NBR	IMX	KMX	NPROC	CLOSEC					
8	NBR: number of branches			1	1	37	47	1	ON				
9	IMX: maximum number of segments including inactive segments												
10	KMX: maximum number of vertical layers including inactive layers (top and bottom)												
11	NPROC: # of processors (INACTIVE at this time)												
12	CLOSEC: close dialog box after executing if =ON												
13	NTR: number of tributaries		NTR	NST	NIWD	NWD	NGT	NSP	NPI	NPU			
14	NST: maximum # of structures in a branch			1	1	0	0	0	0	0			
15	NIWD: # of internal weirs												
16	NWD: # of withdrawals												
17	NGT: # of gates												
18	NSP: # of spillways												
19	NPI: # of pipes												
20	NPU: # of pumps or water level control rules												
21	NGC: # of generic water quality constituents		NGC	NSS	NAL	NEP	NBOD	NMC	NZP				
22				5	1	3	1	10	0	1			
23	NDAY: Maximum number of output dates or timestep related changes		NDAY	SELECTC	HABITATC	ENVIRPC	AERATEC	INITUWL					
24	SELECTC: Turn ON/OFF/USGS automatic port selection from a multiple outlet structure			100/OFF	ON	OFF	OFF	OFF					
25	HABITATC: Turn ON/OFF habitat analyses for fish and eutrophication variables												
26	ENVIRPC: Turn ON/OFF environmental performance criteria												
27	AERATEC: Turn ON/OFF aeration to waterbody with dissolved oxygen probe control		TMSTRT	TMEND	YEAR								
28	INITUWL: Turn ON/OFF initial water surface slope and velocity calculation for a river system			1.0402	240	2000							
29			NDLT	DLTMIN	DLTINTER								
30	Do not change bolded headers in COL C - these are checked by the program			6	0.1	ON							
31													

Figure 3. An example of an existing **w2_con.xlsm** file. Columns A and B are not used in the control file and we will be pasting the **w2_con45.csv** into column C1, not A1.

- i. The easiest option is to open the file **w2_con45.csv** from the converter utility in Excel. Select the columns and rows with data. Do not select the entire sheet

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	CE-QUAL-W2 Version 4.2.1																					
2	Control File version 4.2.1		w2_con.csv																			
3	Title comments: next 10 lines																					
4	"Version 3.7 Long Lake Model"																					
5	"Workshop water balance problem"																					
6	***																					
7	***																					
8	***																					
9	***																					
10	***																					
11	Tom Cole WES; Scot PSU; Rob J PSU; Chris PSU"																					
12	***																					
13	***																					
14																						
15	NWB	NBR	IMX	KMX	NPROC	CLOSEC																
16		1	1	37	47	1 ON																
17																						
18	NTR	NST	NIW	NWD	NGT	NSP	NPI	NPU														
19		1	1	0	0	0	0	0														
20																						
21	NGC	NSS	NAL	NEP	NBOD	NMC	NZP															
22		5	1	3	1	10	0	1														
23																						
24	NDAY	SELECTC	HABTATC	ENVIRPC	AERATEC	INITUWL																
25		100 OFF	OFF	OFF	OFF	OFF																
26																						
27	TMSTRT	TMEND	YEAR																			
28		100	272	2000																		
29																						

- ii. Then Copy the selected cells from **w2_con45.csv** and paste to **w2_con.xlsm** cell C1 using *Paste Values and Number Formatting* so that the colors and other formatting are preserved in the original document.
- b. The next step is to adjust rows for the number of constituents. The template file you used **w2_con.xlsm** had an assumed number of water quality constituents that may be different from the number you will use. Go to approximately row 384 as shown below and delete or add cells **in only column A and B** only to match the specified number in the formula. So, if you need to add cells, then select the correct number of cells in column A and B, right click your mouse and choose 'Insert', then 'shift cells down.' If you need to delete cells, then select the correct number of cells in column A and B, right click your mouse, choose 'Delete', then select 'move cells up'. Now the information in column A and B should line up with what is written in Column C.

	A	B	C	D	E	F	G	H	I	J	K	L	M
604		ANEQN NH4 NO3 preference equation #	2	2	2								
605		ANPR Parameter for ANEQNR2	0.001	0.001	0.001								
606		O2AR Stoichiometric ratio of O2 to algae biomass, for algal respiration (mg O2/mg a	1.1	1.1	1.1								
607		O2AG Stoichiometric ratio of O2 to algae biomass, for algal primary production (mg	1.8	1.8	1.8								
608													
609		EPIPHYTON	WB1	WB2	WB3	WB4	WB5	WB6	WB7	WB8	WB9	WB10	
610		EPIC Turn ON/OFF Periphyton group 1	ON										
611		EPRC Turn ON/OFF print for Periphyton group 1	ON										
612		EPIC INIT Initial areal density Periphyton group 1 g/m2	10										
613		EPIC Turn ON/OFF Periphyton group 2											
614		EPIC Turn ON/OFF print for Periphyton group 2											
615		EPIC INIT Initial areal density Periphyton group 2 g/m2											
616		EPIC Turn ON/OFF Periphyton group 3											
617		EPRC Turn ON/OFF print for Periphyton group 3											
618		EPIC INIT Initial areal density Periphyton group 3 g/m2											
619		EPIC Turn ON/OFF Periphyton group 4											
620		EPRC Turn ON/OFF print for Periphyton group 4											
621		EPIC INIT Initial areal density Periphyton group 4 g/m2											
622		EPIC Turn ON/OFF Periphyton group 5											
623		EPRC Turn ON/OFF print for Periphyton group 5											
624		EPIC INIT Initial areal density Periphyton group 5 g/m2											
625													
626		EPIPHYTON growth rate constants for each periphyton group	EP1	EP2	EP3	EP4	EP5	EP6					
627		EG growth rate day-1	1.2										
628		ER respiration rate day-1	0.04										
629		EF excretion rate day-1	0.04										
630		EM mortality rate day-1	0.1										
631		EB burial rate day-1	0.001										
632		EHSP half saturation constant P: g/m^3	0.003										
633		EHSN half saturation constant N: g/m^3	0.014										
634		EHSSi half saturation constant Si: g/m^3	0										

Figure 7. Location in Excel file where the number of rows needs to be increased in case there are more than 5 epiphyton/periphyton groups..

- iii. For more than 5 algae and 5 zooplankton groups, look in the zooplankton section (search for zooplankton).

	A	B	C	D	E	F	G	H	I	J	K	L	M
655		ZOOPLANKTON RATES	Zoo1	Zoo2	Zoo3	Zoo4	Zoo5	Zoo6	Zoo7				
656		ZG growth rate day-1	1.5										
657		ZR respiration rate day-1	0.1										
658		ZM mortality rate day-1	0.01										
659													
660		ZEFF Zooplankton assimilation efficiency or the proportion of food assimilated to fo	0.5										
661		PREFP Preference factor of zooplankton for detritus or LPOM (dimensionless), from	0.5										
662		ZOOMIN Threshold food concentration at which zooplankton feeding begins, g m-3	0.01										
663		ZS2P Zooplankton half saturation constant for food (includes LPOM, algae, and zoop	0.3										
664		ZT1 Lower temperature for zooplankton growth, oC	0										
665		ZT2 Lower temperature for maximum zooplankton growth, oC	15										
666		ZT3 Upper temperature for maximum zooplankton growth, oC	20										
667		ZT4 Upper temperature for zooplankton growth, oC	36										
668		ZK1 Fraction of maximum zooplankton growth rate at ZT1	0.1										
669		ZK2 Fraction of maximum zooplankton growth rate at ZT2	0.9										
670		ZK3 Fraction of maximum zooplankton growth rate at ZT3	0.98										
671		ZK4 Fraction of maximum zooplankton growth rate at ZT4	0.1										
672		ZP Stoichiometric equivalent between zooplankton biomass and phosphorus	0.015										
673		ZN Stoichiometric equivalent between zooplankton biomass and nitrogen	0.08										
674		ZC Stoichiometric equivalent between zooplankton biomass and carbon	0.45										
675		O2ZR Oxygen stoichiometry for zooplankton respiration (mg O2/mg zooplankton or	1.1										
676		PREFA-Algal Group 1 Preference factor of zooplankton for algae (dimensionless) fro	1										
677		PREFA-Algal Group 2 Preference factor of zooplankton for algae (dimensionless) fro	0.5										
678		PREFA-Algal Group 3 Preference factor of zooplankton for algae (dimensionless) fro	0.5										
679		PREFA-Algal Group 4 Preference factor of zooplankton for algae (dimensionless) from 0 to 1.	0										
680		PREFA-Algal Group 5 Preference factor of zooplankton for algae (dimensionless) from 0 to 1.	0										
681		PREFZ-Group 1 Preference factor of zooplankton for zooplankton (dimensionless) fr	0										
682		PREFZ-Group2 Preference factor of zooplankton for zooplankton (dimensionless) from 0 to 1.											
683		PREFZ-Group3 Preference factor of zooplankton for zooplankton (dimensionless) from 0 to 1.											
684		PREFZ-Group4 Preference factor of zooplankton for zooplankton (dimensionless) from 0 to 1.											
685		PREFZ-Group5 Preference factor of zooplankton for zooplankton (dimensionless) from 0 to 1.											

Figure 8. Location in Excel file where the number of rows needs to be increased in case there are more than 5 zooplankton groups.

- iv. For more than 5 macrophyte groups, look in the macrophyte section (search for macrophyte).

A	B	C	D	E	F	G	H	I	J	K	L	M
687 Initial conditions for 5 macrophyte groups - rows must be added to if more than 5 macrophyte groups	MACROPHYTES	WB1	WB2	WB3	WB4	WB5	WB6	WB7	WB8	WB9	WB10	
688 You must keep the 5 groups even if not used	MAC Waterbody macrophyte 1 computations, ON or OFF	OFF										
689	MAC Waterbody macrophyte 2 computations, ON or OFF	OFF										
690	MAC Waterbody macrophyte 3 computations, ON or OFF	OFF										
691	MAC Waterbody macrophyte 4 computations, ON or OFF	OFF										
692 Increase rows if more than 5 macrophyte groups	MAC Waterbody macrophyte 5 computations, ON or OFF	OFF										
693	MPRWBC Macrophyte 1 concentration print output, ON or OFF	OFF										
694	MPRWBC Macrophyte 2 concentration print output, ON or OFF	OFF										
695	MPRWBC Macrophyte 3 concentration print output, ON or OFF	OFF										
696	MPRWBC Macrophyte 4 concentration print output, ON or OFF	OFF										
697 Increase rows if more than 5 macrophyte groups	MPRWBC Macrophyte 5 concentration print output, ON or OFF	OFF										
698	MACWBCI-Group1 Initial macrophyte concentration for each macrophyte group, gm-3	0										
699	MACWBCI-Group2 Initial macrophyte concentration for each macrophyte group, gm-3											
700	MACWBCI-Group3 Initial macrophyte concentration for each macrophyte group, gm-3											
701	MACWBCI-Group4 Initial macrophyte concentration for each macrophyte group, gm-3											
702 Increase rows if more than 5 macrophyte groups	MACWBCI-Group5 Initial macrophyte concentration for each macrophyte group, gm-3											
703												
704	MAC RATE	MacGroup1	MacGroup	MacGroup	MacGroup	MacGroup	MacGroup	MacGroup	MacGroup	MacGroup	MacGroup	MacGroup
705	MG maximum macrophyte growth rate, day-1	0.3										
706	MR maximum macrophyte respiration rate, day-1	0.05										
707	MM maximum macrophyte mortality rate, day-1	0.05										
708	MSAT light saturation intensity at maximum photosynthetic rate, W m-2	30										
709	MHSP macrophyte half-saturation for phosphorus limited growth, g m-3	0										
710	MHNP macrophyte half-saturation for nitrogen limited growth, g m-3	0										
711	MHSC macrophyte half-saturation for carbon limited growth, g m-3	0										
712	MPOCM Fraction of macrophyte biomass that is converted to particulate organic matter	0.9										
713	LIPMAC Fraction of POC which originates as dead macrophytes becoming labile POC	0.2										
714	PSED Fraction of phosphorus uptake by macrophytes obtained from sediments	0.5										
715	NSED Fraction of nitrogen uptake by macrophytes obtained from sediments	0.5										
716	MBMP Threshold macrophyte concentration for which growth is moved to the above	40										
717	MMAX Maximum macrophyte concentration, g m-3	500										

Figure 9. Location in Excel file where the number of rows needs to be increased in case there are more than 5 macrophyte groups.

- d. You will need to put quotes around the **TITLE** field, **HNAME** field, **CNAME** field, and **CDNAME** field. This can be done by setting up a formula using the existing cells as `'=char(34)&[CELLREF]&char(34)'` where CELLREF is the cell reference number of the original text. Then paste the **values** with the double quotes to the locations in the control file. ***If anyone figures out an easier way to do this let me know! Frustratingly, Excel drops the double quotes on importing them.***
- e. The **w2_con.xlsm** file should now be working. As edits are made in this file, you will push the button on the top of the file in Column B to export it to **w2_con.csv**, which is read in by the W2 model. The preprocessor will 'yell' at you in case there is an error in the set up.

4. GUI Interface for CE-QUAL-W2 for Versions 3.7-4.20

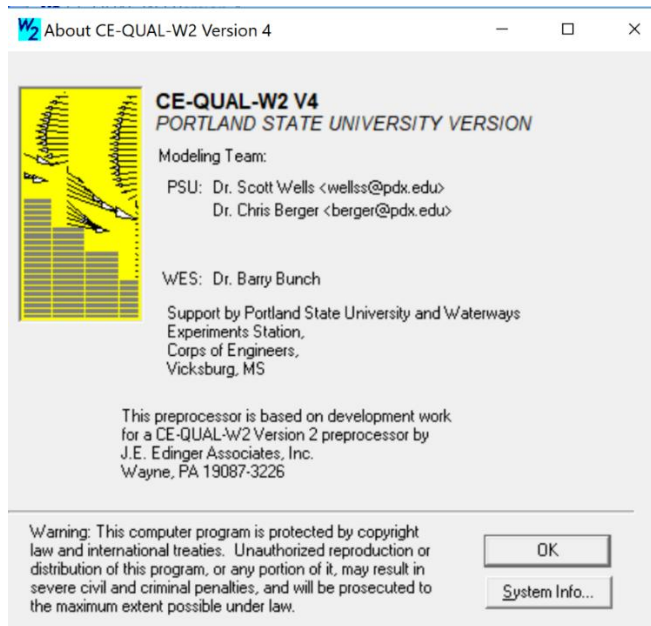


Figure 10. About W2 Control GUI preprocessor.

Introduction

This section describes the GUI interface that is released with CE-QUAL-W2. This works with the old version of the control file, **w2_con.npt** (only up to Version 4.2) but does work with the Excel version of the control file (**w2_con.csv**). The w2_post program shown in the next section has a bathymetry viewer that can be used with the **w2_con.csv** control file.

The manual describes the uses of the GUI for editing the control file (**w2_con.npt**) and the bathymetry file (user-defined filename, typically called **bth.npt** or **bth.csv**). Please note that the model files can be edited using a text editor or in Excel, and the GUI is therefore not required to run the model. The GUI though may be very helpful for the new CE-QUAL-W2 user as is the Excel version of the control file. The bathymetry editor is very useful in viewing the model grid and making changes to the grid (e.g., grid refinement). The bathymetry editor though does not assist in setting up the bathymetry. The model user must have already developed the bathymetry file using other programs. The bathymetry editor only edits an existing bathymetry file. Often the bathymetry is developed using GIS software or programs like SURFER or other 3-D contour plotting software packages.

Installing the GUI Interface

After downloading the GUI interface setup files and unzipping them, from Windows Explorer double-click on 'setup.exe' to guide you in the set-up process.

Opening the GUI Interface

By double-clicking on the W2 icon from the Desktop or from Windows Explorer, the user is presented with the following screen:

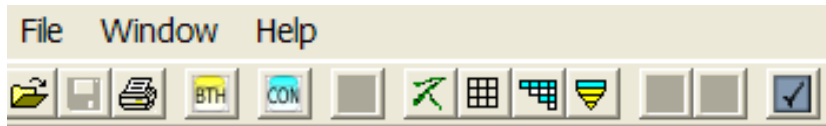


Figure 11. Dialog box menu.

This interface is meant to be simple and uncomplicated.

Menus

The user has 3 menus: File, Window, and Help. The File menu controls which control file to open and to save. The user can only open a CE-QUAL-W2 control file – such as w2_con.npt. Opening any other file type will result in an error. The File save button and file open buttons and menu items only affect the control file and the associated files (bathymetry and graph files). Note that when you save a file, you overwrite the original file with which you started. The GUI program though copies the original output file to a similar name but with a number designator. This allows the user always to go back and track changes made over time to the control file or bathymetry files.

Button Bar

The buttons become active as the user opens the files that affect the buttons. For example, once the bathymetry file is opened, the 4 grid viewing options are then displayed (see more information below). The open file button opens the w2_con.npt file or another file that is in the format of the control file. The save button becomes active once a file is opened and changes are made. The print button prints the current screen (I would not use this since it is not a very useful print out. It is much more convenient to just print the text file directly from Notepad or other text editor.) The next buttons are described below.

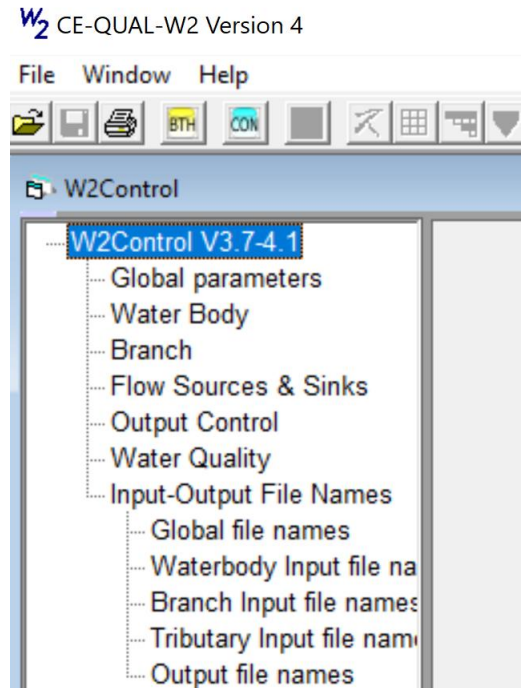


Figure 12. Menu of w2control.

Editing the Control File

Once you have chosen a control file to open (w2_con.npt), open it and adjust the window so that the window options are visible. If the file does not open there may be serious formatting errors in the file or you may have not chosen a CE-QUAL-W2 control file. Check the file for errors using the CE-QUAL-W2 preprocessor (a separate program released with W2).

You will notice that the BTH button is now active, meaning that you can also open the bathymetry file. The left pane now has 7 headings – each subheadings that can be shown by double-clicking on the main heading. For example, several of the subheadings have been expanded, as shown to your right.

By double-clicking on a subheading, the user can edit the information in a cell. Information about each model parameter is shown in Appendix C in the User Manual.

Once you have edited the cells (either by changing a number or changing a menu parameter), there are 2 buttons at the bottom of the page as shown below.

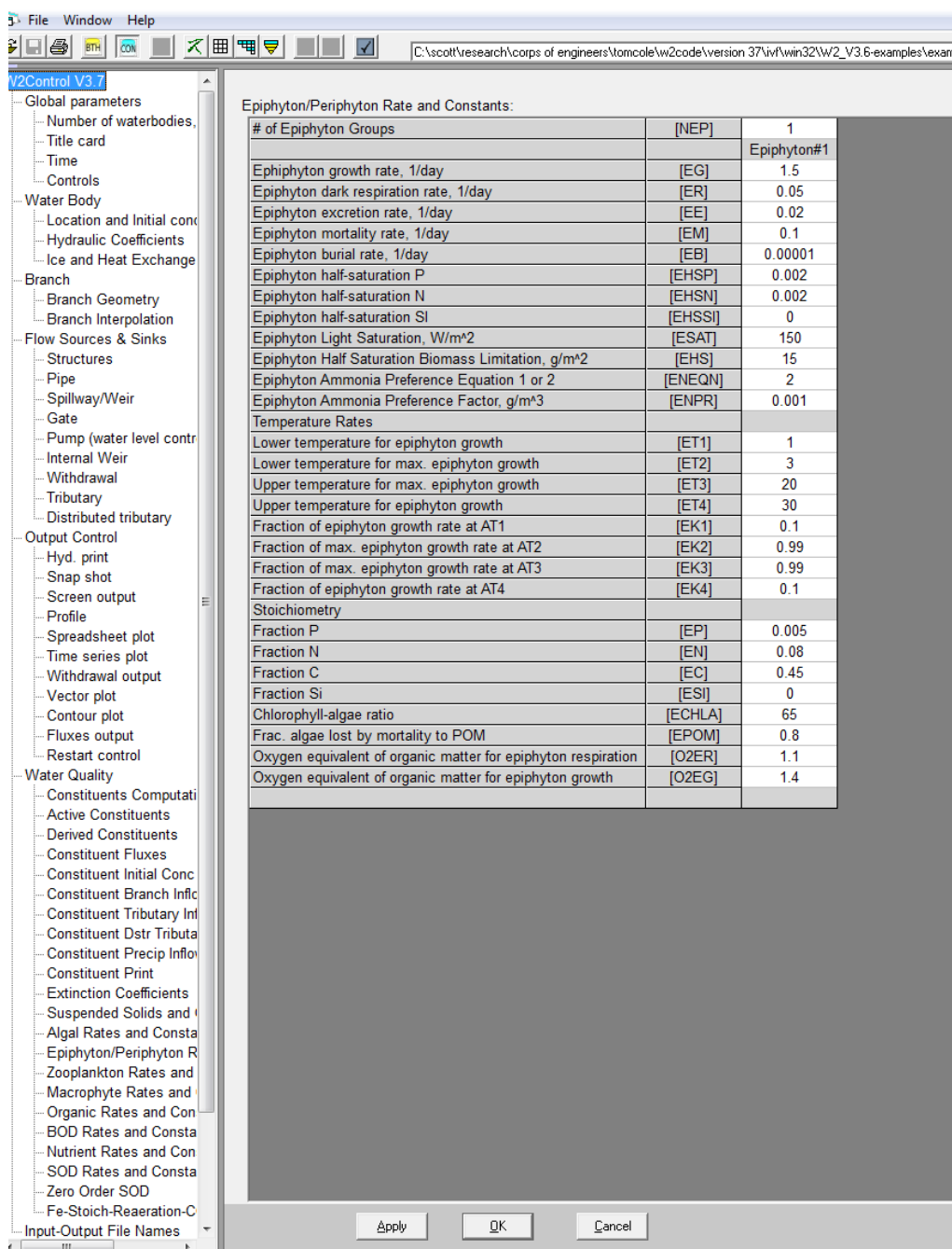


Figure 13. Example of w2control menu.

The Apply button is used to set the parameters of the table. For example above, the # of epiphyton groups is '1'. If one wanted to change this to "2". One would enter 2 in the cell for NEP and then press "Apply". The table will be re-drawn as shown below.

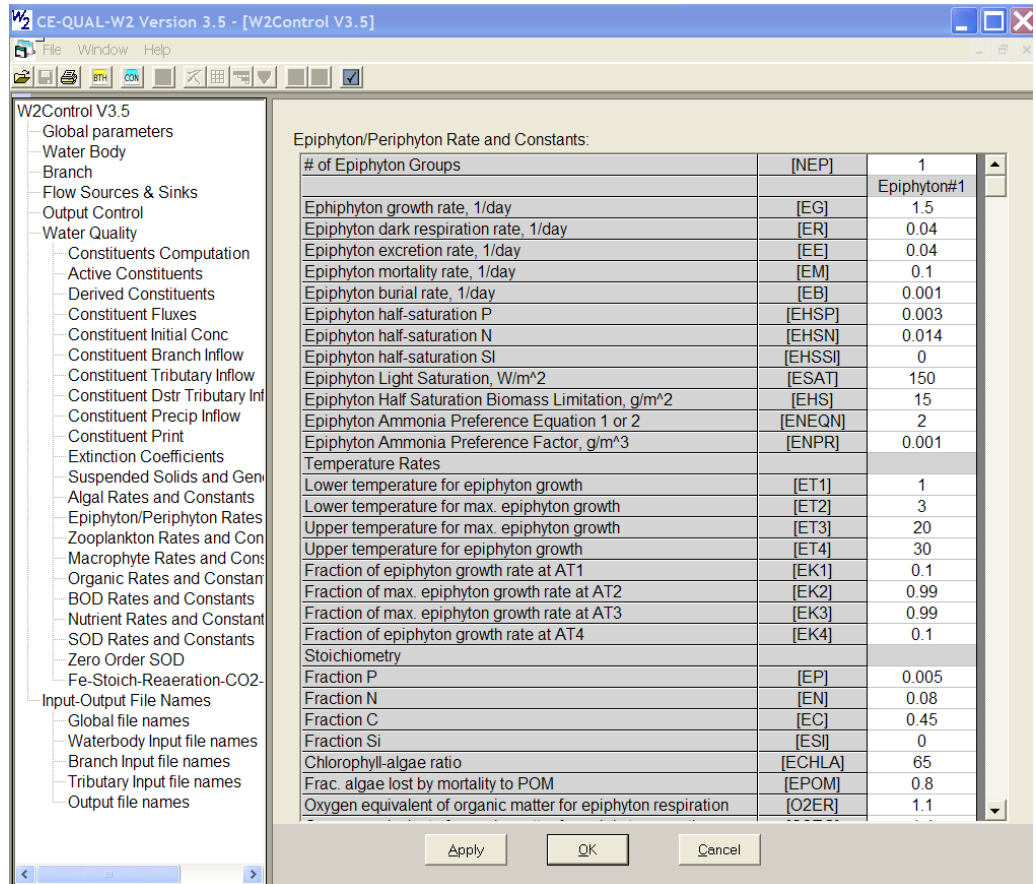


Figure 14. Dialog box example.

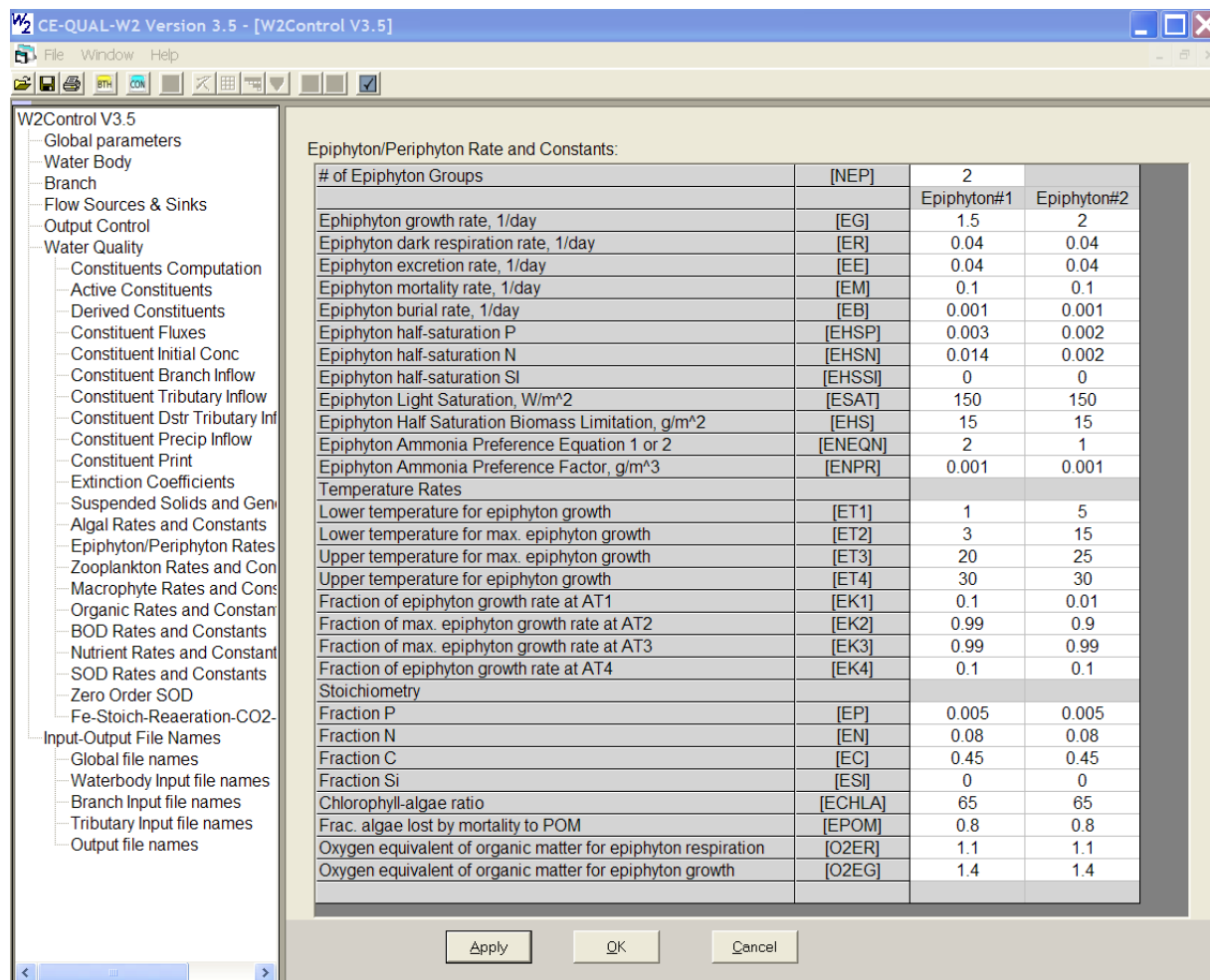


Figure 15. Dialog box example.

Then after making changes, the OK button MUST be pressed. But this does not SAVE the file to the hard disk. The User must also then press the SAVE button on the button bar above (or the menu FILE/SAVE file) to save these changes to the disk. The save button will be highlighted after a selection is changed.

Editing the Bathymetry File

After opening the control file, the bathymetry file can be opened by clicking the BTH button. Note that the bathymetry file name is under the 'INPUT-OUTPUT FILE NAMES' Menu and the submenu 'WATERBODY INPUT FILE NAMES'. At this point there are 4 views that become active: Side View, Top View, Data View, End View.

The buttons on the top allow one to toggle between the different bathymetric views. By expanding the Top View, one can enlarge the view by holding down the left mouse button and scrolling from right to left (see example below). To decrease the image size, hold down the left mouse button and scroll from right to left.

To change the active segment and layer, you must place the mouse cursor in the Data View at a segment and layer of interest and click the 'Refresh View' button. This affects which segment is shown in the End View and which segment is highlighted in the Side View.

To edit the bathymetry, click on the variable and change its value. Then the user must click the 'OK' button to accept these changes. Using the SAVE button saves all the user files.

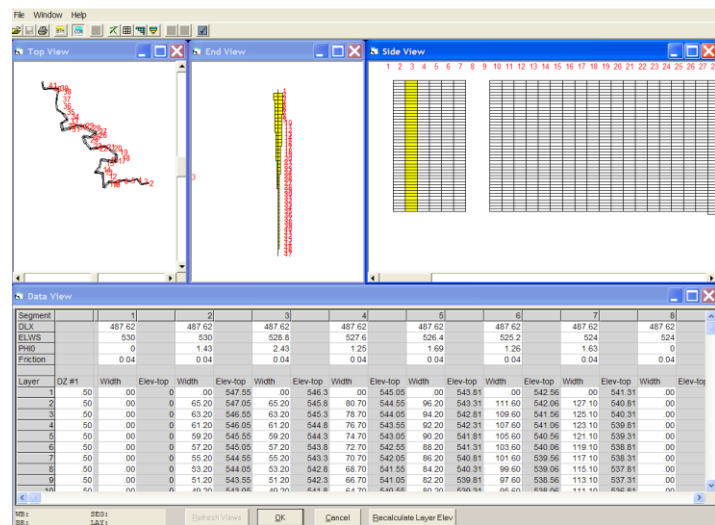


Figure 16. Bathymetry dialog box example.

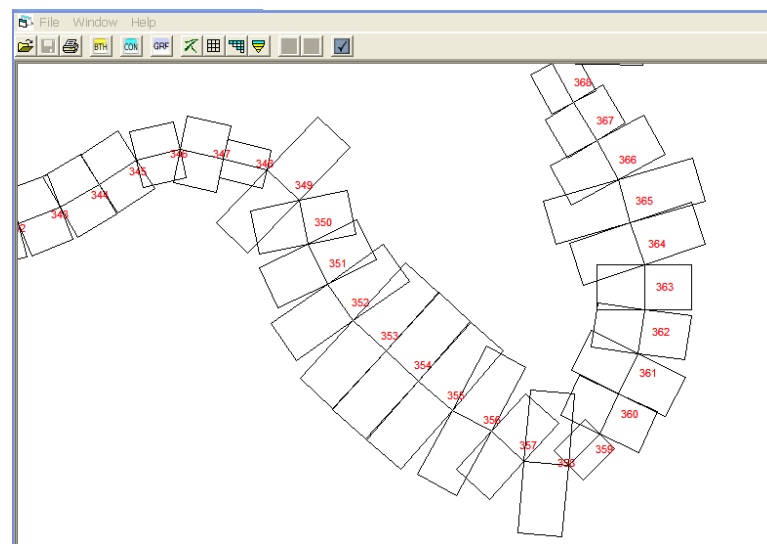


Figure 17. Segments of a model dialog box.

Merging or Splitting Segments or Layers

In order to perform this operation, open the Data View and select the range of layers and/or segments to merge or split by highlighting them with the mouse. You will notice in the left hand corner a box shows what segments and layers have been selected.

In the example below I have selected layers 5-14 and segments 2 and 3. You will also now notice that the layer and segment merge/split buttons are now active. To merge or split segments, click on the Merge/Split segment button. A small dialog box will come up with options for merging or splitting the segments (see below). You then have the option to merge the 2 segments or split them into many more segments. The control file (w2_con.npt) will automatically adjust segment numbers when using this option. But keep in

mind that the following files will need to be edited to adjust them for changes in the number of model segments: the wind sheltering coefficient file (wsc.npt) and the shading file (shade.npt).

A similar process is followed for merging or splitting layers.

Segment	1	2	3	4	5	6	7	8
DLX	250.2	250.2	250.2	250.2	250.2	250.2	250.2	250.2
ELWS	41.63	41.63	41.5	41.37	41.23	41.1	40.97	40.84
PHIO	0	4.21	4.37	4.32	3.8	3.48	3.45	3.41
Friction	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04

Layer	DZ #1	DZ #2	DZ #3	Width	Elev-top	Width	Elev-top	Width	Elev-top	Width	Elev-top	Width	Elev-top	Width	Elev-top	Width	Elev-top
1	1.00	1.00	1.00	.00	0	.00	51.16	.00	51.03	.00	50.9	.00	50.77	.00	50.64	.00	50.51
2	1.00	1.00	1.00	.00	0	284.76	50.16	307.73	50.03	373.35	49.9	236.76	49.77	249.74	49.64	325.37	49.51
3	1.00	1.00	1.00	.00	0	284.76	49.16	307.73	49.03	373.35	48.9	236.76	48.77	249.74	48.64	325.37	48.51
4	1.00	1.00	1.00	.00	0	284.76	48.16	307.73	48.03	373.35	47.9	236.76	47.77	249.74	47.64	325.37	47.51
5	1.00	1.00	1.00	.00	0	284.76	47.16	307.73	47.03	373.35	46.9	236.76	46.77	249.74	46.64	325.37	46.51
6	1.00	1.00	1.00	.00	0	284.76	46.16	307.73	46.03	373.35	45.9	236.76	45.77	249.74	45.64	325.37	45.51
7	1.00	1.00	1.00	.00	0	284.76	45.16	307.73	45.03	373.35	44.9	236.76	44.77	249.74	44.64	325.37	44.51
8	1.00	1.00	1.00	.00	0	284.76	44.16	307.73	44.03	373.35	43.9	236.76	43.77	249.74	43.64	325.37	43.51
9	1.00	1.00	1.00	.00	0	284.76	43.16	307.73	43.03	373.35	42.9	236.76	42.77	249.74	42.64	325.37	42.51
10	1.00	1.00	1.00	.00	0	284.76	42.16	307.73	42.03	373.35	41.9	236.76	41.77	249.74	41.64	325.37	41.51
11	1.00	1.00	1.00	.00	0	284.76	41.16	307.73	41.03	373.35	40.9	236.76	40.77	249.74	40.64	325.37	40.51
12	1.00	1.00	1.00	.00	0	284.76	40.16	307.73	40.03	373.35	39.9	236.76	39.77	249.74	39.64	325.37	39.51
13	1.00	1.00	1.00	.00	0	284.76	39.16	307.73	39.03	373.35	38.9	236.76	38.77	249.74	38.64	325.37	38.51
14	1.00	1.00	1.00	.00	0	282.94	38.16	302.99	38.03	365.51	37.9	227.49	37.77	235.22	37.64	296.11	37.51
15	1.00	1.00	1.00	.00	0	280.54	37.16	296.96	37.03	358.02	36.9	220.05	36.77	226.26	36.64	270.19	36.51
16	1.00	1.00	1.00	.00	0	167.55	36.16	207.53	36.03	343.54	35.9	209.81	35.77	213.84	35.64	240.60	35.51
17	1.00	1.00	1.00	.00	0	122.64	35.16	117.59	35.03	242.47	34.9	188.69	34.77	196.98	34.64	180.43	34.51
18	1.00	1.00	1.00	.00	0	18.33	34.16	17.71	34.03	102.91	33.9	151.01	33.77	113.00	33.64	144.18	33.51
19	1.00	1.00	1.00	.00	0	.00	33.16	.00	33.03	12.78	32.9	24.13	32.77	17.59	32.64	22.53	32.51
20	1.00	1.00	1.00	.00	0	.00	32.16	.00	32.03	.00	31.9	.00	31.77	.00	31.64	.00	31.51
21	1.00	1.00	1.00	.00	0	.00	31.16	.00	31.03	.00	30.9	.00	30.77	.00	30.64	.00	30.51
22	1.00	1.00	1.00	.00	0	.00	30.16	.00	30.03	.00	29.9	.00	29.77	.00	29.64	.00	29.51
23	1.00	1.00	1.00	.00	0	.00	29.16	.00	29.03	.00	28.9	.00	28.77	.00	28.64	.00	28.51
24	1.00	1.00	1.00	.00	0	.00	28.16	.00	28.03	.00	27.9	.00	27.77	.00	27.64	.00	27.51
25	1.00	1.00	1.00	.00	0	.00	27.16	.00	27.03	.00	26.9	.00	26.77	.00	26.64	.00	26.51
26	1.00	1.00	1.00	.00	0	.00	26.16	.00	26.03	.00	25.9	.00	25.77	.00	25.64	.00	25.51
27	1.00	1.00	1.00	.00	0	.00	25.16	.00	25.03	.00	24.9	.00	24.77	.00	24.64	.00	24.51
28	1.00	1.00	1.00	.00	0	.00	24.16	.00	24.03	.00	23.9	.00	23.77	.00	23.64	.00	23.51
29	1.00	1.00	1.00	.00	0	.00	23.16	.00	23.03	.00	22.9	.00	22.77	.00	22.64	.00	22.51
30	1.00	1.00	1.00	.00	0	.00	22.16	.00	22.03	.00	21.9	.00	21.77	.00	21.64	.00	21.51
31	1.00	1.00	1.00	.00	0	.00	21.16	.00	21.03	.00	20.9	.00	20.77	.00	20.64	.00	20.51
32	1.00	1.00	1.00	.00	0	.00	20.16	.00	20.03	.00	19.9	.00	19.77	.00	19.64	.00	19.51
33	1.00	1.00	1.00	.00	0	.00	19.16	.00	19.03	.00	18.9	.00	18.77	.00	18.64	.00	18.51
34	1.00	1.00	1.00	.00	0	.00	18.16	.00	18.03	.00	17.9	.00	17.77	.00	17.64	.00	17.51
35	1.00	1.00	1.00	.00	0	.00	17.16	.00	17.03	.00	16.9	.00	16.77	.00	16.64	.00	16.51
36	1.00	1.00	1.00	.00	0	.00	16.16	.00	16.03	.00	15.9	.00	15.77	.00	15.64	.00	15.51
37	1.00	1.00	1.00	.00	0	.00	15.16	.00	15.03	.00	14.9	.00	14.77	.00	14.64	.00	14.51
38	1.00	1.00	1.00	.00	0	.00	14.16	.00	14.03	.00	13.9	.00	13.77	.00	13.64	.00	13.51
39	1.00	1.00	1.00	.00	0	.00	13.16	.00	13.03	.00	12.9	.00	12.77	.00	12.64	.00	12.51

Figure 18. Splitting or merging layers example.

Note that in the bottom left corner of the above screen shot, the layers and segments that were selected is shown.

Merge/Split Segments

Starting Segment: 2

End Segment: 3

Current Number of Segments: 2

Total DLX: 500.4

Merge/Split into: segments

Interpolation Style:

- ☐ Average
- ☐ Linear
- ☒ Piecewise Linear

OK Cancel Help

Figure 19. Merge/split dialog box.

Running the Preprocessor within the GUI

The preprocessor which checks all input files can be executed from within the GUI by clicking on the CHECK button. At which point a dialogue box will pop up with information about running the preprocessor. The preprocessor executable must be in the root directory with the input files.

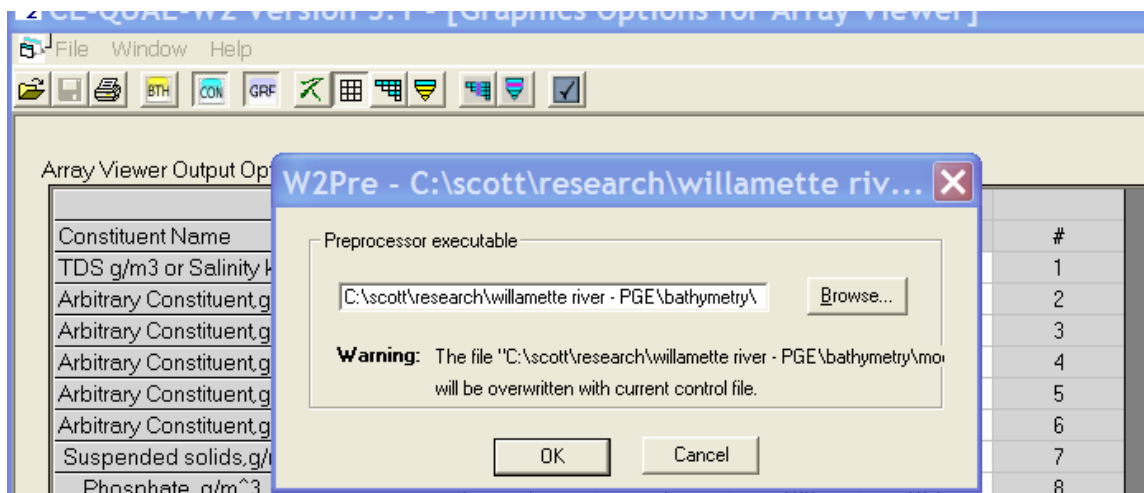


Figure 20. Running dialog box within w2control.

A Note on Saving Files

The GUI interface makes a backup copy of each file opened during a session. It will add an integer number to the files opened. This integer is based on whether previous backup files have been created or not. For example, the 1st time the GUI opens the files, w2_con.npt, graph.npt, and bth.npt; the program will write as backup copies of the unaltered original files: w2_con.npt2, graph.npt2, and bth.npt2. The next time the program opens these files, the files w2_con.npt3, graph.npt3, and bth.npt3 will be created (if you have already deleted the *.*2 files from your hard disk, the GUI will again use the integer '2' rather than '3').

Typical Errors with the GUI

Sometimes the GUI cannot read a control file because of the format of the text. UNIX/Linux systems often use a line-feed at the end of a line, while Windows text files include a carriage-return and a line-feed. You must convert the file to a Windows text file for the GUI to work properly. In a text editor you can view all the characters and verify you have the proper Windows format if the GUI is having problems reading the file. Many text editors automatically convert files from one type to the other (such as the text editor 'Notepad ++').

It is recommended that you use the W2 preprocessor to catch errors in the set-up files first before using the GUI. The W2 preprocessor though will not catch the line-feed error noted above and will read the files correctly whether in UNIX/Linux or Windows format.

Command Line Processing

The GUI also can be controlled by command line passing of the working directory and file. In a batch program or from the command line in a DOS box you can execute the GUI as follows:

```
"C:\scott\research\corps of engineers\tomcole\w2code\GUI36\w2control\w2control4.exe"  
C:\scott\w2workshop\2009 workshop\waterqual\problem1\w2_con.npt
```

The first string in quotes executes the GUI (this shows the path to the GUI – your path to the GUI will be different). The command line argument is NOT in quotes. This program was developed in VB6 and does not take quotes around the command line. Note that this is different than the FORTRAN command line argument. So the above command will open the GUI and load the control file automatically from the directory “C:\scott\w2workshop\2009 workshop\waterqual\problem1”.

Using the GUI W2Control on Touchscreen Laptops and Monitors

The software, W2Control, is a GUI preprocessor for the W2 model. It works fine on non-touch enabled monitors. But for touch screen monitors, like many of the latest laptops, the opening “treeview” menu does not work because of a software incompatibility with VB6, the source code. W2Control can though be used on a touch screen laptop by doing the following:

Go to Services by typing ‘Services’ in the command line or Cortana line. In the list choose ‘Touch keyboard and Handwriting Panel Service’. Right click your mouse and choose Properties and change ‘Automatic’ to ‘Disabled’. Then click STOP in the Service Status to stop the service. Click APPLY. The W2Control then works as expected.

5. Post-Processor for CE-QUAL-W2 by DSI, Inc.

Paul Craig at DSI, Inc. has generously provided a software system for evaluating model results. There is a short user’s manual produced by DSI that documents the software utility and this is a separate pdf file download. This post-processor currently works for Version 3.7 to the current version. This post-processor, as shown in Figure 21 and Figure 22, allows the user to look at model bathymetry, show time series, profile plots, contour plots, velocity vectors, and animations with up to 4 simultaneous state variables.

Quick Guide for the CE-QUAL-W2 Post Processor W2_Post

July 2012 Updates June 2016

W2_Post

A post-processor for CE-QUAL-W2 Version3/4 that provides the user with a broad range of visualization and analyses of the model results. W2_Post provides for rapid visualization and assessment of W2 model results. W2_Post uses a binary file generated by the CE-QUAL-W2 (i.e. the "W2L" file extension) for all of its model data analysis. No need to output multiple types of output from W2. The post-processor provides extensive model calibration / measured data comparison tools and statistics. The following are summaries of each major type of post-processing available.

Figure 21. Splash screen for W2_Post.

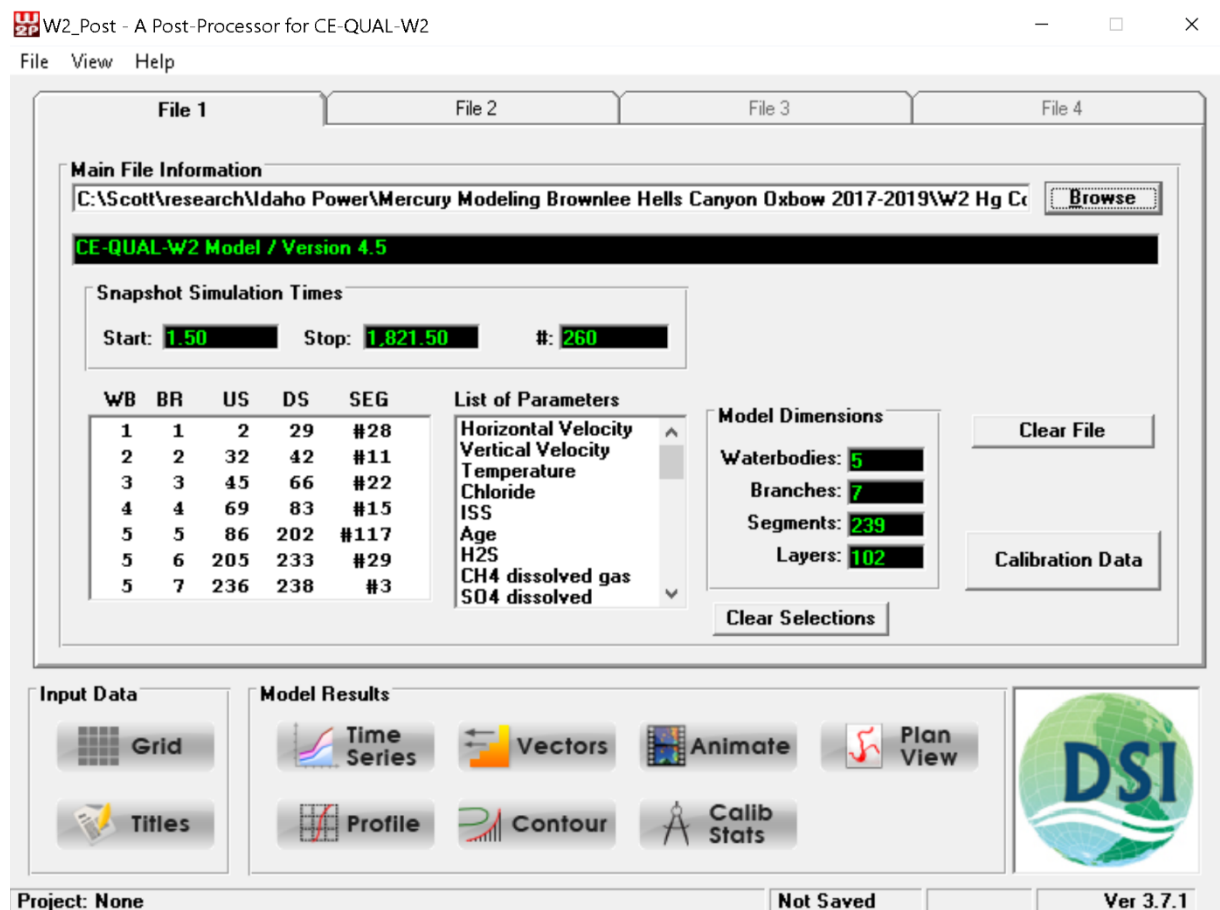


Figure 22. Menu for w2_post.exe post-processing tool from DSI.

6. Excel Macro Utility for CE-QUAL-W2

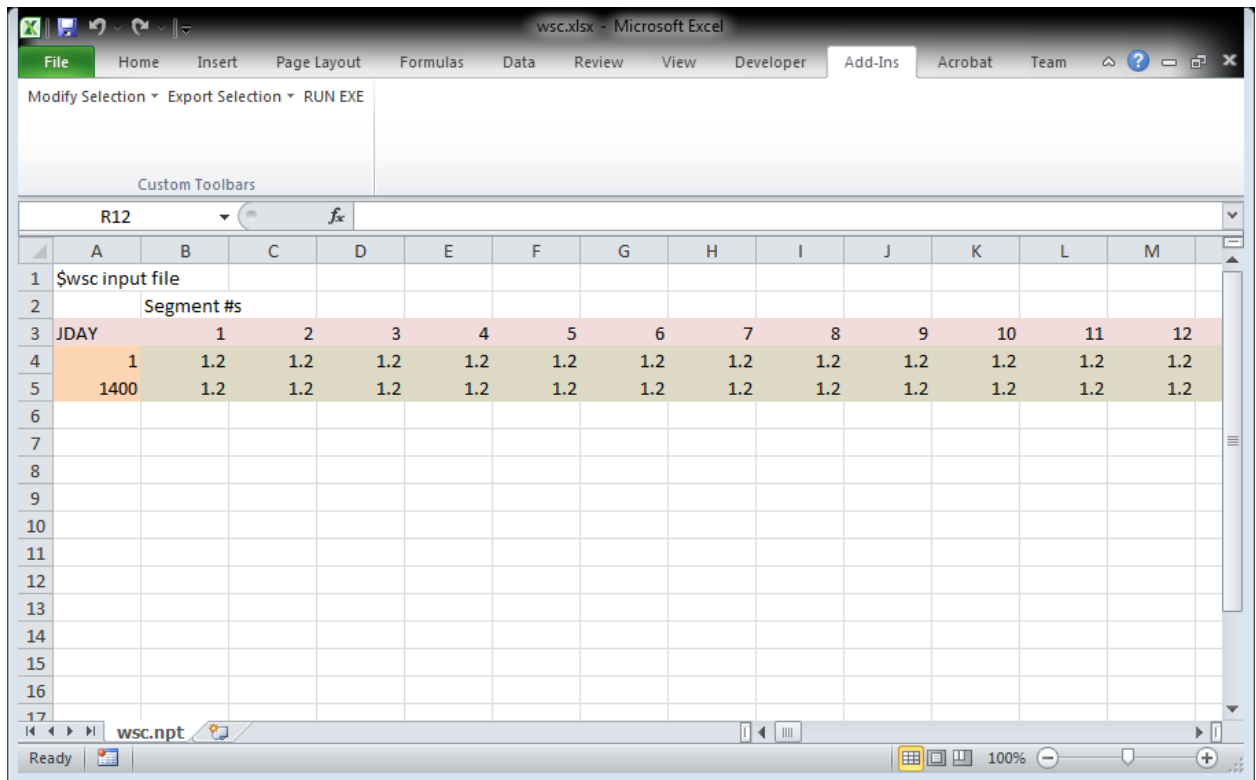
This Excel macro was written by Jeffrey Gregory, P.E., Civil Engineer, USACE, Nashville District and updated by Scott Wells, PSU.

The Excel macro, “w2tools_examples.xlsm”, can be used to develop and write out input files for the CE-QUAL-W2 model. This can be especially useful if the model user develops an Excel workbook with separate tabs for the different input files for the model.

For example, consider the input file for wind sheltering using the new free format (or csv) formatting option. You develop a workbook for this as shown below. The coloring in the workbook is optional and is just an aid to developing your work.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	\$wsc input file												
2		Segment #s											
3	JDAY	1	2	3	4	5	6	7	8	9	10	11	12
4	1	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
5	1400	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
6													
7													
8													
9													
10													
11													
12													
13													
14													
15													
16													
17													

Now one opens “w2tools_examples2.xlsm” and ‘Enables Macros’. Once this file is opened, the user will notice that the sheet now has a header termed “Add-Ins”. Clicking this header one obtains the following options as shown below: “Modify Selection”, “Export Selection”, or “Run EXE”.



The user then selects by highlighting the portion of the file to export or modify. By selecting “Export Selection” then “Any selection to csv format”. The file is written as a free-format csv file with the file name based on the worksheet name, i.e. in this case “wsc.npt”. There are many other export file formats that can be used that are compatible with CE-QUAL-W2. Also, the worksheets in the “w2tools_examples.xlsx” generally show how to use the different macro features. These sheets are only used to show examples of the use of the Add-In macros.

This set of tools also allow you to modify a selection by multiplying, adding, interpolating data gaps, rounding, and transposing. Select the area of the spreadsheet that you want to modify, then select Add-Ins, Modify Selection, and one of the tools, such as Interpolate (either the X column or the Y column of a 2 column selection).

7. W2 V4.5/v4.5.5 Bug Fixes, Enhancements, and User Manual Changes

Table 1 shows a list of model bug fixes since the release of CE-QUAL-W2 Version 4.5. Table 2 shows the enhancements for Version 4.5.5 and any further fixes.

Table 1. List of bug fixes and enhancement code changes since CE-QUAL-W2 Version 4.5 was released. Note that shaded areas toggle between release versions.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
1	W2/PREW2	Release	Initial code release	7/26/2021
2	W2/PREW2	Update	Re-release fixing minor read error in sediment diagenesis	7/30/2021
3	W2	Sediment diagenesis bug fix – critical fix	The sediment temperature for the aerobic layer and anaerobic layers were re-written. There were a couple mistakes in the earlier version since Version 4.0. These have been tested and are working fine now. Also 2 diagenesis files are now produced showing the model temperature in the aerobic layer and the anaerobic layers over time and segment.	8/30/2021
4	Converter	Bug fix	The converter utility output formats were adjusted to avoid character overflow in the output file and a bug in reading the w2_con.npt file was corrected in case epiphyton were set to zero.	8/30/2021
5	W2	Withdrawal.f90	A parentheses error was found by Stewart Rounds that has been corrected. Old code: IF ((WSEL-EL(KBOT,ID)-ELR) /= 0.0) THEN Updated code: IF (WSEL-(EL(KBOT,ID)-ELR) /= 0.0) THEN This code error would not have affected any simulation with zero slope which includes all lakes and reservoirs.	11/19/2021
6	W2	Sediment diagenesis	More output files were added to evaluate the computations within the sediment diagenesis model, these include porewater concentrations of NO ₃ , NH ₃ , PO ₄ , and SO ₄ in both aerobic and anaerobic layers. The user manual was also updated.	11/19/2021
7	Converter	Bug fix	There was a bug fix for older bathymetry file formats. This has been fixed.	12/17/2021

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
8	W2	C vs Organic matter	The input variables when all organics are in terms of C basis was adjusted in the code to mimic the existing organic matter decay paths. Some of the additional functionality of this will be added in a subsequent release when the organic matter and C species are done similarly.	12/17/2021
9	Waterbalance	Csv file type output	A few minor changes were made to prevent spurious error messages and changing the GUI version to write out csv files rather than fixed format. Also, the console version had a couple errors that were fixed in variable type.	12/20/2021
10	PreW2	Preprocessor water balance	An error was fixed in the output for pre.opt file showing the waterbalance flows if there were multiple withdrawals and tributaries	12/28/2021
11	W2	Gates	There was no bug fix, just a refinement in the gate algorithm when the gamma variable was set equal to zero in the rating curve. In this case the rating curve is similar to a spillway rather than a gate where the term B raised to the gamma power is assumed to be '1'. In the code this was forced to '1' rather than being calculated from the exponentiation. This avoids a problem if the gate opening is zero raised to the zero power. Fortran will equate this to zero, not 1.	12/28/2021
12	W2	ICE=ONWB	When Ice=ONWB, the volume balance calculation was not correct in the SNP output and in the last column of the flowbalance.csv output file. Changed line of code for ice balance from temperature subroutine to main program code since it applies to the next rather than current time step: !VolIce(jb)=VolIce(jb)+iceqss(i)*dlt ! since this flow is not exercised until the next time step - moved code to main program calculation of qss	1/13/2022
13	W2	DO for gates and spillways	Added another option for DO at gates and spillways. Reaeration is increased by a fraction of the input dissolved oxygen. In many cases dams have turbine venting or other processes that introduce air into the turbine discharge. The model user can specify when this occurs by submitting a time series file of fractions that are multiplied by the incoming dissolved oxygen concentrations. The W2 code, preprocessor and User Manual has been updated.	1/18/2022

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
14	Prew2	pre.opt	More variables were output to pre.opt such as the gate and spillway gas equations to allow for a more thorough review of input variables.	1/18/2022
15	W2	Restart	Restart input files may have had an incorrect Julian day if the number was greater than 8 digits. That has been fixed for all restart time series.	2/15/2022
16	W2	Restart with ONH	A variable was not initialized properly for cases where the WITHDRAWAL OUTPUT file was set to ONH implying hourly or ONS implying seconds for the output frequency.	3/12/2022
17	W2	Sediment Diagenesis	Fixed output for PO4 and SO4 from sediment diagenesis output files. Also, added new output files for reduced Fe, Mn, and CH4 in the aerobic and anaerobic layers.	5/24/2022
18	W2	CH4	The sediment model had CH4 in O2 equivalents while the water column model had CH4 in as C units. The water column concentrations were adjusted to a more common reporting unit of mg/l as CH4 and the link between the sediment and water column was fixed.	5/24/2022
19	W2	Lowering layer output of SED in TSR file	The model has an algorithm to create a deep slot in a segment in case the water layer is too low. This keeps the segment hydrated during low flow events. When using the first order sediment model in a shallow stream, the values of SED, SEDC, SEDP, and SEDN were set to zero for this new lower layer. The output for SED in the tsr file was updated using the value for the current bottom layer rather than the fictitious lower layer.	5/26/2022
20	W2	Constriction	The input file User Manual description was inconsistent. Fixed code and User Manual to agree with spreadsheet template for constriction.	7/12/2022
21	W2	Water Quality	The Generic constituent's new feature of release based on SOD was not implemented correctly. This was fixed. The MNII reduction computation used the FeII reduction rate rather than the MnII reduction rate. This was fixed. Many thanks to Cory McDonald at Hydros in Boulder for finding those!	7/21/2022
22	W2	W2_POST	Post-processing using W2_post has been enhanced. Now all derived variables are able to be viewed as time-series, profiles, contours, and animations. Prior to this, only state variables were able to be viewed.	10/13/2022
28	PreW2	Water balance	The water balance summary in pre.opt had a bug in computing and displaying the tributary and withdrawal flows. This has been fixed.	10/13/2022

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
29	W2	ENVR performance	The Environmental performance Julian day range was implemented as a yearly criterion implying that if one chose JD 20-60 over a multiple year scenario, it would use JD 20-60 each year to compile statistics. Now the user can specify yearly or a discrete set of Julian days. The User Manual has been updated. Now the Julian day range is only applied to the given range. If the first Julian day is negative, it is applied yearly.	10/13/2022
30	W2	PARTP	A bug has been fixed in the PARTP algorithm where P sorbs onto inorganic SS. This fix will somewhat reduce the P loss since it was overestimated before. This may result in using a higher value of PARTP to get similar loss of Pas in the earlier model.	10/13/2022
31	W2	Example problems	The Excel macro for writing out the example problems was updated to account for systems where the estimated number of columns was under-estimated. For each example, the Excel macro input file macro code was updated.	10/13/2022
32	W2	Sediment Diagenesis	The updated sediment diagenesis code was called – not every time step – but every CUF time steps. The Time for the sediment diagenesis model was computed as $SD_tc = dlt * CUF / DAY$ Which is the time step in days for the sediment diagenesis algorithm. This is approximately correct, but not exact since the time step can vary over time and is not constant. The exact time between calls for Update Kinetics was computed from $SD_tc = JDAY - JDAY_INIT$ Where JDAY_INIT was the JDAY at the beginning of the last call to Update Kinetics.	10/15/2022
33	W2	Flux output	The columns for flux output for DOEP (epiphyton production) and DOAR (algae respiration) were accidentally switched from the order in the control file. This has been fixed.	10/20/2022
34	W2	Tsr output	If the tsr output with derived concentration variables followed an 'Add Layer' event, the derived variables in the output were often zero or incorrectly initialized in the layer add subroutine. All state variables were correctly initialized. On the next time step the model recomputed derived variables correctly. This only affected output.	11/10/2022
35	W2	TSR output	Further improvements for #34. APLIM, ANLIM output fixed also for special output conditions.	11/17/2022

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
36	W2	FLX output file	The flux output file similar to the SNP file had a limitation of 200 segments output. That has been corrected to 2000 in case one has more than 200 active segments defined in one waterbody.	1/12/2023
37	W2	Sediment Diagenesis: Fe and Mn	The Fe and Mn routines were re-written. The prior routines did not interact with the water column (oops!). These have been corrected as well as adding temperature effects for oxidation and reduction in the sediments and water column using an Arrhenius approach with a theta value of 1.05.	1/12/2023
38	W2	Sediment Diagenesis output	More sediment diagenesis output files are now written out including the transfer velocity (m/d) for each sediment layer.	1/12/2023
39	W2	Sediment Diagenesis temperature	Another term has been added to the sediment diagenesis input file – at the end – that tells the diagenesis model to either compute temperature of the aerobic and anaerobic layers dynamically or use adjacent water temperature for aerobic and anaerobic layers.	1/12/2023
40	W2	Habitat output	The model user can now use a new control, DOVOL, to specify the output of volume below 1 mg/l over time. See the User Manual Part 3 on Habitat volume.	2/1/2023
41	W2	Deallocate variables for restart	Many new model variables needed to be deallocated to have restart to work properly.	4/13/2023
42	W2	Manual updates	Several minor manual updates were made	4/13/2023
43	W2	Preprocessor	Additional checks added	4/13/2023
44	W2	Gates	Added functionality for gates. User Manual updated. Gates now can specify FLOW_ZGT where both a flow rate and a variable elevation can be entered.	4/13/2023

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
45	W2	CUF	CUF is an integer that the user can use to delay updates of the kinetic rates and WQ source/sink terms for simulating water-quality constituents. It skips the hydrodynamic time step by CUF time steps. In order to have more control over the update frequency of water quality kinetics, CUF will now be a real number. If it is positive it will have the same functionality as before (skipping CUF time steps between kinetic updates). If CUF is negative, then this is the Julian day interval for updating kinetics. Hence, a value of -0.02 means that the kinetics will be updated only every 0.01 days or every 14.4 min. This allows more precise control of the interval.	7/2/2023
46	W2	New output file	If one uses Stewart Rounds new post-processor W2Anim, https://github.com/sa-rounds/w2anim/releases/latest , the W2 model automatically outputs a file for viewing the outlet dynamics (flow, temperature and water quality) for any outflow. The output file is written out whenever there is a WDO (withdrawal output) specified.	7/2/2023
47	W2	Diagenesis	Updated and added another transfer velocity output file from sediment diagenesis model.	7/7/2023
48	W2	WDO	Withdrawal output for cwo and dwo files now reads the format specifier in the w2_con.csv file. Earlier this format only applied to the SNP output file.	7/7/2023
49	User Manual	Part 2	Updated heat transfer section adding longwave atmospheric formula which were missing from the text.	7/10/2023
50	W2	SPR	Spreadsheet output format precision has been improved as well as names and units of state and derived variables.	7/27/2023
51	W2	SPR	Added horizontal velocity and horizontal layer flow to the SPR output; revised User Manual	8/2/2023
52	W2	Withdrawal output	Added code to prevent gaps in withdrawal output files after a model restart	8/10/2023
53	W2	Auto-port	The number of elevations for a multiple structure withdrawal was increased from 11 to 100. Revised User Manual.	8/16/2023

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
54	W2	Input file	A series of minor updates for the code were made for reading the input file, one of which had switched one of the inputs for BACT (bacteria). The old code had: <pre> READ (CON,*) (BACTLDK(JW), JW=1,NWB) READ (CON,*) (BACTS(JW), JW=1,NWB) </pre> But this order was not consistent with the Excel template nor the User manual. This was changed to <pre> READ (CON,*) (BACTS(JW), JW=1,NWB) READ (CON,*) (BACTLDK(JW), JW=1,NWB) </pre> Also, if one still used the w2_con.npt text file, there were errors in reading that file which have been corrected. Also, the pH-buffering input file input warning was fixed. Thanks to Stewart Rounds for finding these issues.	9/7/2023
55	W2	Silica and Epi-phyton	The variable PSIEM in was not initialized to zero causing it to increase each time step. Added PSIEM(:,IU:ID) = 0.0 to Particulate_Silica subroutine in water_quality.f90. Also, there was a minor issue in computing Epiphyton decay due to an incorrect subscript for the variable H1: Old code: <pre> EGR(K,I,J) = MIN(ETRM(K,I,J)*EG(J)*LIMIT*BLIM,FDPO4(K,I)*PO4(K,I)/(EP(J)*DLT*EPD(K,I,J)/H1(KT,I)+NONZERO), & (NH4(K,I)+NO3(K,I))/(EN(J)*DLT*EPD(K,I,J)/H1(K,I)+NONZERO)) </pre> New code: <pre> EGR(K,I,J) = MIN(ETRM(K,I,J)*EG(J)*LIMIT*BLIM,FDPO4(K,I)*PO4(K,I)/(EP(J)*DLT*EPD(K,I,J)/H1(K,I)+NONZERO), & (NH4(K,I)+NO3(K,I))/(EN(J)*DLT*EPD(K,I,J)/H1(K,I)+NONZERO)) </pre> Thanks again to Stewart Rounds for finding these issues!	9/7/2023

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
56	W2	Density when below zero	Currently the density computation for water goes below zero making the water lighter and lighter. As suggested by Stewart Rounds, perhaps the density for water should be computed at 0°C rather than at negative temperature values. The W2 model without ICE cover can compute negative temperatures as the water keeps losing heat. Turning Ice ON helps keep water temperatures above 0°C. Hence, an IF test was added to make the density for temperatures below 0°C to be at 0°C. Fix below: <pre> REAL(R8) FUNCTION DENSITY (T,TDS,SS) use PREC USE LOGICC, ONLY: SUSP_SOLIDS, FRESH_WATER, SALT_WATER; USE GLOBAL, ONLY:JW IMPLICIT NONE REAL(R8) :: T,TDS,SS IF(T < 0.0) T = 0.0 ! Suggested by Stewart Rounds </pre>	9/7/2023
57	W2	W2 Error Dump file	Another output variable to show whether a segment was active or not was added to the “W2Errordump.csv” output file to help interpret the error dump file information. This suggestion was made by Stewart Rounds.	9/7/2023
58	W2	SNP file	Some updates to the format of the SNP file were implemented following suggestions from Stewart Rounds. The branch inflows and distributed tributaries are identified as INACTIVE if they are not active.	9/7/2023
59	W2	Layer Add Sub	Stewart Rounds suggested many code improvements to improve the robustness of the code. Many of these have been implemented.	9/7/2023
60	W2	WDO output	The wdo output files did not allow a subdirectory specification even though the User Manual said there was one. This has been fixed now. Hence, if the wdo file specification is ‘.\wdofiles\wdo.csv’ then all the wdo files will be written to the subdirectory off the working directory of ‘wdofiles’ with a file suffix of ‘.csv’.	9/18/2023
61	W2	Restart	Restart was broken if one used Sediment Diagenesis based on earlier enhancements. Restart was fixed. [A later fix, see #69, fixed this if one used the dialog box restart.]	9/20.2023
62	W2	USGS Selective	The USGS selective routine had several bugs that were fixed by Stewart Rounds.	10/1/2023

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
63	W2	TSR and Epiphyton	The TSR output did not list the limiting nutrient for epiphyton. This output was added for each epiphyton group near the limiting nutrient for algae. This already was shown in the SNP output file if the print control was ON for epiphyton.	11/2/2023
64	W2	NH3 volatilization	The wind speed used to compute the water vapor transfer rate for NH3 volatilization was changed from 0.5 X Wind at 2 m to the wind at 2 m effectively doubling the gas transfer rate. This occurred as a result of uncertainty in the wind speed height in the original formulation for water vapor gas transfer as it affects ammonia volatilization. (See User Manual Part 3 under Ammonium where the equation has been updated.)	11/8/2023
65	W2	WDO output	The qwo and two files for the mixed flow and temperatures did not have all the flows or temperatures in columns to the right of the total flow when there was a mix of downstream outflows and lateral outflows. This has been fixed.	11/14/2023
66	W2	DZMAX	During periods of density instabilities in W2, the code used a value of DZ=DZMAX which was hardwired at 1000 m2/s. This is incredibly high. There is a possibility that this could lead to too fast mixing that could carry up through a hypolimnion. Now DZMAX is an input variable to check if the temperature regime is sensitive to this and if stability issues may arise. This may be more often an issue if there is sediment bed heating of the water column. If DZMAX is entered as a positive value, it is used directly as the value of DZ when there is an instability. If entered as a negative number, it is a multiplier on the actual value of DZ. Hence if the user enters '-100'. Then during an instability the value of DZ=DZ(computed) X 100. This has led to smoother and not so dramatic changes in density when there are instabilities in the hypolimnion. The User Manual has Part 3 been updated and is found on the MISCELLANEOUS input row starting with NDAY in the Excel macro (search for DZMAX).	11/20/2023

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
67	PreW2	pre.opt +pre.wrn	Several additional variables were output to the pre.opt file; several more checks were made for common issues; moved the error for cloud cover below 10% to a warning; the error for light extinction below 0.25 m-1 was moved to a warning; the code was updated for the new variable DZMAX mentioned above (#66); several code fixes were made as suggested by Stewart Rounds for using the text file control file and a couple other variable fixes.	12/14/2023
68	W2	Met data	A model enhancement has been added for reading variable meteorological data in regions of segments rather than tied to waterbodies. This allows a reservoir to be one waterbody and add different meteorological conditions on different parts of the reservoir. A new tab has been added to the master Excel spreadsheet and checks have been added to the preprocessor. The User Manual has also been updated.	12/14/2023
69	W2	Restart	Restart was fixed by making sure some new variables were properly deallocated when a restart was performed from the dialog box. Also, similar bugs were fixed with Sediment Diagenesis restart using the dialog box.	12/14/2023
70	W2	Input/output	Several updates to output files were made allowing for a subdirectory specification for diagenesis files (updated User Manual since there is a new input variable in the sediment diagenesis file), consistent format for habitat output files, and cleaned up snp.opt file (eliminating code for line printers!). These are courtesy of Dr. Zhong who is setting up HDF5 output.	12/28/2023
71	W2	W2	Stewart Rounds found a few bugs in the initial conditions that may have affected the first time step if you have a constriction, output variables DEPTHM and DEPTHB (used in model output) for sloping sections, and computation of the current upper layer in case the water surface was exactly equal to the top of a grid elevation (extremely rare). A few other assorted bugs were fixed for waterbody-waterbody connections. A big shout out to Stewart for going through the code!!	1/17/2024

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
72	W2	PCO2	Partial pressure of CO2 gas in the atmosphere was computed incorrectly for years prior to 1980 because of an integer overflow. Many thanks to Cory McDonald of Hydros for pointing that out. Old Code: $PCO2 = (0.000041392 * REAL(YEAR * YEAR * YEAR) - 0.231409975 * REAL(YEAR * YEAR) + 430.804190829 * REAL(YEAR) - 266735.857433224) * PALT(DS(BE(1))) * 1.0E-6$ New Code: $PCO2 = (0.000041392 * REAL(YEAR) * real(YEAR) * real(YEAR) - 0.231409975 * REAL(YEAR) * real(YEAR) + 430.804190829 * REAL(YEAR) - 266735.857433224) * PALT(DS(BE(1))) * 1.0E-6$	2/29/2024
73	W2	Input.f90	Possible array shape mismatches fixed by Stewart Rounds: Updated code: <pre> ISO_EPIPHYTON(JW,1:NEPT) = EPICI(JW,1:NEPT) >= 0 .AND. EPIC(JW,1:NEPT) == ' ON' VERT_EPIPHYTON(JW,1:NEPT) = EPICI(JW,1:NEPT) == -1.0 .AND. EPIC(JW,1:NEPT) == ' ON' LONG_EPIPHYTON(JW,1:NEPT) = EPICI(JW,1:NEPT) < -1.0 .AND. EPIC(JW,1:NEPT) == ' ON' ISO_MACROPHYTE(JW,1:NMCT) = MACWBCI(JW,1:NMCT) >= 0 .AND. MACWBC(JW,1:NMCT) == ' ON' VERT_MACROPHYTE(JW,1:NMCT) = MACWBCI(JW,1:NMCT) == -1.0 .AND. MACWBC(JW,1:NMCT) == ' ON' LONG_MACROPHYTE(JW,1:NMCT) = MACWBCI(JW,1:NMCT) < -1.0 .AND. MACWBC(JW,1:NMCT) == ' ON' IF (NEP > 0 .AND. ANY(VERT_EPIPHYTON(JW,1:NEPT))) VERT_PROFILE(JW) = .TRUE. IF (NEP > 0 .AND. ANY(LONG_EPIPHYTON(JW,1:NEPT))) LONG_PROFILE(JW) = .TRUE. IF (NMC > 0 .AND. ANY(VERT_MACROPHYTE(JW,1:NMCT))) VERT_PROFILE(JW) = .TRUE. IF (NMC > 0 .AND. ANY(LONG_MACROPHYTE(JW,1:NMCT))) LONG_PROFILE(JW) = .TRUE. </pre>	5/27/2024

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
74	PREW2	Water balance	Depending on how the model user set up the withdrawals and tributaries (for example out-of-order where tribs are not increasing as segment # is increasing is one example), the water balance summary in the pre.opt file may not account for all the tributaries. Also, the precipitation was in units of m/s not in m3/s and rarely was printed since the value was so small. The precipitation was changed to m3/s based on the area of the surface of the water body at the start of the simulation and is now comparable to other flow inputs. And if an input file only had 2 rows and the 2 nd entry was beyond the end date, this flow may not have been summarized in the pre.opt file. Many thanks to Nabin Basnet for noticing this discrepancy.	5/31/2024

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
75	PREW2	Assorted	(1) The mass loading array dimensions for synchronizing flow and concentration were increased. Under rare conditions, the array dimension is exceeded if the flow and concentration input file time series lengths are vastly different or when using a flow file from a water balance. (2) The preprocessor flagged as an error when the internal weir was located at a downstream branch segment. That was changed as a warning to alert the user. But this is not an error and can be useful to eliminate flows on the right-hand face of a downstream segment. (3) The preprocessor flagged an error for lack of a VPR or LPR input concentration file even if CCC=OFF (no constituent computation). Now when CCC=OFF, the initial condition for water quality does not trigger an error. (4) Another check for grid geometry was updated in the preprocessor so that processing would stop and the user could examine the pre.err file rather than continue and crash due to internal errors caused by the error condition. (5) False warning for constituent input for distributed tributaries when no ON was eliminated. (6) When there are coordinate calculations for writing out segment coordinates that have a glitch, processing is stopped but preprocessor output checking continues. (7) Added file checks for output file subdirectories. Since the model in writing output files needs to have these subdirectories defined prior to running, the preprocessor now will check their existence and create them if necessary. (8) Added more checks to the particle transport input file to catch errors in set-up.	7/10/2024
76	User Manual	Part 3	Updated figures and text in Balances and Grid Geometry in the control file section.	7/10/2024

77	W2	Various Fixes	1. Zooplankton when ORGC CALC is ON had a bug fix for CAC(NLPOM) in computing zooplankton uptake of LPOM since NLPOM is zero. The fix is shown below with old code commented out): <pre> IF (.NOT .ORGC_CALC) THEN ! SW 7/8/2024 ! IF (CAC(NLPOM) == ' ON') THEN LPZOOINC(K,I) = LPZOOINC(K,I) + ZOO(K,I,JZ)*PREFP(JZ)*ZMU(K,I,JZ)*LPOM (K,I)/TGRAZE(K,I,JZ)*ZC(JZ) ELSE LPZOOINC(K,I) = LPZOOINC(K,I) + ZOO(K,I,JZ)*PREFP(JZ)*ZMU(K,I,JZ)*LPOM C(K,I)/ORGC(JW)/TGRAZE(K,I,JZ)*ZC(JZ) END IF </pre> 2. Stewart Rounds (USGS) found that during a time step violation, the ELWS was not returned to its original value at the beginning of a time step. This may have affected those hydraulic structures like spillways and pipes that are dependent on ELWS. This has been fixed by implementing a new array SELWS that is the saved variable that is reset after a violation. As a result, the Restart file format has changed. So old Restart files will not be able to be read by this executable and the model will need to be rerun to create new restart files. 3. Restart for SPR files. A preprocessor warning asks the user to eliminate forward slashes in a state variable definition (usually for units) and this interferes with reading in the SPR file during a restart. Error trapping was added to the W2 code to allow restart to proceed without processing the SPR file and an error is written out to the w2.err file in case this occurs. 4. Added ERROR_OPEN to several diagnostic outputs. Without this the w2.err file	7/10/2024
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			would not be written with these messages. 5. Stewart Rounds (USGS): IF AZC==NICK, the value of DEPTH was not computed correctly. This was corrected. 6. Stewart Rounds suggested a fix for the TKE routine PDE time dependent solution. The fix was to use the value of BH1 and BH2 as they were before the update routine set them equal. This is all because of the effort the original coders made to use as little computer memory as possible. 7. Stewart Rounds: The w2_constriction file was kept open but has now been closed after being read. 8. The enhanced vertical mixing from hypolimnetic aeration was moved to after the water surface computation and was integrated into the w2_4_win.f90 code rather than in aerate.f90 code. This necessitated adding a module for the aerate.f90 variables and using that module with the main program code. 9. Stewart Rounds suggested fixes on calculating AVHR in the case when in a sloping river channel the bottom elevation is lowered creating a deep notch to allow the model to stay hydrated. These fixes are for consistency across the code. 10. Layer Add Subtract subroutine when using dynamic scaling of DX and AX with velocity, there was a subscript error in H(K,JW) 11. Initialized QERR=0.0 12. When the Bubble algorithm was turned ON, the Sediment Diagenesis model did not properly output the Bubble files from the diagenesis model. These files and file headers were implemented.	
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#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
			Also, cleaned up some initialization issues with the gas bubbles and changed an array error to an array warning.	
78	W2	Particle transport	3 bugs were fixed: (1) The monitoring stations were not logging when particles passed by correctly. If the XLOC was set equal to 0 m, then particles were not recorded at the monitoring station. (2) If there were uneven DLX segment spacing, the particles could get 'stuck' in a segment if the DLX was increasing. (3) Added in particle transport output for the histograms the file number CONE which earlier was undefined	7/10/2024
79	PreW2	More checks	Several more checks and corrections of false warnings were made as well as additional checks to guide the user to the issue. Several of these checks made sure the selective withdrawal limits included the centerline of the hydraulic structure for withdrawals, pumps, spillways and gates.	8/5/2024
80	W2	Allocate/Deal-locate	Fixed several issues with deallocating arrays that were not initially allocated to allow a smooth model end simulation and restart. There were no reported issues on this other than new diagnostic messages from Intel/Microsoft Visual Studio.	8/5/2024
81	W2	Withdrawal output file name	The wdo output files were named internally by file names such as qwo_XX.csv where XX was the segment number and '.csv' was provided in the wdo file description. But this might create problems if you are using downstream models to read in upstream input files and the downstream model also writes out a wdo file at that same segment. Now the model user can specify for each model the output name of the wdo file. For example if in the w2_con.csv control file the wdo name is: '.\withdrawalout\wdo_Model1.csv' then the wdo files will be written to the subdirectory 'withdrawalout' and will have file names such as 'q_wdo_Model1_XX.csv' where the XX is the segment number. Hence, the model user has more control over the output name of the file.	8/30/2024
82	W2	WL output file	The wl.csv output file now contains the flag '-999' when the water level is printed from an inactive segment. Stewart Rounds suggested this update.	12/10/2024

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
83	W2	TSR output file	The output for EPI column was supposed to be the g/m2 epiphyton density. It was actually the epiphyton P limitation value from 0 to 1 – a regression from an earlier fix (#63). This has been updated to the correct output of epiphyton density in g/m2.	1/20/2025
84	W2	Screen Dialog Box	Revised parameters for CREATETHREAD so that the W2 code dialog box is compatible with new Intel IFX compiler. This does not affect model execution.	1/20/2025
85	W2	TSS	Derived variable TSS now includes FeOOH and MnO ₂ – prior to this these were not included in the derived variable output for TSS.	1/29/2025
86	W2	Dyn Gate Elev	In the file, dynselevGT.npt, if the order of the gates was not Gate 1 and Gate 2, the file misapplied the flow rate to gate 1 and 2, not the gate specified in the input file dynselevGT.npt. This was found by Mohammad Gohardoust at DWR (many thanks!!). This has been fixed in the TVDS subroutine: Old code: EGT(J)=NXEGT(J) New code: EGT(JJG(J))=NXEGT(JJG(J))	2/14/2025
87	W2	PreW2	A few extra checks added for external extinction coefficient and fixed a bug in calculation of WSC averages and minimum and maximum that is output to the pre.opt file	2/14/2025
88	W2	TSR	Time series output (TSR) had an incorrect header if NEP (# epiphyton groups) was > 0 and the Epiphyton group was not active. This was corrected.	3/16/2025
89	User Manual	Part 2, 3	User Manual Part 2 had an addition of material on how to determine the exchange coefficient between the sediment and the water column. Part 3 text on CBHE was also slightly revised.	4/25/2025
90	W2	PreW2	Some updates were added to the Selective USGS checks and a couple other updates. [from Stewart Rounds]	6/20/2025
91	W2	Withdrawal	The USGS Selective withdrawal algorithm added the MINWL criterion and a few other minor updates. [from Stewart Rounds]	6/20/2025

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
92	W2	PreW2	Added more checks on the preprocessor based on feedback during the W2 workshop 6/2025. These include error trapping before the preprocessor 'bombs' and does not provide good direction on where the error occurred. Also added more checks on csv vs fixed format. Improved format for tributary temperatures in pre.opt file.	7/7/2025
93	W2	Restart	Fixed a couple allocation/deallocation issues that may crop up depending on the state variables used in the simulation. Checked that it now works properly for a specific model setup that was causing issues. There is currently still a bug with Restart when using macrophytes. This will be addressed in the next code update.	7/7/2025
94	W2	Sed Diag output	An extra column was added to one of the sediment diagenesis output files. This was fixed.	7/7/2025
95	W2	Spreadsheet output	Changed the spreadsheet output format to make it easier for restart to read the file. The first column is now of fixed length and in quotes for Excel to interpret correctly. No longer is there a warning to get rid of commas and '/' characters which caused Excel to interpret these as extra columns.	7/7/2025
96	W2	Water quality	In the subroutine for algae toxins was a redundant 'RETURN' which was ignored by the IFORT compiler but flagged an error with the new IFX compiler. Fixed.	11/7/2025
97	PreW2	Preprocessor	Updates were made for improving the preprocessor output messages. Also the preprocessor was updated to handle the new gas generation regression equation for powerhouse output.	12/4/2035

Table 2. Enhancements for Version 4.5.5 and bug fixes and future enhancements since CE-QUAL-W2 Version 4.5.5 was released. Note that shaded areas toggle between release versions. Note that the Version 4.5.5 is file compatible with Version 4.5. The first shaded area lists the enhancements in Version 4.5.5.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
1	W2	Water quality	In the subroutine for algae toxins was a redundant 'RETURN' which was ignored by the IFORT compiler but flagged an error with the new IFX compiler. Fixed.	1/15/2026

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
2	PreW2	Preprocessor	Updates were made for improving the preprocessor output messages.	1/15/2026
3	W2	Longwave radiation options	The model user can now specify a different longwave radiation model than used in the W2 model V4.5 and earlier. Some of these formulations are better able to accurately predict atmospheric longwave radiation. These are described in the User Manual Part 2 and in Part 3 where a new additional meteorological input file is described.	1/15/2026
4	W2	Shortwave solar radiation	The model user can now specify a different model of short-wave solar radiation if shortwave radiation is not measured. This new formulation includes the effect of elevation, atmospheric absorbance and scattering, and atmospheric turbidity. These are described in the User Manual Part 2 and in Part 3 where a new additional meteorological input file is described.	1/15/2026
5	W2	Atmospheric constants	The new input file for meteorological variables mentioned in #3 and #4 above also includes 2 constants for the effect of cloud cover on shortwave and longwave atmospheric radiation.	1/15/2026
6	W2	Gate and spillway oxygen	An additional option was added to a gate being used to add dissolved oxygen based on a regression equation for flow rate, temperature and incoming dissolved oxygen. This is described in the User Manual Part 2 and in Part 3 where Gate and Spillway dissolved oxygen equations are mentioned.	1/15/2026
7	W2	Code cleanup	The code was purged of all unused variables in many different subroutines. This was a general code cleanup to make the code easier to read. Also old commented out code that was not being used was also deleted.	1/15/2026
8	W2	IFX compiler	Version 4.5 was compiled using a now deprecated Fortran compiler. The new Fortran compiler is called 'IFX' from Intel. The Version 4.5.5 code base was compiled using IFX. This leads to slight differences in model predictions from the prior compiler. Most of the changes though are often only in the 3 rd significant digit of a state variable. The IFX compiler for some applications is slightly slower and for others slightly faster than the Version 4.5 compiler.	1/15/2026

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
9	W2	Flow balance file	The output file, flowbal.csv, now has 2 more columns for output as a function of waterbody: surface area and volume. Oftentimes users have asked for an easy way to see how the volume and surface area change in a waterbody over time. It used to be that a user would have to use the water level time series and then back compute the volume and surface area (assuming a flat surface).	1/15/2026
10	PreW2	Updates	The preprocessor now checks the new meteorological input file and also checks other enhancements such as the new gas generation regression equation in the GATE section of the main control file, w2_con.csv. Also, the preprocessor has been compiled using the IFX compiler.	1/15/2026
11	User Manual	Updates	Updates have been made for the User Manual describing the new features.	1/15/2026
12	W2	WDO output	The format descriptors have been updated to be more general avoiding the rare event where a state variable exceeds the output format.	2/11/2026
13	PREW2	Updates	Updated checks for format specifiers in w2_con.csv control file: FMTH, FMTC, and FMTCd	2/11/2026

8. Frequently Asked Questions

This is a list of common questions and answers for debugging your model applications.

Question topic	Question	Response
Numerical Stability	I'm modeling a canal system with high flows (~80 cms) that has a series of gates and pumps to control the flows. I don't have flow rating data on the gates, so I'm using structures to control the flows. I have data on flow rates at the pumps and infer flows at the gates. At higher flows, the model gets unstable and crashes. Is there something with the structure algorithms that would cause a crash	I would force DLTMAX lower during the high velocity period – there is no way a priori to exactly predict the numerical stability for a non-linear PDE – it is just an estimate. Also make sure you don't set all inflows to be placed by density (PQC) and use the implicit turbulence closure scheme rather than explicit (AZSLC).

	or would you infer that it's just a model construct?	

9. W2 Known Code Limitations

Table 3 shows known bugs and issues with the current release of the code:

Table 3. Known bugs in code.

#	Item	Description
1	Water levels in a "bowl"	If water levels decrease in a waterbody shaped like a "bowl", the removal of model layers as the water level decreases will cause the model to bomb if an upstream segment dries up.
2	Pipes under high head	The pipes algorithm does not handle well high-head, high-speed, dynamic flow conditions in a pipe as a result of numerical stability.
3	Time step limitation in a complex system model	The time step for stability in a system model is governed by the lowest time step for numerical stability. If you have a very dynamic river with several reservoirs, the time step for the river will control. This can result in very long run times. One can still break apart the model and run the pieces separately using the WDOUT files to provide boundary conditions for downstream waterbodies.
4	Partitioning	The partitioning coefficient for sorption is currently constant for all organic and inorganic compartments
5	Internal weir at a Dam segment	Putting an internal weir at a Dam segment does not affect the outflow from the selective withdrawal structure. One must limit selective withdrawal rather than use an internal weir at the dam segment. Remember the internal weir works for the right-hand-face of a model layer.
6	W2 multiple file error check	If the model user accidentally enters duplicate file names for an input file, the w2 executable will "bomb" because it will try to read the file in more than once. The first use of the file will lock its availability for the second instance. The W2 error message that comes on the screen (traceback error) should mention the file name that has problems. The W2 preprocessor should catch this potential error.

#	Item	Description
7	Raising level of spillway/weir above grid	The preprocessor will say there is an error if the user raises the weir, spillway, gate, water level control or any other hydraulic element above the current top-of-the-grid. The w2 code will still run properly though. But more correctly, the model user should increase the DZ of the upper-most layer to a value that would eliminate this problem. Keep in mind that the segment widths from the top layer then extend upward at that same width.
8	Internal weirs	The internal weir algorithm does not work when all vertical layers of a segment are blocked by the weir.
9	Multiple dams into one downstream reach	Currently, the code will allow one dam inflow to a downstream branch by a user-specified outflow file. The code though does allow multiple dams inflowing to a common downstream branch if the outflow is specified as a hydraulic structure.
10	Problems reading file in GUI or in W2 pre-processor or in W2 model	Sometimes the control file or bathymetry file or an input file cannot be read properly. This can be a result of the text editor used to produce the file or file conversions that occur when transferring files from workstations running Linux or from email. There may be a problem with the end of line character in the file. For Windows files, the standard end of line is a carriage return followed by a line feed: <CR><LF>. For UNIX systems it is usually only a Line Feed <LF>. To convert this from a UNIX system to a Windows system text file, use Notepad++ (a free windows text editor), go to EDIT/EOL Conversion and select Windows. Another issue common in reading text files is that the editor adds 'tabs'. All 'tabs' must be converted to 'spaces' for the file to be read properly.

10. Appendix A: Differences between CE-QUAL-W2 Versions

Differences between Version 4.22 and 4.5

In a large departure from keeping the version form almost static, Version 4.5 has many differences in the fixed format text file and the Excel xlsx file that has the w2_con.csv version of the control file.

Below are the changes in the w2_con.npt file, but comparable ones are found in the Excel input file version. Only areas with new variables are highlighted:

MISCELL	NDAY	SELECTC	HABTATC	ENVIRPC	AERATEC	INITUWL	ORGCC	SEDIAG
	100	OFF	ON	ON	OFF	OFF	OFF	OFF

Eliminated the unused variables UQB and DOB:

BRANCH G	US	DS	UHS	DHS	NLMIN	SLOPE	SLOPEC
BR1	2	31	0	0	2	0.00000	0.000

TSR SEG	ITSR	ITSR	ITSR	ITSR	ITSR	ITSR	ITSR	ITSR	ITSR
	31								

TSR LAYE	ETSR	ETSR	ETSR	ETSR	ETSR	ETSR	ETSR	ETSR	ETSR
	0.00000								

WLOUT	WLC	WLFREQ
	ON	0.5

FLOWBAL	FLOWBC	FBFREQ
	ON	7.0

NPBAL	NPBALC	NPBFREQ
	ON	7.0

WITH OUT	WDOC	NWDO	NIWDO
	ON	1	1

RSO FREQ	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF
----------	------	------	------	------	------	------	------	------	------

CST COMP	CCC	LIMC	CUF	CO2PPM	CO2YRLY
	ON	ON	3	400.	ON

ATMDEP	ATMDPC	ATMDPIN
WB1	OFF	ON

CST ACTIVE	CAC
TDS	ON
Gen1	ON
Gen2	ON

CST_ACTIVE	CAC
TDS	ON
Gen1	ON
Gen2	ON
Gen3	ON
ISS1	ON
WATERAGE	OFF
BACTERIA	OFF
DGP	OFF
N2	OFF
H2S	OFF
CH4	OFF
SO4	OFF
FEII	OFF
FEOOH	OFF
MNII	OFF
MNO2	OFF
PO4	ON
NH4	ON
NO3	ON
DSI	OFF
PSI	OFF
FE	ON
LDOM	ON
RDOM	ON
LPOM	ON
RPOM	OFF
ALG1	ON
DO	ON
TIC	ON
ALK	ON
ZOO1	OFF
LDOM_P	OFF
RDOM_P	OFF
LPOM_P	OFF
RPOM_P	OFF
LDOM_N	OFF
RDOM_N	OFF
LPOM_N	OFF
RPOM_N	OFF
MICROCY	OFF
CYLINDR	OFF
ANATOXIN	OFF
SAXITOXN	OFF

CST_DERI	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC
DOC	OFF								
POC	OFF								
TOC	ON								
DON	OFF								
PON	OFF								
TON	OFF								
TKN	OFF								
TN	ON								
NH3	ON								
DOP	OFF								
POP	OFF								
TOP	OFF								
TP	OFF								
APR	OFF								
CHLA	OFF								
ATOT	OFF								
%DO	OFF								
TDG	ON								
TURBIDITY	OFF								
TSS	OFF								

TISS	OFF								
CBOD	OFF								
pH	OFF								
CO2	OFF								
HCO3	OFF								
CO3	OFF								
SECCHI	OFF								
CST FLUX	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC
TISSIN	OFF								
TISSOUT	OFF								
PO4AR	OFF								
PO4AG	OFF								
PO4AP	OFF								
PO4ER	OFF								
PO4EG	OFF								
PO4EP	OFF								
PO4POM	OFF								
PO4DOM	OFF								
PO4OM	OFF								
PO4SED	OFF								
PO4SOD	OFF								
PO4SET	OFF								
NH4NITR	OFF								
NH4AR	OFF								
NH4AG	OFF								
NH4AP	OFF								
NH4ER	OFF								
NH4EG	OFF								
NH4EP	OFF								
NH4POM	OFF								
NH4DOM	OFF								
NH4OM	OFF								
NH4SED	OFF								
NH4SOD	OFF								
NH3GAS	ON								
NO3DEN	OFF								
NO3AG	OFF								
NO3EG	OFF								
NO3SED	OFF								
DSIAG	OFF								
DSIEG	OFF								
DSIPIS	OFF								
DSISED	OFF								
DSISOD	OFF								
DSISET	OFF								
PSIAM	OFF								
PSINET	OFF								
PSIDK	OFF								
LDOMDK	OFF								
LRDOM	OFF								
RDOMDK	OFF								
LDOMAP	OFF								
LDOME P	OFF								
LPOMDK	OFF								
LRPOM	OFF								
RPOMDK	OFF								
LPOMAP	OFF								
LPOME P	OFF								
LPOMSET	OFF								
RPOMSET	OFF								
CBODDK	OFF								
DOAP	OFF								
DOEP	OFF								
DOAR	OFF								
DOER	OFF								
DOPOM	OFF								

DODOM	OFF
DOOM	OFF
DONITR	OFF
DOCBOD	OFF
DOREAR	ON
DOSED	ON
DOSOD	OFF
TICAG	OFF
TICEG	OFF
SEDDK	OFF
SEDAS	OFF
SEDLPOM	OFF
SEDSET	OFF
SODDK	OFF

CST ICON	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB
TDS	51.0000								
Gen1	100.000								
Gen2	0.00000								
Gen3	10.0000								
ISS1	2.00000								
WATERAGE	0.0								
BACTERIA	0.0								
DGP	0.0								
N2	0.0								
H2S	0.0								
CH4	0.0								
SO4	0.0								
FEII	0.0								
FEOOH	0.0								
MNII	0.0								
MNO2	0.0								
PO4	0.00100								
NH4	0.00200								
NO3	0.14000								
DSI	0.00000								
PSI	0.00000								
LDM	0.70000								
RDM	2.02200								
LPOM	0.10000								
RPOM	0.00000								
ALG1	-1.0000								
DO	-1.0000								
TIC	11.9100								
ALK	31.0000								
ZOO1	0.1000								
LDM_P	0.0005								
RDM_P	0.0005								
LPOM_P	0.0005								
RPOM_P	0.0005								
LDM_N	0.0080								
RDM_N	0.0080								
LPOM_N	0.0080								
RPOM_N	0.0080								
MICROCYS	0.0								
CYLINDRO	0.0								
ANATOXIN	0.0								
SAXITOXN	0.0								

CST PRIN	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC
TDS	ON								
Gen1	ON								
Gen2	OFF								
Gen3	OFF								
ISS1	ON								
WATERAGE	OFF								
BACTERIA	OFF								
DGP	OFF								

N2	OFF
H2S	OFF
CH4	OFF
SO4	OFF
FEII	OFF
FEOOH	OFF
MNII	OFF
MNO2	OFF
PO4	ON
NH4	ON
NO3	ON
DSI	OFF
PSI	OFF
LDOM	ON
RDOM	ON
LPOM	ON
RPOM	OFF
ALG1	ON
DO	ON
TIC	OFF
ALK	OFF
ZOO1	OFF
LDOM_P	OFF
RDOM_P	OFF
LPOM_P	OFF
RPOM_P	OFF
LDOM_N	OFF
RDOM_N	OFF
LPOM_N	OFF
RPOM_N	OFF
MICROCYSTIN	OFF
CYLINDROSP	OFF
ANATOXIN	OFF
SAXITOXIN	OFF

CIN CON	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC
TDS	ON								
Gen1	ON								
Gen2	OFF								
Gen3	ON								
ISS1	ON								
WATERAGE	OFF								
BACTERIA	OFF								
DGP	OFF								
N2	OFF								
H2S	OFF								
CH4	OFF								
SO4	OFF								
FEII	OFF								
FEOOH	OFF								
MNII	OFF								
MNO2	OFF								
PO4	ON								
NH4	ON								
NO3	ON								
DSI	OFF								
PSI	OFF								
LDOM	ON								
RDOM	ON								
LPOM	ON								
RPOM	OFF								
ALG1	ON								
DO	ON								
TIC	ON								
ALK	ON								
ZOO1	OFF								
LDOM_P	OFF								

RDOM_P	OFF
LPOM_P	OFF
RPOM_P	OFF
LDOM_N	OFF
RDOM_N	OFF
LPOM_N	OFF
RPOM_N	OFF
MICROCYSTIN	OFF
CYLINDROSP	OFF
ANATOXIN	OFF
SAXITOXIN	OFF

CTR CON	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC
TDS	ON	OFF								
Gen1	ON	OFF								
Gen2	OFF	OFF								
Gen3	ON	OFF								
ISS1	ON	OFF								
WATERAGE	OFF	OFF								
BACTERIA	OFF	OFF								
DGP	OFF	OFF								
N2	OFF	OFF								
H2S	OFF	OFF								
CH4	OFF	OFF								
SO4	OFF	OFF								
FEII	OFF	OFF								
FEOOH	OFF	OFF								
MNII	OFF	OFF								
MNO2	OFF	OFF								
PO4	ON	OFF								
NH4	ON	OFF								
NO3	ON	OFF								
DSI	OFF	OFF								
PSI	OFF	OFF								
LDOM	ON	OFF								
RDOM	ON	OFF								
LPOM	ON	OFF								
RPOM	OFF	OFF								
ALG1	ON	OFF								
DO	ON	OFF								
TIC	ON	OFF								
ALK	ON	OFF								
ZOO1	OFF	OFF								
LDOM_P	OFF	OFF								
RDOM_P	OFF	OFF								
LPOM_P	OFF	OFF								
RPOM_P	OFF	OFF								
LDOM_N	OFF	OFF								
RDOM_N	OFF	OFF								
LPOM_N	OFF	OFF								
RPOM_N	OFF	OFF								
MICROCYSTIN	OFF	OFF								
CYLINDROSP	OFF	OFF								
ANATOXIN	OFF	OFF								
SAXITOXIN	OFF	OFF								

CDT CON	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC
TDS	ON	OFF								
Gen1	ON	OFF								
Gen2	OFF	OFF								
Gen3	ON	OFF								
ISS1	ON	OFF								
WATERAGE	OFF	OFF								
BACTERIA	OFF	OFF								
DGP	OFF	OFF								
N2	OFF	OFF								
H2S	OFF	OFF								
CH4	OFF	OFF								

SO4	OFF	OFF
FEII	OFF	OFF
FEOOH	OFF	OFF
MNII	OFF	OFF
MNO2	OFF	OFF
PO4	ON	OFF
NH4	ON	OFF
NO3	ON	OFF
DSI	OFF	OFF
PSI	OFF	OFF
LDOM	ON	OFF
RDOM	ON	OFF
LPOM	ON	OFF
RPOM	OFF	OFF
ALG1	ON	OFF
DO	ON	OFF
TIC	ON	OFF
ALK	ON	OFF
ZOO1	OFF	OFF
LDOM_P	OFF	OFF
RDOM_P	OFF	OFF
LPOM_P	OFF	OFF
RPOM_P	OFF	OFF
LDOM_N	OFF	OFF
RDOM_N	OFF	OFF
LPOM_N	OFF	OFF
RPOM_N	OFF	OFF
MICROCYSTIN	OFF	OFF
CYLINDROSP	OFF	OFF
ANATOXIN	OFF	OFF
SAXITOXIN	OFF	OFF

CPR CON	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC
TDS	ON	ON								
Gen1	ON	OFF								
Gen2	OFF	OFF								
Gen3	ON	OFF								
ISS1	ON	OFF								
WATERAGE	OFF	OFF								
BACTERIA	OFF	OFF								
DGP	OFF	OFF								
N2	OFF	OFF								
H2S	OFF	OFF								
CH4	OFF	OFF								
SO4	OFF	OFF								
FEII	OFF	OFF								
FEOOH	OFF	OFF								
MNII	OFF	OFF								
MNO2	OFF	OFF								
PO4	ON	OFF								
NH4	ON	OFF								
NO3	ON	OFF								
DSI	OFF	OFF								
PSI	OFF	OFF								
LDOM	ON	OFF								
RDOM	ON	OFF								
LPOM	ON	OFF								
RPOM	OFF	OFF								
ALG1	ON	OFF								
DO	ON	OFF								
TIC	ON	OFF								
ALK	ON	OFF								
ZOO1	OFF	OFF								
LDOM_P	OFF	OFF								
RDOM_P	OFF	OFF								
LPOM_P	OFF	OFF								
RPOM_P	OFF	OFF								

LDOM_N	OFF	OFF
RDOM_N	OFF	OFF
LPOM_N	OFF	OFF
RPOM_N	OFF	OFF
MICROCYSTIN	OFF	OFF
CYLINDROSP	OFF	OFF
ANATOXIN	OFF	OFF
SAXITOXIN	OFF	OFF

GENERIC	CGQ10	CG0DK	CG1DK	CGS	CGLDK	CGKLF	CGCS	CGR
CG 1	0.00000	0.00000	0.00000	0.00000	0.0	0.0	0.0	0.0
CG 2	0.00000	-1.00000	0.00000	0.00000	0.0	0.0	0.0	0.0
CG 3	1.04000	0.00000	1.40000	0.00000	0.0	0.0	0.0	0.0

S SOLIDS	SSS	SEDRC	TAUCR	SSCS
SS# 1	1.00000	OFF	0.00000	0.0

BACTERIA	BACTQ10	BACT1DK	BACTS	BACTLDK
WB1	1.025	0.100	0.00	0.050

H2S	H2SR	H2SQ10	H2S1DK	SO4R
WB1	0.005	1.04	0.10	0.050

CH4	CH4R	CH4Q10	CH41DK
WB1	0.004	1.0400	0.05

FE	FEIIR	KFEOXID	KFERED	HalfSat	SetVel
WB1	0.004	1.0400	0.05	0.10	0.10

MN	MNIIR	KMNOXID	KMNRED	HalfSat	SetVel
WB1	0.004	1.0400	0.05	0.10	0.10

ALGAL RATE	AG	AR	AE	AM	AS	AHSP	AHSN	AHSSI	ASAT
ALG1	2.00000	0.04000	0.04000	0.10000	0.10000	0.00300	0.01400	0.00000	100.000

ALG STOI	ALGP	ALGN	ALGC	ALGSI	ACHLA	ALPOM	ANEQN	ANPR	AVERT_M
ALG1	0.00500	0.08000	0.45000	0.00000	0.06500	0.80000	1	0.00100	OFF

OM STOIC	ORGP	ORGN	ORGC	ORGS
WB 1	0.00500	0.08000	0.45000	0.18000

OM RATE	OMT1	OMT2	OMK1	OMK2
WB 1	4.00000	30.0000	0.10000	0.99000

TURBSEC	COEFFA	COEFFB	SECCHI
WB1	1.10	0.05	1.5

CBOD	KBOD	TBOD	RBOD	CBODS
BOD 1	0.25000	1.01500	1.85000	0.0

REAERATION	TYPE	EQN#	COEF1	COEF2	COEF3	COEF4	DGPO2	MINKL
WB 1	LAKE	2	0.00000	0.00000	0.00000	0.00000	1.027	0.6

MET FILE.....	METFN.....
WB 1	met.npt

EXT FILE.....	EXTFN.....
WB 1	ext_1.npt

ATD FILE.....	ATDFN.....
WB 1	atm_dep_wb1.csv

VPR FILE.....	VPRFN.....
WB 1	vpr.npt

Differences between Version 4.2.2 and Version 4.2.1

RSO DATE	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD
RSO FREQ	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF
CST COMP	CCC ON	LIMC OFF	CUF 10							
CST ACTIVE	CAC									
TDS	ON									
Gen1	ON									
ISS1	ON									
ISS2	ON									

RSO DATE	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD	RSOD
RSO FREQ	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF	RSOF
CST COMP	CCC ON	LIMC ON	CUF 10	CO2PPM 400.	CO2YRLY ON					
CST ACTIVE	CAC									
TDS	ON									
Gen1	ON									
ISS1	ON									
ISS2	ON									

[illegible]

Differences between Version 4.2.1 and Version 4.2

Version 4.2.1 is file compatible except that the **multiple_wb.npt** format has changed. Also, a new input file, **w2_con.csv**, can be read by the model. This file is developed using the Excel file **w2_con.xslm**.

Differences between Version 4.2 and Version 4.1

Version 4.2 is file compatible with Version 4.1. There are no changes in the main control file. There are new control files though that control new processes: (1) Multiple processor simulation for cascade of waterbodies (**multiple_wb.npt**), (2) SYSTDG Total Dissolved Gas algorithm for spillways (**systdg.npt**).

Differences between Version 4.1 and Version 4.0

Version 4.1 is file compatible with Version 4.0. There are no changes in the main control file. There is only 1 new input file, '**particle.csv**', that serves as an input file to the particle tracking algorithm.

Differences between Version 4.0 and Version 3.72

Version 4 is file compatible with Version 3.72, even though there are new options in the main control file, **w2_con.npt**, and new input files whose presence or absence is detected by the model. For example, for ICEC control the options now include ON, ONWB, and OFF, where ONWB is a new option. New input files include a file for sediment diagenesis, '**W2_CEMA_Input.npt**', and a file for the dynamic alkalinity calculation, '**pH_buffering.npt**'

Control file differences are in the Generic Constituent Section of the Code where new variables were added to the control file to allow for phot-degradation and the new N2 state variable for TDG:

GENERIC	CGQ10	CG0DK	CG1DK	CGS	CGLDK	CGKLF	CGS	
CG 1	0.00000	0.00000	0.00000	0.00000	0.00000	1.03400	-1.0000	! TDG
CG 2	0.00000	-1.0000	0.00000	0.00000	0.00000	0.00000	0.00000	
CG 3	1.04000	0.00000	1.40000	0.00000	0.00000	0.00000	0.00000	

Differences between Version 3.72 and Version 3.71

These 2 codes are file compatible. Besides a few bug fixes since the last release of Version 3.71, Version 3.72 includes the USGS automatic port selection code. This can be activated by setting SELECTC='USGS' in the control file w2_con.npt. In Version 3.71, only 'ON' or 'OFF' were input variables for SELECTC. If one sets SELECTC='USGS', the format of the file w2_selective.npt is also changed from Version 3.71. Details of this and examples are provided in the User's Manual and on-line.

Differences between Version 3.71 and Version 3.7

There is only one change in the control file between Version 3.7 and 3.71. There is a new option for outlet structures – dynamic centerline elevation. In the control file, there is an ON/OFF option after declaring the # of structures for each branch:

```
EDDY VISC    AZC    AZSLC    AZMAX    FBC    E    ARODI STRCKLR BOUNDFR TKECAL
WB 1         TKE    IMP 1.00000    3 9.53500 0.43100 0.00000 0.00000 IMP

N STRUC      NSTR DYNELEV
BR1          17    ON
BR2           0    OFF
BR3           0    OFF

STR INT      STRIC  STRIC  STRIC  STRIC  STRIC  STRIC  STRIC  STRIC  STRIC
```

If these fields are missing the model will assume that DYNELEV=OFF.

Differences between Version 3.7 and Version 3.6

Even though there are some cases where a Version 3.7 executable will run Version 3.6 and Version 3.5 files fine, there are updates required to the w2_con.npt file that need to be made. The preprocessor will catch these errors.

Control file changes: w2_con.npt

The main changes to the W2 control file are additional flags to turn ON/OFF new control file options and the addition of new state variables for water quality, BOD-N and BOD-P for each BOD group.

Below is a list of changes in the control file with the card image header for each line changed (highlighted options are new in V3.7). Descriptions of these new features are in the W2 User's Manual.

1. MISCELL

```
MISCELL      NDAY SELECTC HABTATC ENVIRPC AERATEC INITUWL
            100    OFF      ON      ON      ON      OFF
```

Five new variables, SELECTC, HABITATC, ENVIRPC, AERATEC, and INITUWL, are 5 new control variables that turn ON/OFF the use of automatic selective withdrawal, fish habitat volumes, environmental performance criteria, artificial aeration, and the initial water surface and velocity computations, respectively. If using an old Version 3.6 control file, all of these would default to 'OFF' if they were left blank. Also the model preprocessor would flag these are missing variables.

2. DLT CON

```
DLT CON      NDT  DLTMIN  DLTINTR
1 1.00000    OFF
```

where DLTINTR is a control for interpolating the the time step DLTMAX and DLTF rather than use as a step function

3. BRANCH G

```
BRANCH G      US      DS      UHS      DHS      UQB      DQB      NLMIN      SLOPE  SLOPEC
Br 1          2       59       0       0       0       0       1       0.0    0.0
```

where SLOPEC is the hydraulic equivalent slope for a river channel that affects the momentum equation.

4. GATE WEIR

```
GATE WEIR     GTA1     GTB1     GTA2     GTB2     DYNVAR     GTIC
Gate1         1.00000  1.50000  1.00000  1.50000    FLOW      ON
```

where GTIC is an interpolation control for the specified DYNVAR for the GATE-WEIR.

5. Dynamic pipe

```
PIPES         IUPI      IDPI      EUPI      EDPI      WPI      DLXPI      FPI      FMINPI  LATPIC  DYNPIPE
Pi 1          24       28       28.0     27.0     0.5     230.0    0.065    0.1     DOWN    ON
```

where DYNPIPE controls whether the pipe is controlled by time series of an ON/OFF or partially open gate

6. Dynamic pump

```
PUMPS 1       IUPU      IDPU      EPU      STRTPU     ENDP      EONPU     EOFFPU     QPU      WTHLC  DYNPUMP
111           0       440.     1.00     366.     441.0    435.0     1.0     DOWN    ON
```

where DYNPUMP controls the EPU, EONPU, EOFFPU, and QPU over time by reading in a time series file

7. INIT CND

```
INIT CND      TEMPI      ICEI      WTYPEC      GRIDC
WB 1          -1.0000  0.00000    FRESH      RECT
```

where GRIDC controls whether the grid is interpreted as rectangular in depth or trapezoidal.

8. CST ACTIVE [Note that this change only appears if NBOD>0]

CST ACTIVE	CAC
TDS	ON
Gen1	ON
Gen2	OFF
Gen3	OFF
Gen4	OFF
Gen5	OFF
ISS1	ON
PO4	ON
NH4	ON
NO3	ON
DSI	OFF
PSI	OFF
FE	OFF
LDM	ON
RDM	ON
LPOM	ON
RPOM	ON
1CBOD	ON
2CBOD	ON
3CBOD	ON
4CBOD	ON
5CBOD	ON
6CBOD	ON
7CBOD	ON
8CBOD	ON
9CBOD	ON
10CBOD	ON
1CBODP	ON
2CBODP	ON
3CBODP	ON
4CBODP	ON
5CBODP	ON
6CBODP	ON
7CBODP	ON
8CBODP	ON
9CBODP	ON
10CBODP	ON
1CBODN	ON
2CBODN	ON
3CBODN	ON
4CBODN	ON
5CBODN	ON
6CBODN	ON
7CBODN	ON
8CBODN	ON
9CBODN	ON
10CBODN	ON
ALG1	ON
ALG2	ON
ALG3	ON
DO	ON
TIC	ON
ALK	ON
ZOO1	OFF
LDM_P	ON
RDM_P	ON
LPOM_P	ON
RPOM_P	ON
LDM_N	ON
RDM_N	ON
LPOM_N	ON
RPOM_N	ON

9. CST ICON, CST PRIN, CIN CON,CTR CON, CDT CON and CPR CON

CST ICON	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB
TDS	0.0								
AGE	0.0								
TRACER	0.0								
COL1	0.0								
Conduct	0.0								
Chlorine	0.0								
ISS1	0.0								
PO4	0.03								
NH4	0.01								
NOx	0.3								
DSi	0.0								
PSi	0.0								
TFe	0.0								
LDOM	0.1								
RDOM	0.1								
LPOM	0.1								
RPOM	0.1								
1CBOD	0.0								
2CBOD	0.0								
3CBOD	0.0								
4CBOD	0.0								
5CBOD	0.0								
6CBOD	0.0								
7CBOD	0.0								
8CBOD	0.0								
9CBOD	0.0								
10CBOD	0.0								
1CBODP	0.0								
2CBODP	0.0								
3CBODP	0.0								
4CBODP	0.0								
5CBODP	0.0								
6CBODP	0.0								
7CBODP	0.0								
8CBODP	0.0								
9CBODP	0.0								
10CBODP	0.0								
1CBODN	0.0								
2CBODN	0.0								
3CBODN	0.0								
4CBODN	0.0								
5CBODN	0.0								
6CBODN	0.0								
7CBODN	0.0								
8CBODN	0.0								
9CBODN	0.0								
10CBODN	0.0								
ALG1	0.1								
ALG2	0.1								
ALG3	0.1								
DO	12.0								
TIC	5.0								
ALK	19.8								
ZOO1	0.0								
LDOM_P	0.0005								
RDOM_P	0.0005								
LPOM_P	0.0005								
RPOM_P	0.0005								
LDOM_N	0.0080								
RDOM_N	0.0080								
LPOM_N	0.0080								
RPOM_N	0.0080								
CST PRIN	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC

TDS	ON										
AGE	ON										
TRACER	ON										
COL1	ON										
Conduct	ON										
Chlorine	ON										
ISS1	ON										
PO4	ON										
NH4	ON										
NOx	ON										
DSi	OFF										
PSi	OFF										
TFe	OFF										
LDOM	ON										
RDOM	ON										
LPOM	ON										
RPOM	ON										
1CBOD	ON										
2CBOD	ON										
3CBOD	ON										
4CBOD	ON										
5CBOD	ON										
6CBOD	ON										
7CBOD	ON										
8CBOD	ON										
9CBOD	ON										
10CBOD	ON										
1CBODP	ON										
2CBODP	ON										
3CBODP	ON										
4CBODP	ON										
5CBODP	ON										
6CBODP	ON										
7CBODP	ON										
8CBODP	ON										
9CBODP	ON										
10CBODP	ON										
1CBODN	ON										
2CBODN	ON										
3CBODN	ON										
4CBODN	ON										
5CBODN	ON										
6CBODN	ON										
7CBODN	ON										
8CBODN	ON										
9CBODN	ON										
10CBODN	ON										
ALG1	ON										
ALG2	ON										
ALG3	ON										
DO	ON										
TIC	ON										
ALK	ON										
ZOO1	OFF										
LDOM_P	ON										
RDOM_P	ON										
LPOM_P	ON										
RPOM_P	ON										
LDOM_N	ON										
RDOM_N	ON										
LPOM_N	ON										
RPOM_N	ON										
CIN CON	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	
TDS	ON	ON									
AGE	OFF	OFF									
TRACER	OFF	OFF									

COL1	OFF	OFF
Conduct	ON	ON
Chlorine	OFF	OFF
ISS1	ON	ON
PO4	ON	ON
NH4	ON	ON
NOx	ON	ON
DSi	OFF	OFF
PSi	OFF	OFF
TFe	OFF	OFF
LDOM	ON	ON
RDOM	ON	ON
LPOM	ON	ON
RPOM	ON	ON
1CBOD	ON	ON
2CBOD	ON	ON
3CBOD	ON	ON
4CBOD	ON	ON
5CBOD	ON	ON
6CBOD	ON	ON
7CBOD	ON	ON
8CBOD	ON	ON
9CBOD	ON	ON
10CBOD	ON	ON
1CBODP	ON	ON
2CBODP	ON	ON
3CBODP	ON	ON
4CBODP	ON	ON
5CBODP	ON	ON
6CBODP	ON	ON
7CBODP	ON	ON
8CBODP	ON	ON
9CBODP	ON	ON
10CBODP	ON	ON
1CBODN	ON	ON
2CBODN	ON	ON
3CBODN	ON	ON
4CBODN	ON	ON
5CBODN	ON	ON
6CBODN	ON	ON
7CBODN	ON	ON
8CBODN	ON	ON
9CBODN	ON	ON
10CBODN	ON	ON
ALG1	ON	ON
ALG2	ON	ON
ALG3	ON	ON
DO	ON	ON
TIC	ON	ON
ALK	ON	ON
ZOO1	OFF	OFF
LDOM_P	ON	ON
RDOM_P	ON	ON
LPOM_P	ON	ON
RPOM_P	ON	ON
LDOM_N	ON	ON
RDOM_N	ON	ON
LPOM_N	ON	ON
RPOM_N	ON	ON

CTR	CON	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC
TDS		ON	ON							
AGE		OFF	OFF							
TRACER		ON	ON							
COL1		ON	ON							
Conduct		ON	ON							
Chlorine		ON	ON							
ISS1		ON	ON							

CDT CON	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC
TDS	ON	ON							
AGE	OFF	OFF							
TRACER	ON	ON							
COL1	ON	ON							
Conduct	ON	ON							
Chlorine	ON	ON							
ISS1	ON	ON							
PO4	ON	ON							
NH4	ON	ON							
NOx	ON	ON							

DSi	OFF	OFF								
PSi	OFF	OFF								
TFe	OFF	OFF								
LDOM	ON	ON								
RDOM	ON	ON								
LPOM	ON	ON								
RPOM	ON	ON								
1CBOD	ON	ON								
2CBOD	ON	ON								
3CBOD	ON	ON								
4CBOD	ON	ON								
5CBOD	ON	ON								
6CBOD	ON	ON								
7CBOD	ON	ON								
8CBOD	ON	ON								
9CBOD	ON	ON								
10CBOD	ON	ON								
1CBODP	ON	ON								
2CBODP	ON	ON								
3CBODP	ON	ON								
4CBODP	ON	ON								
5CBODP	ON	ON								
6CBODP	ON	ON								
7CBODP	ON	ON								
8CBODP	ON	ON								
9CBODP	ON	ON								
10CBODP	ON	ON								
1CBODN	ON	ON								
2CBODN	ON	ON								
3CBODN	ON	ON								
4CBODN	ON	ON								
5CBODN	ON	ON								
6CBODN	ON	ON								
7CBODN	ON	ON								
8CBODN	ON	ON								
9CBODN	ON	ON								
10CBODN	ON	ON								
ALG1	ON	ON								
ALG2	ON	ON								
ALG3	ON	ON								
DO	ON	ON								
TIC	ON	ON								
ALK	ON	ON								
ZOO1	OFF	OFF								
LDOM_P	ON	ON								
RDOM_P	ON	ON								
LPOM_P	ON	ON								
RPOM_P	ON	ON								
LDOM_N	ON	ON								
RDOM_N	ON	ON								
LPOM_N	ON	ON								
RPOM_N	ON	ON								
CPR CON	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC
TDS	ON	ON								
AGE	OFF	OFF								
TRACER	ON	ON								
COL1	ON	ON								
Conduct	ON	ON								
Chlorine	ON	ON								
ISS1	ON	ON								
PO4	ON	ON								
NH4	ON	ON								
NOx	ON	ON								
DSi	OFF	OFF								
PSi	OFF	OFF								
TFe	OFF	OFF								
LDOM	ON	ON								

RDOM	ON	ON
LPOM	ON	ON
RPOM	ON	ON
1CBOD	ON	ON
2CBOD	ON	ON
3CBOD	ON	ON
4CBOD	ON	ON
5CBOD	ON	ON
6CBOD	ON	ON
7CBOD	ON	ON
8CBOD	ON	ON
9CBOD	ON	ON
10CBOD	ON	ON
1CBODP	ON	ON
2CBODP	ON	ON
3CBODP	ON	ON
4CBODP	ON	ON
5CBODP	ON	ON
6CBODP	ON	ON
7CBODP	ON	ON
8CBODP	ON	ON
9CBODP	ON	ON
10CBODP	ON	ON
1CBODN	ON	ON
2CBODN	ON	ON
3CBODN	ON	ON
4CBODN	ON	ON
5CBODN	ON	ON
6CBODN	ON	ON
7CBODN	ON	ON
8CBODN	ON	ON
9CBODN	ON	ON
10CBODN	ON	ON
ALG1	ON	ON
ALG2	ON	ON
ALG3	ON	ON
DO	ON	ON
TIC	ON	ON
ALK	ON	ON
ZOO1	OFF	OFF
LDM_P	ON	ON
RDM_P	ON	ON
LPOM_P	ON	ON
RPOM_P	ON	ON
LDM_N	ON	ON
RDM_N	ON	ON
LPOM_N	ON	ON
RPOM_N	ON	ON

New control files

Based on the options the user turns ON or OFF, new control files are required. These new control files are named:

1. w2_selective.npt – new variables controlling the selective withdrawal algorithm to select temperature targets
2. w2_habitat.npt – new variables controlling fish habitat limits for temperature and dissolved oxygen and surface and segment volume weighted eutrophication state variables
3. w2_envirpf.npt – new variables controlling setting environmental performance criteria
4. w2_aerate.npt – variables describing use of dissolved oxygen addition to enhance dissolved oxygen levels through diffusers

Details of these new control files are in the CE-QUAL-W2 User Manual.

Differences between Version 3.6 and Version 3.5

Version 3.6 can be run without changing any of the input files, even though the preprocessor will identify errors in the control file because of missing variables. Below is a highlighted list of locations in the file w2_con.npt where additional variables have been added. There are no other changes in the input files for Version 3.6.

The TKE algorithm has been updated with new algorithms that match experimental tank data for kinetic energy and dissipation. This is based on a Master's degree project by Sam Gould at Portland State University. A new user option is the TKE1 algorithm, in add addition to the legacy algorithm TKE. This results in several new input variables on the following line of the w2_con.npt file that are only active if TKE1 is chosen for AZC:

EDDY VISC	AZC	AZSLC	AZMAX	FBC	E	ARODI	STRCKLR	BOUNDFR	TKECAL
WB 1	W2	IMP	1.00000	3	9.535	0.430	24.0	10.00	IMP

The roughness height of the water for correction of the vertical velocity wind profile is now a user-defined input, z_0 . Prior to this the model had hardwired the value of $z_0=0.003$ m for wind speed correction at 2m (for evaporation where wind height at 2 m is typical) and $z_0=0.01$ m for wind at 10 m (for shear stress calculations where wind height of 10 m is typical). For consistency, both conversions now use the same value of roughness height. If the user does not specify the value of z_0 (for example if he/she leaves the spaces blank for z_0 using a V3.5 control file), the code uses 0.001 m.

HYD COEF	AX	DX	CBHE	TSED	FI	TSEDF	FRICC	Z0
WB 1	1.00000	1.00000	0.30000	11.5000	0.01000	1.00000	MANN	0.001

A new option for output is in the format required for TECPLOT. For TECPLOT animation there is only a flag in the CPL output line. This allows for easy model animation of the variables U, W, T, RHO, and all active constituents at the frequency specified by the CPL file as a function of distance and elevation.

CPL PLOT	CPLC	NCPL	TECPLOT
WB 1	ON	1	ON

A new variable for determining the fraction of NO₃-N that is diffused into the sediments that becomes organic matter, or SED-N was introduced. According to one study, only about 37% of NO₃-N that diffuses into the sediments becomes incorporated into organic matter in the sediments. The rest is denitrified.

NITRATE	NO3DK	NO3S	FNO3SED
Wb 1	0.05	0.0	0.37
Wb 2	0.05	0.0	0.37

In V3.5 the model computed an average decay coefficient of the sediments based on what was deposited. The user now has the option to dynamically compute that decay rate or to have it fixed and controlled by the model user. A new variable was introduced called DYNSEDK which is either ON/OFF to allow or not allow dynamic computation of the sediment decay rate.

SEDIMENT	SEDC	PRNSC	SEDCI	SEDK	SEDS	FSOD	FSED	SEDBR	DYNSEDK
Wb 1	ON	ON	0.0	0.1	0.0	1.0	1.0	0.001	OFF
Wb 2	ON	ON	0.0	0.1	0.0	1.0	1.0	0.001	OFF

The User can now specify the # of processors to use on the host computer. Most users find that setting NPROC=2 gets the best results. Sometimes setting this greater than 2 results in slower model performance. Also, the CLOSEC control closes the windows dialog box after the model completes its simulation. This is useful in using the windows version of the release code in batch simulations. These are specified in the control file as follows:

GRID	NWB	NBR	IMX	KMX	NPROC	CLOSEC
	1	4	66	117	2	ON

Differences between Version 3.2 and Version 3.5

The differences in V3.5 and V3.2 input files are found in the control file: **w2_con.npt** and in the **graph.npt** file. All other files are the same between the 2 versions.

w2_con.npt

Below is an example of parts of the control file from V3.5 where all new variables are highlighted. Most of these changes have to do with the new zooplankton, macrophyte, and new state variables added to the model. See the User Manual for a list of changes between V3.2 and V 3.5 in the version history. Also there were some deletions from the V3.2 w2_con.npt file. These are shown below.

New variables added to the control file are highlighted

```
.
IN/OUTFL      NTR      NST      NIW      NWD      NGT      NSP      NPI      NPU
              1        1        0        0        0        0        0        0

CONSTITU      NGC      NSS      NAL      NEP      NBOD      NMC      NZP
              5        1        1        1        5        0        1

MISCELL      NDAY
              100

.
.
CST COMP      CCC      LIMC      CUF
              ON      ON      10

CST ACTIVE    CAC
TDS           OFF
Gen1          ON
Gen2          OFF
Gen3          OFF
Gen4          OFF
Gen5          OFF
ISS1          OFF
PO4           OFF
NH4           OFF
NO3           OFF
DSI           OFF
PSI           OFF
FE            OFF
LDOM          OFF
RDOM          OFF
LPOM          OFF
RPOM          OFF
BOD1          OFF
BOD2          OFF
BOD3          OFF
BOD4          OFF
BOD5          OFF
ALG1          OFF
DO            OFF
TIC           OFF
ALK           OFF
ZOO1          OFF
```

LDOM_P	OFF
RDOM_P	OFF
LPOM_P	OFF
RPOM_P	OFF
LDOM_N	OFF
RDOM_N	OFF
LPOM_N	OFF
RPOM_N	OFF

CST	DERI	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC	CDWBC
DOC		OFF								
POC		OFF								
TOC		OFF								
DON		OFF								
PON		OFF								
TON		OFF								
TKN		OFF								
TN		OFF								
DOP		OFF								
POP		OFF								
TOP		OFF								
TP		OFF								
APR		OFF								
CHLA		OFF								
ATOT		OFF								
%DO		OFF								
TSS		OFF								
TISS		OFF								
CBOD		OFF								
pH		OFF								
CO2		OFF								
HCO3		OFF								
CO3		OFF								

CST	FLUX	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC	CFWBC
TISSIN		OFF								
TISSOUT		OFF								
PO4AR		OFF								
PO4AG		OFF								
PO4AP		OFF								
PO4ER		OFF								
PO4EG		OFF								
PO4EP		OFF								
PO4POM		OFF								
PO4DOM		OFF								
PO4OM		OFF								
PO4SED		OFF								
PO4SOD		OFF								
PO4SET		OFF								
NH4NITR		OFF								
NH4AR		OFF								
NH4AG		OFF								
NH4AP		OFF								
NH4ER		OFF								
NH4EG		OFF								
NH4EP		OFF								
NH4POM		OFF								
NH4DOM		OFF								
NH4OM		OFF								
NH4SED		OFF								
NH4SOD		OFF								
NO3DEN		OFF								
NO3AG		OFF								
NO3EG		OFF								
NO3SED		OFF								
DSIAG		OFF								
DSIEG		OFF								

DSIPIS	OFF
DSISED	OFF
DSISOD	OFF
DSISET	OFF
PSIAM	OFF
PSINET	OFF
PSIDK	OFF
FESET	OFF
FESED	OFF
LDOMDK	OFF
LRDOM	OFF
RDOMDK	OFF
LDOMAP	OFF
LDOMEF	OFF
LPOMDK	OFF
LRPOM	OFF
RPOMDK	OFF
LPOMAP	OFF
LPOMEF	OFF
LPOMSET	OFF
RPOMSET	OFF
CBODDK	OFF
DOAP	OFF
DOAR	OFF
DOEP	OFF
DOER	OFF
DOPOM	OFF
DODOM	OFF
DOOM	OFF
DONITR	OFF
DOCBOD	OFF
DOREAR	OFF
DOSED	OFF
DOSOD	OFF
TICAG	OFF
TICEG	OFF
SEDDK	OFF
SEDAS	OFF
SEDLPOM	OFF
SEDSET	OFF
SODDK	OFF

CST	ICON	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB	C2IWB
TDS		0.00000								
Gen1		0.00000								
Gen2		0.00000								
Gen3		0.00000								
Gen4		0.00000								
Gen5		0.00000								
ISS1		0.00000								
PO4		0.03000								
NH4		0.01000								
NO3		0.30000								
DSI		0.00000								
PSI		0.00000								
FE		0.00000								
LDOM		0.10000								
RDOM		0.10000								
LPOM		0.10000								
RPOM		0.10000								
BOD1		0.00000								
BOD2		0.00000								
BOD3		0.00000								
BOD4		0.00000								
BOD5		0.00000								
ALG1		0.10000								
DO		12.0000								
TIC		5.00000								

ALK	19.8000
ZOO1	0.1000
LDOM_P	0.0005
RDOM_P	0.0005
LPOM_P	0.0005
RPOM_P	0.0005
LDOM_N	0.0080
RDOM_N	0.0080
LPOM_N	0.0080
RPOM_N	0.0080

CST	PRIN	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC	CPRWBC
TDS		OFF									
Gen1		ON									
Gen2		OFF									
Gen3		OFF									
Gen4		OFF									
Gen5		OFF									
ISS1		OFF									
PO4		OFF									
NH4		OFF									
NO3		OFF									
DSI		OFF									
PSI		OFF									
FE		OFF									
LDOM		OFF									
RDOM		OFF									
LPOM		OFF									
RPOM		OFF									
BOD1		OFF									
BOD2		OFF									
BOD3		OFF									
BOD4		OFF									
BOD5		OFF									
ALG1		OFF									
DO		OFF									
TIC		OFF									
ALK		OFF									
ZOO1		OFF									
LDOM_P		OFF									
RDOM_P		OFF									
LPOM_P		OFF									
RPOM_P		OFF									
LDOM_N		OFF									
RDOM_N		OFF									
LPOM_N		OFF									
RPOM_N		OFF									

CIN	CON	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC	CINBRC
TDS		ON									
Gen1		OFF									
Gen2		ON									
Gen3		ON									
Gen4		ON									
Gen5		ON									
ISS1		ON									
PO4		ON									
NH4		ON									
NO3		ON									
DSI		OFF									
PSI		OFF									
FE		OFF									
LDOM		ON									
RDOM		ON									
LPOM		ON									
RPOM		ON									
BOD1		ON									

BOD2	ON
BOD3	ON
BOD4	ON
BOD5	ON
ALG1	ON
DO	ON
TIC	ON
ALK	ON
ZOO1	OFF
LDOM_P	OFF
RDOM_P	OFF
LPOM_P	OFF
RPOM_P	OFF
LDOM_N	OFF
RDOM_N	OFF
LPOM_N	OFF
RPOM_N	OFF

CTR CON	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC	CTRTRC
TDS	ON	ON								
Gen1	OFF	OFF								
Gen2	ON	ON								
Gen3	ON	ON								
Gen4	ON	ON								
Gen5	ON	ON								
ISS1	ON	ON								
PO4	ON	ON								
NH4	ON	ON								
NO3	ON	ON								
DSI	OFF	OFF								
PSI	OFF	OFF								
FE	OFF	OFF								
LDOM	ON	ON								
RDOM	ON	ON								
LPOM	ON	ON								
RPOM	ON	ON								
BOD1	ON	ON								
BOD2	ON	ON								
BOD3	ON	ON								
BOD4	ON	ON								
BOD5	ON	ON								
ALG1	ON	ON								
DO	ON	ON								
TIC	ON	ON								
ALK	ON	ON								
ZOO1	OFF	OFF								
LDOM_P	OFF	OFF								
RDOM_P	OFF	OFF								
LPOM_P	OFF	OFF								
RPOM_P	OFF	OFF								
LDOM_N	OFF	OFF								
RDOM_N	OFF	OFF								
LPOM_N	OFF	OFF								
RPOM_N	OFF	OFF								

CDT CON	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC	CDTBRC
TDS	ON									
Gen1	OFF									
Gen2	ON									
Gen3	ON									
Gen4	ON									
Gen5	ON									
ISS1	ON									
PO4	ON									
NH4	ON									
NO3	ON									
DSI	OFF									
PSI	OFF									

FE	OFF
LDOM	ON
RDOM	ON
LPOM	ON
RPOM	ON
BOD1	ON
BOD2	ON
BOD3	ON
BOD4	ON
BOD5	ON
ALG1	ON
DO	ON
TIC	ON
ALK	ON
ZOO1	OFF
LDOM_P	OFF
RDOM_P	OFF
LPOM_P	OFF
RPOM_P	OFF
LDOM_N	OFF
RDOM_N	OFF
LPOM_N	OFF
RPOM_N	OFF

CPR	CON	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC	CPRBRC
TDS		ON								
Gen1		OFF								
Gen2		ON								
Gen3		ON								
Gen4		ON								
Gen5		ON								
ISS1		ON								
PO4		ON								
NH4		ON								
NO3		ON								
DSI		OFF								
PSI		OFF								
FE		OFF								
LDOM		ON								
RDOM		ON								
LPOM		ON								
RPOM		ON								
BOD1		ON								
BOD2		ON								
BOD3		ON								
BOD4		ON								
BOD5		ON								
ALG1		ON								
DO		ON								
TIC		ON								
ALK		ON								
ZOO1		OFF								
LDOM_P		OFF								
RDOM_P		OFF								
LPOM_P		OFF								
RPOM_P		OFF								
LDOM_N		OFF								
RDOM_N		OFF								
LPOM_N		OFF								
RPOM_N		OFF								

EX	COEF	EXH2O	EXSS	EXOM	BETA	EXC	EXIC
WB	1	0.45000	0.01000	0.40000	0.45000	OFF	OFF
ALG	EX	EXA	EXA	EXA	EXA	EXA	EXA
		0.10000					

ZOO EX	EXZ	EXZ	EXZ	EXZ	EXZ	EXZ			
	0.2	0.2	0.2						

MACRO EX	EXM	EXM	EXM	EXM	EXM	EXM			
	0.0100								

GENERIC	CGQ10	CG0DK	CG1DK	CGS					
CG 1	0.00000	-1.00000	0.00000	0.00000					
CG 2	0.00000	0.00000	0.00000	0.00000					
CG 3	1.04000	0.00000	0.50000	0.00000					
CG 4	0.00000	0.00000	0.00000	0.00000					
CG 5	0.00000	0.00000	0.00000	0.00000					

S SOLIDS	SSS	SEDRC	TAUCR						
SSI	1.50000	OFF	0.00						

ALGAL RATE	AG	AR	AE	AM	AS	AHSP	AHSN	AHSSI	ASAT
ALG1	2.00000	0.12000	0.02000	0.05000	0.04000	0.00500	0.00500	0.00000	50.0000

ALGAL TEMP	AT1	AT2	AT3	AT4	AK1	AK2	AK3	AK4	
ALG1	5.00000	12.0000	20.0000	30.0000	0.10000	0.99000	0.99000	0.10000	

ALG STOI	ALGP	ALGN	ALGC	ALGSI	ACHLA	ALPOM	ANEQN	ANPR	
ALG1	0.00500	0.08000	0.45000	0.00000	65.0000	0.80000	1	0.00100	

EPIPHYTE	EPIC	EPIC	EPIC	EPIC	EPIC	EPIC	EPIC	EPIC	EPIC
EPI1	OFF								

EPI PRIN	EPRC	EPRC	EPRC	EPRC	EPRC	EPRC	EPRC	EPRC	EPRC
EPI1	OFF								

EPI INIT	EPICI	EPICI	EPICI	EPICI	EPICI	EPICI	EPICI	EPICI	EPICI
EPI1	10.0000								

EPI RATE	EG	ER	EE	EM	EB	EHSP	EHSN	EHSSI	
EPI1	2.00000	0.05000	0.02000	0.05000	0.01000	0.00200	0.00200	0.00000	

EPI HALF	ESAT	EHS	ENEQN	ENPR					
EPI1	50.0000	40.0000	2	0.00200					

EPI TEMP	ET1	ET2	ET3	ET4	EK1	EK2	EK3	EK4	
EPI1	2.00000	5.00000	20.0000	30.0000	0.10000	0.99000	0.99000	0.10000	

EPI STOI	EP	EN	EC	ESI	ECHLA	EPOM			
EPI1	0.00500	0.08000	0.45000	0.00000	65.0000	0.80000			

ZOOP RATE	ZG	ZR	ZM	ZEFF	PREFP	ZOOMIN	ZS2P		
Zoo1	1.50	0.10	0.010	0.50	0.50	0.0100	0.30		

ZOOP ALGP	PREFA	PREFA	PREFA	PREFA	PREFA	PREFA	PREFA	PREFA	PREFA
Zoo1	1.00	0.50	0.50						

ZOOP ZOOP	PREFZ	PREFZ	PREFZ	PREFZ	PREFZ	PREFZ	PREFZ	PREFZ	PREFZ
Zoo1	0.00	0.00	0.00						

ZOOP TEMP	ZT1	ZT2	ZT3	ZT4	ZK1	ZK2	ZK3	ZK4	
	0.0	15.0	20.0	36.0	0.1	0.9	0.98	0.100	

ZOOP STOI	ZP	ZN	ZC						
	0.01500	0.08000	0.45000						

MACROPHYT	MACWBC	MACWBC	MACWBC	MACWBC	MACWBC	MACWBC	MACWBC	MACWBC	MACWBC
Mac1	ON	OFF	OFF						

MAC PRINT	MPRWBC	MPRWBC	MPRWBC	MPRWBC	MPRWBC	MPRWBC	MPRWBC	MPRWBC	MPRWBC
Mac1	ON	OFF	OFF						

MAC INI	MACWBCI	MACWBCI	MACWBCI	MACWBCI	MACWBCI	MACWBCI	MACWBCI	MACWBCI	MACWBCI
---------	---------	---------	---------	---------	---------	---------	---------	---------	---------

Mac1	0.00000	0.1	0.5
------	---------	-----	-----

MAC RATE	MG	MR	MM	MSAT	MHSP	MHSN	MHSC	MPOM	LRPMAC
Mac 1	0.30	0.05	0.05	30.0	0.0	0.0	0.0	0.9	0.2

MAC SED	PSED	NSED
Mac 1	0.5	0.5

MAC DIST	MBMP	MMAX
Mac 1	40.0	500.0

MAC DRAG	CDSTEM	DWV	DMSA	ANORM
Mac 1	2.0	7e4	8.00	0.80

MAC TEMP	MT1	MT2	MT3	MT4	MK1	MK2	MK3	MK4
Mac 1	7.0	15.0	24.0	34.0	0.1	0.99	0.99	0.01

MAC STOICH	MP	MN	MC
Mac 1	0.005	0.08	0.45

DOM	LDOMDK	RDOMDK	LRDDK
WB 1	0.10000	0.00100	0.00100

POM	LPOMDK	RPOMDK	LRPDK	POMS
WB 1	0.08000	0.00100	0.00100	0.10000

OM STOIC	ORGP	ORGN	ORGC	ORGS
WB 1	0.00500	0.08000	0.45000	0.18000

OM RATE	OMT1	OMT2	OMK1	OMK2
WB 1	4.00000	30.0000	0.10000	0.99000

CBOD	KBOD	TBOD	RBOD	CBODS
BOD 1	0.04180	1.01470	1.00000	0.0
BOD 2	0.13020	1.01470	1.00000	0.0
BOD 3	0.04690	1.01470	1.00000	0.0
BOD 4	0.08800	1.01470	1.00000	0.0
BOD 5	0.05000	1.01470	1.00000	0.0

CBOD STOIC	BODP	BODN	BODC
BOD 1	0.00500	0.08000	0.45000
BOD 2	0.00500	0.08000	0.45000
BOD 3	0.00500	0.08000	0.45000
BOD 4	0.00500	0.08000	0.45000
BOD 5	0.00500	0.08000	0.45000

PHOSPHOR	PO4R	PARTP
WB 1	0.00100	0.00000

AMMONIUM	NH4R	NH4DK
WB 1	0.00100	0.50000

NH4 RATE	NH4T1	NH4T2	NH4K1	NH4K2
WB 1	5.00000	25.0000	0.10000	0.99000

NITRATE	NO3DK	NO3S
WB 1	0.05000	0.00000

NO3 RATE	NO3T1	NO3T2	NO3K1	NO3K2
WB 1	5.00000	25.0000	0.10000	0.99000

SILICA	DSIR	PSIS	PSIDK	PARTSI
WB 1	0.10000	0.00000	0.30000	0.20000

IRON	FER	FES
WB 1	0.10000	0.00000

```
SED CO2      CO2R
WB 1         0.10000
```

```
STOICH 1     O2NH4     O2OM
WB 1         4.57000 1.40000
```

```
STOICH 2     O2AR      O2AG
ALG1         1.10000 1.40000
```

```
STOICH 3     O2ER      O2EG
EPI1         1.10000 1.40000
```

```
STOICH 4     O2ZR
ZOO1         1.10000
```

```
STOICH 5     O2MR      O2MG
MAC1         1.1      1.4
```

```
O2 LIMIT     KDO
              0.10000
```

```
SEDIMENT     SEDC  SEDPRC  SEDCI   SEDK   SEDS   FSOD   FSED   SEDBR
WB 1          ON      ON  0.00000 0.10000 0.1   1.00000 1.00000 0.2
```

```
SOD RATE     SODT1   SODT2   SODK1   SODK2
WB 1          4.00000 30.0000 0.10000 0.99000
```

```
S DEMAND      SOD      SOD      SOD      SOD      SOD      SOD      SOD      SOD      SOD
              0.6      0.6      0.6      0.6      0.6      0.6      0.6      0.6      0.6
              0.6      0.6      0.6      0.6      0.6      0.6      0.6      0.6      0.6
              0.6      0.6      0.6      0.6      0.6      0.6      0.6      0.6      0.6
              0.6
```

```
REAERATION   TYPE     EQN#    COEF1    COEF2    COEF3    COEF4
WB1          LAKE      6
```

Lines removed from the V3.2 control file: These are a result of eliminating the pumpback and line printer settings.

Here is the part of the V3.2 control file that was deleted:

```
DST TRIB     DTRC
Br 1         ON
Br 2         ON
Br 3         OFF
Br 4         OFF
Br 5         OFF
```

```
PUMPBACK     JBC      KTC      KBC      JBP      KTP      KBP
              0
```

```
PRINTER      LJC
              IV
```

```
HYD PRINT    HPRWBC   HPRWBC   HPRWBC   HPRWBC   HPRWBC   HPRWBC   HPRWBC   HPRWBC   HPRWBC
NVIOL        OFF      OFF
U            ON       ON
```

Graph.npt file changes. These changes are a result of the new state variables in W2 and are highlighted below.

Hydrodynamic, constituent, and derived constituent names, formats, multipliers, and array viewer controls

.....HNAME.....	FMTH	HMULT	HMIN	HMAX	HPLTC
#					
Timestep violations [NVIOL]	(I10)	1.0	-1.0	1.0	OFF
1					
Horizontal velocity [U], m/s	(1PE10.1)	1.0	-.1000	0.15	OFF
2					
Vertical velocity [W], m/s	(1PE10.1)	1.0	-.1E-6	-0.01	OFF
3					
Temperature [T1], <o/>C	(F10.2)	1.0	-10.0	-26.0	ON
4					
Density [RHO], g/m^3	(F10.3)	1.0	997.0	1005.0	OFF
5					
Vertical eddy viscosity [AZ], m^2/s	(F10.3)	1.0	-1E-08	0.01	OFF
6					
Velocity shear stress [SHEAR], 1/s^2	(F10.3)	1.0	-1E-08	0.01	OFF
7					
Internal shear [ST], m^3/s	(F10.3)	1.0	-1E-08	0.01	OFF
8					
Bottom shear [SB], m^3/s	(F10.3)	1.0	-1E-08	0.01	OFF
9					
Longitudinal momentum [ADMX], m^3/s	(F10.3)	1.0	-1E-08	0.01	OFF
10					
Longitudinal momentum [DM], m^3/s	(F10.3)	1.0	-1E-08	0.01	OFF
11					
Horizontal density gradient [HDG], m^3/s	(F10.3)	1.0	-1E-08	0.01	OFF
12					
Vertical momentum [ADMZ], m^3/s	(F10.3)	1.0	-1E-08	0.01	OFF
13					
Horizontal pressure gradient [HPG], m^3/s	(F10.3)	1.0	-1E-08	10.0	OFF
14					
Gravity term channel slope [GRAV], m^3/s	(F10.3)	1.0	0.0	0.0	OFF
15					
.....CNAME.....	FMTC	CMULT	CMIN	CMAX	CPLTC
#					
TDS, g/m^3	(F10.3)	1.0	-1.0	200.0	OFF
1					
Age, days	(F10.3)	1.0	-1.0	-200.0	ON
2					
Tracer, g/m^3	(F10.3)	1.0	-20.000	100.0	OFF
3					
Bacteria, col/100ml	(F10.3)	1.0	-20.000	100.0	OFF
4					
Conductivity, mhos	(F10.3)	1.0	-20.000	100.0	OFF
5					
Chloride, mg/l	(F10.3)	1.0	-20.000	100.0	OFF
6					
ISS, g/m^3	(F10.3)	1.0	-20.000	100.0	OFF
7					
Phosphate, g/m^3	(F10.3)	1000.0	-1.0	500.0	OFF
8					
Ammonium, g/m^3	(F10.3)	1000.0	-0.1000	300.0	OFF
9					
Nitrate-Nitrite, g/m^3	(F10.3)	1.0	-0.1000	5.0	OFF
10					
Dissolved silica, g/m^3	(F10.3)	1.0	-1.0	10.0	OFF
11					
Particulate silica, g/m^3	(F10.3)	1.0	-0.2000	15.0	OFF
12					
Total iron, g/m^3	(F10.3)	1.0	-0.1000	2.0	OFF
13					
Labile DOM, g/m^3	(F10.3)	1.0	-0.1000	-3.0	OFF
14					
Refractory DOM, g/m^3	(F10.3)	1.0	-0.1000	-4.0	OFF
15					
Labile POM, g/m^3	(F10.3)	1.0	-0.1000	-3.0	OFF
16					

Refractory POM, g/m ³	(F10.3)	1.0	-0.1000	-4.0	OFF
17					
CBOD1, g/m ³	(F10.3)	1.0	-0.0100	3.0	OFF
18					
CBOD2, g/m ³	(F10.3)	1.0	-0.0100	3.0	OFF
19					
CBOD3, g/m ³	(F10.3)	1.0	-0.0100	3.0	OFF
20					
CBOD4, g/m ³	(F10.3)	1.0	-0.0100	3.0	OFF
21					
CBOD5, g/m ³	(F10.3)	1.0	-0.0100	3.0	OFF
22					
Algae, g/m ³	(F10.3)	1.0	-0.0100	3.0	OFF
23					
Dissolved oxygen, g/m ³	(F10.3)	1.0	-0.0100	-1.0	OFF
24					
Inorganic carbon, g/m ³	(F10.3)	1.0	-0.0100	3.0	OFF
25					
Alkalinity, g/m ³	(F10.3)	1.0	-0.0100	3.0	OFF
26					
zooplankton1, mg/m ³	(g10.3)	1000.0	-0.0100	1.0	OFF
27					
LDOM P, mg/m ³	(g10.3)	1000.0	0.0	1.0	OFF
28					
RDOM P, mg/m ³	(g10.3)	1000.0	0.0	1.0	OFF
29					
LPOM P, mg/m ³	(g10.3)	1000.0	0.0	1.0	OFF
30					
RPOM P, mg/m ³	(g10.3)	1000.0	0.0	1.0	OFF
31					
LDOM N, mg/m ³	(g10.3)	1000.0	0.0	1.0	OFF
32					
RDOM N, mg/m ³	(g10.3)	1000.0	0.0	1.0	OFF
33					
LPOM N, mg/m ³	(g10.3)	1000.0	0.0	1.0	OFF
34					
RPOM N, mg/m ³	(g10.3)	1000.0	0.0	1.0	OFF
35					
.....CDNAME.....					
#	FMTCD	CDMULT	CDMIN	CDMAX	CDPLTC
Dissolved organic carbon, g/m ³	(F10.3)	1.0	-1.0	25.0	OFF
1					
Particulate organic carbon, g/m ³	(F10.3)	1.0	-1.0	50.0	OFF
2					
Total organic carbon, g/m ³	(F10.3)	1.0	-1.0	25.0	OFF
3					
Dissolved organic nitrogen, g/m ³	(F10.3)	1.0	-1.0	25.0	OFF
4					
Particulate organic nitrogen, g/m ³	(F10.3)	1.0	-1.0	25.0	OFF
5					
Total organic nitrogen, g/m ³	(F10.3)	1.0	-1.0	50.0	OFF
6					
Total Kheldahl Nitrogen, g/m ³	(F10.3)	1.0	-1.0	15.0	OFF
7					
Total nitrogen, g/m ³	(F10.3)	1.0	-1.0	15.0	OFF
8					
Dissolved organic phosphorus, mg/m ³	(F10.3)	1000.0	-1.0	25.0	OFF
9					
Particulate organic phosphorus, mg/m ³	(F10.3)	1000.0	-1.0	-1.0	OFF
10					
Total organic phosphorus, mg/m ³	(F10.3)	1000.0	-1.0	5.0	OFF
11					
Total phosphorus, mg/m ³	(F10.3)	1000.0	-1.0	20.0	OFF
12					
Algal production, g/m ² /day	(F10.3)	1.0	-1.0	5.0	OFF
13					

Chlorophyll a, mg/m ³	(F10.3)	1.0	-5.0	145.0	OFF
14					
Total algae, g/m ³	(F10.3)	1.0	-1.0	60.0	OFF
15					
Oxygen % Gas Saturation	(F10.3)	1.0	-1.0	50.0	OFF
16					
Total suspended Solids, g/m ³	(F10.3)	1.0	-1.0	5.0	OFF
17					
Total Inorganic Suspended Solids, g/m ³	(F10.3)	1.0	-1.0	20.0	OFF
18					
Carbonaceous Ultimate BOD, g/m ³	(F10.3)	1.0	5.0	9.0	OFF
19					
pH	(F10.3)	1.0	-1.0	10.0	OFF
20					
CO2	(F10.3)	1.0	-1.0	10.0	OFF
21					
HCO3	(F10.3)	1.0	-1.0	10.0	OFF
22					
CO3	(F10.3)	0.0	0.0	0.0	OFF
23					

Differences between Version 3.1 and Version 3.2

There are minor differences in 2 input files between the 2 versions: **w2_con.npt** and the **graph.npt** file. All other files are the same between the 2 versions.

w2_con.npt

The only section where there is a slight difference in the control file is in the section where the inorganic suspended solids group settling velocities are entered. In Version 3.1, this section looks like this:

```

ALG EX      EXA      EXA      EXA      EXA      EXA      EXA
      0.10000

GENERIC      CGQ10    CG0DK    CG1DK      CGS
CG 1      0.00000 -1.0000 0.00000 0.00000
CG 2      0.00000 0.00000 0.00000 0.00000
CG 3      1.04000 0.00000 0.50000 0.00000
CG 4      0.00000 0.00000 0.00000 0.00000
CG 5      0.00000 0.00000 0.00000 0.00000

S SOLIDS      SSS      SSS      SSS      SSS      SSS      SSS      SSS      SSS      SSS
      1.50000

ALGAL RATE      AG      AR      AE      AM      AS      AHSP      AHSN      AHSSI      ASAT
ALG1      2.00000 0.12000 0.02000 0.05000 0.04000 0.00500 0.00500 0.00000 50.0000

```

In Version 3.2, there is now a sediment resuspension capability for wind driven resuspension along the shores of lakes and reservoirs. The Version 3.2 control file has the following lines in this same section of the control file:

```

ALG EX      EXA      EXA      EXA      EXA      EXA      EXA
      0.10000

GENERIC      CGQ10    CG0DK    CG1DK      CGS
CG 1      0.00000 -1.0000 0.00000 0.00000
CG 2      0.00000 0.00000 0.00000 0.00000
CG 3      1.04000 0.00000 0.50000 0.00000
CG 4      0.00000 0.00000 0.00000 0.00000
CG 5      0.00000 0.00000 0.00000 0.00000

S SOLIDS      SSS      SEDRC      TAUCR
SS1      1.50000      OFF      0.00

ALGAL RATE      AG      AR      AE      AM      AS      AHSP      AHSN      AHSSI      ASAT
ALG1      2.00000 0.12000 0.02000 0.05000 0.04000 0.00500 0.00500 0.00000 50.0000

```

For Version 3.2, SSS is the settling velocity for particle group 1, SEDRC is the control which turns ON or OFF sediment resuspension, and TAUCR is the critical shear stress at which resuspension occurs. For Version 3.2, each line represents 1 SS group, while in Version 3.1, each group settling velocity is in the next 8 columns moving across the page.

graph.npt

The graph file controls output formatting and the graphing parameters used in Array Viewer (only for the PC platform). The files have been rearranged significantly. A Version 3.1 graph file is shown below:

Constituent, hydrodynamic, and derived constituent names, formats, multipliers, and array viewer controls

.....CNAME.....	CMULT	CMIN	CMAX	CPLTC	#
TDS g/m^3 or Salinity kg/m^3	1.00000	-1.0000	200.000	OFF	1
Generic Constituent,g/m^3, #1	1.00000	-1.0000	-200.00	ON	2
Generic Constituent,g/m^3, #2	1.00000	-1.0000	1000.00	OFF	3
Generic Constituent,g/m^3, #3	1.00000	-1.0000	5.00000	OFF	4
Generic Constituent,g/m^3, #4	1.00000	-1.0000	-300.00	OFF	5
Generic Constituent,g/m^3, #5	1.00000	-1.0000	-3.0000	OFF	6
Suspended solids,g/m^3, #1	1.00000	-1.0000	15.0000	OFF	7
Phosphate, g/m^3	1000.00	-1.0000	-50.000	OFF	8
Ammonium, g/m^3	1000.00	-0.1000	-300.00	OFF	9
Nitrate-Nitrite, g/m^3	1.00000	-0.1000	-5.0000	OFF	10
Dissolved silica, g/m^3	1.00000	-1.0000	10.0000	OFF	11
Particulate silica, g/m^3	1.00000	-0.2000	15.0000	OFF	12
Total iron, g/m^3	1.00000	-0.1000	2.00000	OFF	13
Labile DOM, g/m^3	1.00000	-0.1000	-3.0000	OFF	14
Refractory DOM, g/m^3	1.00000	-0.1000	4.00000	OFF	15
Labile POM, g/m^3	1.00000	-0.1000	3.00000	OFF	16
Refractory POM, g/m^3	1.00000	-0.1000	4.00000	OFF	17
CBOD, g/m^3, #1	1.00000	-0.1000	10.0000	OFF	18
CBOD, g/m^3, #2	1.00000	-0.1000	10.0000	OFF	19
CBOD, g/m^3, #3	1.00000	-0.1000	10.0000	OFF	20
CBOD, g/m^3, #4	1.00000	-0.1000	10.0000	OFF	21
CBOD, g/m^3, #5	1.00000	-0.1000	10.0000	OFF	22
Algae, g/m^3, #1	1.00000	-0.0100	-3.0000	OFF	23
Dissolved oxygen, g/m^3	1.00000	-2.0000	15.0000	OFF	24
Inorganic carbon, g/m^3	1.00000	-1.0000	10.0000	OFF	25
Alkalinity, g/m^3	1.00000	-1.0000	200.000	OFF	26
.....HNAME.....	HFMT	HMIN	HMAX	HPLTC	#
Timestep violations [NVIOL]	(F10.0)	-1.0000	100000	OFF	1
Horizontal velocity [U], m/s	(1PE10.1)	-0.0100	0.10000	ON	2
Vertical velocity [W], m/s	(1PE10.1)	-.10E-06	0.01000	OFF	3
Temperature [T1], <o/>C	(F10.2)	-2.0000	-30.000	ON	4
Density [RHO], g/m^3	(F10.2)	997.000	1005.00	OFF	5
Vertical eddy viscosity [AZ], m^2/s	(1PE10.1)	-1E-08	0.00100	OFF	6
Velocity shear stress [SHEAR], 1/s^2	(1PE10.1)	-1E-08	0.01000	OFF	7
Internal shear [ST], m^3/s	(1PE10.1)	-1E-08	0.01000	OFF	8
Bottom shear [SB], m^3/s	(1PE10.1)	-1E-08	0.01000	OFF	9
Longitudinal momentum [ADMX], m^3/s	(1PE10.1)	-1E-08	0.01000	OFF	10
Longitudinal momentum [DM], m^3/s	(1PE10.1)	-1E-08	0.01000	OFF	11
Horizontal density gradient [HDG], m^3/s	(1PE10.1)	-1E-08	0.01000	OFF	12
Vertical momentum [ADMZ], m^3/s	(1PE10.1)	-1E-08	0.01000	OFF	13
Horizontal pressure gradient [HPG], m^3/s	(1PE10.1)	-1E-08	0.01000	OFF	14
Gravity term channel slope [GRAV], m^3/s	(1PE10.1)	-1E-08	10.0000	OFF	15
.....CDNAME.....	CDMULT	CDMIN	CDMAX	CDPLTC	#
Dissolved organic carbon, g/m^3	1.00000	-1.0000	3.00000	OFF	1
Particulate organic carbon, g/m^3	1.00000	-1.0000	25.0000	OFF	2
Total organic carbon, g/m^3	1.00000	-1.0000	50.0000	OFF	3
Dissolved organic nitrogen, g/m^3	1.00000	-1.0000	25.0000	OFF	4
Particulate organic nitrogen, g/m^3	1.00000	-1.0000	25.0000	OFF	5
Total organic nitrogen, g/m^3	1.00000	-1.0000	25.0000	OFF	6
Total Kheldahl Nitrogen, g/m^3	1.00000	-1.0000	5.00000	OFF	7
Total nitrogen, g/m^3	1.00000	-1.0000	50.0000	OFF	8
Dissolved organic phosphorus, mg/m^3	1000.00	-1.0000	15.0000	OFF	9
Particulate organic phosphorus, mg/m^3	1000.00	-1.0000	15.0000	OFF	10
Total organic phosphorus, mg/m^3	1000.00	-1.0000	25.0000	OFF	11
Total phosphorus, mg/m^3	1000.00	-1.0000	-1.0000	OFF	12
Algal production, g/m^2/day	1.00000	-1.0000	5.00000	OFF	13

Chlorophyll a, mg/m ³	1000.00	-1.0000	-70.000	OFF	14
Total algae, g/m ³	1.00000	-1.0000	5.00000	OFF	15
Oxygen % Gas Saturation	1.00000	-5.0000	145.000	OFF	16
Total suspended Solids, g/m ³	1.00000	-1.0000	60.0000	OFF	17
Total Inorganic Suspended Solids, g/m ³	1.00000	-1.0000	50.0000	OFF	18
Carbonaceous Ultimate BOD, g/m ³	1.00000	-1.0000	20.0000	OFF	19
pH	1.00000	6.00000	9.00000	OFF	20
CO2	1.00000	-1.0000	10.0000	OFF	21
HCO3	1.00000	-1.0000	10.0000	OFF	22
CO3	1.00000	-1.0000	10.0000	OFF	23

An example of the same graph file but for Version 3.2 is shown below:

Hydrodynamic, constituent, and derived constituent names, formats, multipliers, and array viewer controls

.....HNAME.....	FMTH	HMULT	HMIN	HMAX	HPLTC
#					
Timestep violations [NVIOL]	(I10)	1.0	-1.0	1.0	OFF
1					
Horizontal velocity [U], m/s	(Z10.8)	1.0	-.1000	0.15	ON
2					
Vertical velocity [W], m/s	(Z10.8)	1.0	-.1E-6	-0.01	OFF
3					
Temperature [T1], <o/>C	(Z10.8)	1.0	-10.0	-26.0	ON
4					
Density [RHO], g/m ³	(Z10.8)	1.0	997.0	1005.0	OFF
5					
Vertical eddy viscosity [AZ], m ² /s	(Z10.8)	1.0	-1E-08	0.01	OFF
6					
Velocity shear stress [SHEAR], 1/s ²	(Z10.8)	1.0	-1E-08	0.01	OFF
7					
Internal shear [ST], m ³ /s	(Z10.8)	1.0	-1E-08	0.01	OFF
8					
Bottom shear [SB], m ³ /s	(Z10.8)	1.0	-1E-08	0.01	OFF
9					
Longitudinal momentum [ADMX], m ³ /s	(Z10.8)	1.0	-1E-08	0.01	OFF
10					
Longitudinal momentum [DM], m ³ /s	(Z10.8)	1.0	-1E-08	0.01	OFF
11					
Horizontal density gradient [HDG], m ³ /s	(Z10.8)	1.0	-1E-08	0.01	OFF
12					
Vertical momentum [ADMZ], m ³ /s	(Z10.8)	1.0	-1E-08	0.01	OFF
13					
Horizontal pressure gradient [HPG], m ³ /s	(Z10.8)	1.0	-1E-08	10.0	OFF
14					
Gravity term channel slope [GRAV], m ³ /s	(Z10.8)	1.0	0.0	0.0	OFF
15					
.....CNAME.....	FMTC	CMULT	CMIN	CMAX	CPLTC
#					
TDS, g/m ³	(Z10.8)	1.0	-1.0	200.0	OFF
1					
Age, days	(Z10.8)	1.0	-1.0	-200.0	ON
2					
Tracer, g/m ³	(Z10.8)	1.0	-20.000	100.0	OFF
3					
Bacteria, col/100ml	(Z10.8)	1.0	-20.000	100.0	OFF
4					
Conductivity, mhos	(Z10.8)	1.0	-20.000	100.0	OFF
5					
Chloride, mg/l	(Z10.8)	1.0	-20.000	100.0	OFF
6					
ISS, g/m ³	(Z10.8)	1.0	-20.000	100.0	OFF
7					
Phosphate, g/m ³	(Z10.8)	1000.0	-1.0	500.0	OFF
8					

Ammonium, g/m ³	(Z10.8)	1000.0	-0.1000	300.0	OFF
9					
Nitrate-Nitrite, g/m ³	(Z10.8)	1.0	-0.1000	5.0	OFF
10					
Dissolved silica, g/m ³	(Z10.8)	1.0	-1.0	10.0	OFF
11					
Particulate silica, g/m ³	(Z10.8)	1.0	-0.2000	15.0	OFF
12					
Total iron, g/m ³	(Z10.8)	1.0	-0.1000	2.0	OFF
13					
Labile DOM, g/m ³	(Z10.8)	1.0	-0.1000	-3.0	OFF
14					
Refractory DOM, g/m ³	(Z10.8)	1.0	-0.1000	-4.0	OFF
15					
Labile POM, g/m ³	(Z10.8)	1.0	-0.1000	-3.0	OFF
16					
Refractory POM, g/m ³	(Z10.8)	1.0	-0.1000	-4.0	OFF
17					
CBOD1, g/m ³	(Z10.8)	1.0	-0.0100	3.0	OFF
18					
CBOD2, g/m ³	(Z10.8)	1.0	-0.0100	3.0	OFF
19					
CBOD3, g/m ³	(Z10.8)	1.0	-0.0100	3.0	OFF
20					
CBOD4, g/m ³	(Z10.8)	1.0	-0.0100	3.0	OFF
21					
CBOD5, g/m ³	(Z10.8)	1.0	-0.0100	3.0	OFF
22					
Algae, g/m ³	(Z10.8)	1.0	-0.0100	3.0	OFF
23					
Dissolved oxygen, g/m ³	(Z10.8)	1.0	-0.0100	-1.0	OFF
24					
Inorganic carbon, g/m ³	(Z10.8)	1.0	-0.0100	3.0	OFF
25					
Alkalinity, g/m ³	(Z10.8)	1.0	-0.0100	3.0	OFF
26					
.....CDNAME.....	FMTCD	CDMULT	CDMIN	CDMAX	CDPLTC
#					
Dissolved organic carbon, g/m ³	(F10.3)	1.0	-1.0	25.0	OFF
1					
Particulate organic carbon, g/m ³	(F10.3)	1.0	-1.0	50.0	OFF
2					
Total organic carbon, g/m ³	(F10.3)	1.0	-1.0	25.0	OFF
3					
Dissolved organic nitrogen, g/m ³	(F10.3)	1.0	-1.0	25.0	OFF
4					
Particulate organic nitrogen, g/m ³	(F10.3)	1.0	-1.0	25.0	OFF
5					
Total organic nitrogen, g/m ³	(F10.3)	1.0	-1.0	50.0	OFF
6					
Total Kheldahl Nitrogen, g/m ³	(F10.3)	1.0	-1.0	15.0	OFF
7					
Total nitrogen, g/m ³	(F10.3)	1.0	-1.0	15.0	OFF
8					
Dissolved organic phosphorus, mg/m ³	(F10.3)	1000.0	-1.0	25.0	OFF
9					
Particulate organic phosphorus, mg/m ³	(F10.3)	1000.0	-1.0	-1.0	OFF
10					
Total organic phosphorus, mg/m ³	(F10.3)	1000.0	-1.0	5.0	OFF
11					
Total phosphorus, mg/m ³	(F10.3)	1000.0	-1.0	20.0	OFF
12					
Algal production, g/m ² /day	(F10.3)	1.0	-1.0	5.0	OFF
13					
Chlorophyll a, mg/m ³	(F10.3)	1.0	-5.0	145.0	OFF
14					

Total algae, g/m ³	(F10.3)	1.0	-1.0	60.0	OFF
15					
Oxygen % Gas Saturation	(F10.3)	1.0	-1.0	50.0	OFF
16					
Total suspended Solids, g/m ³	(F10.3)	1.0	-1.0	5.0	OFF
17					
Total Inorganic Suspended Solids, g/m ³	(F10.3)	1.0	-1.0	20.0	OFF
18					
Carbonaceous Ultimate BOD, g/m ³	(F10.3)	1.0	5.0	9.0	OFF
19					
pH	(F10.3)	1.0	-1.0	10.0	OFF
20					
CO ₂	(F10.3)	1.0	-1.0	10.0	OFF
21					
HCO ₃	(F10.3)	1.0	-1.0	10.0	OFF
22					
CO ₃	(F10.3)	0.0	0.0	0.0	OFF
23					

In Version 3.2, the user has format control of all output variables, as well as MULT control (see User Manual). In Version 3.1, some groups had one but not the other. Also, in Version 3.2, the groups (HNAME, CNAME, CDNAME) were reordered.

11. Appendix B: BUG FIXES AND ENHANCEMENTS BETWEEN VERSIONS

There have been many updates and bug fixes between model versions. Even though some model updates have not been documented, we have tried to be diligent in outlining code updates since Version 3.7 between model versions. We have included below a series of tables with code fixes for multiple versions of CE-QUAL-W2 as a reference to earlier versions.

W2 V4.2 Bug Fixes, Enhancements, and User Manual Changes

Table 4 shows a list of model bug fixes since the prior release of CE-QUAL-W2 Version 4.1. Note that some of these fixes were documented before the first release of Version 4.2.

Table 4. List of bug fixes and enhancement code changes since CE-QUAL-W2 Version 4.2 was released. Note that shaded areas toggle between release versions.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
1	W2	Particle tracking	Fixed a bug when a particle was in the “air” above the water surface when in a river a ‘deep slot’ was added to the river bathymetry to keep it hydrated. Also, fixed a reflection error off the surface for vertical movement of the particle.	12/26/2018
2	W2	Particle tracking	For file output for finalparticle.csv, adjusted header so that it would be general for any number of monitoring stations. Previously the header was hard-wired to write only 3 monitoring locations in the title. This does not affect prior model output information only the title header.	1/3/2019
3	PREW2	Pipe invert	The preprocessor gave a ‘false’ error for an invert elevation being below the grid in some rare cases. This has been fixed. The model was not affected.	1/21/2019
4	W2	TN	There was a bug in the calculation of TN when large concentrations of zooplankton were present. If zooplankton were not simulated or they were a small population, there would be no impact of this TN bug. Bug fixed.	1/29/2019

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
5	W2	Epiphyton periphyton burial and conversion to LPOM and sediments	The User Manual showed that the epiphyton parameter EB, burial rate, was in units of m/day. The model implemented this and also computed the cell layer burial rate as $EB \cdot EC$ (concentration of biomass per cell volume) * Surface area/Volume of a cell or $EB \cdot EC/H$ where H is the layer thickness. In reality though it should be the thickness of the periphyton layer. But that layer thickness is not predicted by the model. Hence, the decision was to change the burial rate to units of 1/day such that the loss of epiphyton by burial would follow a first order decay process (similar to the first order sediment model) as $EB \cdot EC$. Hence, this burial rate is no longer a function of the grid. Also, there was the conversion of organic matter and nutrients at death to the sediment P, N and C compartments whereas they should go directly to LPOM first. Hence only burial goes directly into the sediment compartments.	3/5/2019
6	W2	Output files	Code was written to print the initial condition in output files at time $t=0$ at the start of the simulation.	3/5/2019
7	W2	Restart with Sediment Diagenesis	Prior to this date, the restart option did not work with sediment diagenesis turned ON. This enhancement has been coded and debugged and is now working.	7/1/2019
8	W2	Dynamic loading	This is a new feature in CE-QUAL-W2 and applies to linear waterbodies in series. For example, if one has a river system with 9 waterbodies, the model user can break the model into 9 separate models and run them all in parallel where the downstream model dynamically updates its boundary conditions as the upstream model moves forward in time. This has resulted in significant time savings for model runs (90%).	7/1/2019
9	W2	Sediment Diagenesis	Output files were updated to write csv files that can be plotted easily in Excel. The older output files for sediment diagenesis were hard to graph formatted text output files. All internal writes for warnings are now written to the W2 warning file, w2.wrn, and all errors are written to the w2.err output file.	7/1/2019
10	W2	Sediment Diagenesis	Bug fixes were made to the scour section of the model. The code had several errors based on the documentation from CEMA. These have been fixed and synchronized with the description in the CEMA sediment diagenesis report. Thanks to Lindsay Bearup at the USBR! Also, the scour model was not set-up for both Chezy and Manning's friction factors. This code was generalized.	7/1/2019
11	W2	Output files	Output files were not initialized properly for w2.wrn during a restart. This was fixed allowing the w2.wrn file to be appended to rather than over-written.	7/1/2019

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
12	W2	Code updates	Pointers were allocated memory in the ALLOCATE command. This was non-standard Fortran usage and was fixed. There were extra calls to Kinetic_Rates and Temperature_Rates subroutines that were unnecessary during layer addition and in the INIT subroutine.	7/1/2019
13	W2	TDG	A new algorithm for TDG production at spillways, SYSTDG, was implemented in Version 4.2. This is a more detailed TDG correlation equation compared with the existing algorithms in Version 4.1 and before. The primary application was for the Columbia and Snake River dams. A separate technical memo is included with the User Manual on this new process.	7/1/2019
14	W2	opt to csv	Flux output files were changed from a file type opt to csv to facilitate graphing in Excel. Structure output files for the auto port selection were changed from opt to csv.	7/1/2019
15	W2	Contour output	The Tecplot and regular Contour output was not working for cases when branches went inactive. This has been fixed.	7/1/2019
16	W2	Sediment diagenesis	The input file for sediment diagenesis has been changed from w2_CEMA_input.npt to w2_diagenesis.npt for clarity of naming files. All output files have also been changed to include 'Diagenesis' as the first word and eliminating 'CEMA'. For example the file CEMALogFile.opt was changed to DiagenesisLog-File.opt.	7/1/2019
17	W2	Withdrawal	In the withdrawal subroutine, the kt index needed to be defined twice. Below is an example of the fix in 2 places: (1) <pre> ELR = SINA(JB)*DLX(ID)*0.5 ! CB 10/14/11 WSEL = ELWS(ID)-ELR ! CB 10/14/11 kt=ktwb(jw) ! cb 07/24/19 CALL DOWNSTREAM_WITHDRAWAL_ESTIMATE(JTSPLT(J,1),TEMPTOP,ESTR(JTSPLT(J,1),TSPLTJB(J))) </pre> (2) <pre> WSEL = ELWS(IWD(JWD))-ELR ! CB 10/14/11 I=IWD(JWD) kt=ktwb(jjw) ! cb 07/24/19 CALL LATERAL_WITHDRAWAL_ESTIMATE(JTSPLT(J,1),TEMPTOP,EWD(JTSPLT(J,1))) </pre>	8/1/2019
18	W2	Branch inactive	The BR_INACTIVE flag that skips output in a branch that was inactive was dimensioned with the wrong variable for the CPL output (Tecplot output). This has been corrected.	8/27/2019
19	W2	Tecplot Branch	There is a new input file to specify what branches the user wants for Tecplot output, called TecplotBr.csv. This allows the user to only pick some of the branches rather than all of them for Tecplot output.	8/27/2019

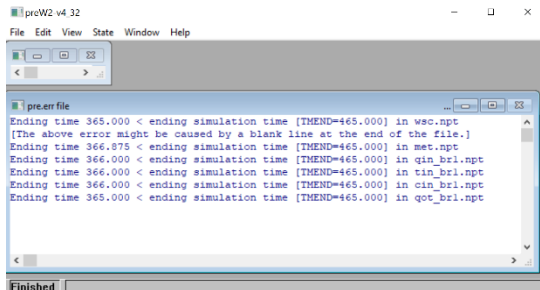
#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
20	Prew2	False warning	A false warning was sometimes activated for pipes for DYNPIPE when in fact was ON or OFF as required. This was fixed. Below is final code fix: <pre> IF ((DYNPIPE(JP) /= 'ON') .AND. (DYNPIPE(JP) /= 'OFF')) THEN CALL ERRORS WRITE (ERR,FMTI) 'Pipe DYNAMIC PIPE control [DYNPIPE='//DYNPIPE(JP)(4:8)//'] must be either "ON" or "OFF" for pipe </pre>	8/27/2019
21	W2	str_brX.opt	Structure output file, str_brX.opt, for automatic port selection was changed from a text delimited *.opt file to a comma delimited *.csv file	8/28/2019
22	W2	Auto Port Selection	The split algorithm for the auto port selection based on temperature now can read an input file of dynamic temperatures rather than relying on multiple rules with different fixed temperatures.	8/30/2019
23	W2	Tecplot	Tecplot output under the CPL output was not set up for restart properly. Tecplot output files now should work for cases where there are model restarts.	9/4/2019
24	W2	Water age	The code below for water age correction as a result of evaporation (a very minor adjustment) was incorrect: The old code was <pre> IF(AERATEC == "ON")CALL AERATEMASS IF (EVAPORATION(JW) .AND. WATER_AGE_ACTIVE)THEN ! CORRECT WATER AGE FOR EVAPORATION SR 7/27/2017 DO I=IU,ID JC=NGCS+JG_AGE-1 CSSB(KT,I,JC)=CSSB(KT,I,JC)-EV(I)*CG(KT,I,JC) ENDDO ENDIF </pre> The updated code is <pre> IF(AERATEC == "ON")CALL AERATEMASS IF (EVAPORATION(JW) .AND. WATER_AGE_ACTIVE)THEN ! CORRECT WATER AGE FOR EVAPORATION SR 7/27/2017 DO I=IU,ID JC=NGCS+JG_AGE-1 CSSB(KT,I,JC)=CSSB(KT,I,JC)- EV(I)*CG(KT,I,JG_AGE) ENDDO ENDIF </pre>	10/17/2019
25	W2	Pipe algorithm	The pipe algorithm was revised to reduce issues with pipe algorithm instabilities. There is error trapping to reduce the prevalence of growing instabilities from a water surface slope that becomes unstable.	11/1/2019

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
26	W2	W2 fixes	Small fixes that do not affect the model user: Eliminated the unused variable PHSET and changed INTEGER*8 variables in the SYSTDG algorithm to INTEGER*4. Thanks to Stewart Rounds USGS for noting these items.	11/13/2019
27	W2	Turbidity	Turbidity can be used through the sediment diagenesis model as a derived variable. This will shortly be moved to a derived constituent in W2 and taken out of the sediment diagenesis subroutine. There was a coding error in the turbidity routine that was fixed: <pre> !CellTSSValue = C1(K,SegNumI,6) !CellTSSValue = TOTSS(K,SegNumI) !SW 11/15/2019 C1(K,SegNumI,nturb) ! cb 2/18/13 !C1(K,SegNumI,6) = exp(CoeffA_Turb*log(CellTSSValue) + CoeffB_Turb) !C2(K,SegNumI,6) = exp(CoeffA_Turb*log(CellTSSValue) + CoeffB_Turb) !C1(K,SegNumI,nturb) = exp(CoeffA_Turb*log(CellTSSValue) + CoeffB_Turb) ! cb 2/18/13 !C2(K,SegNumI,nturb) = exp(CoeffA_Turb*log(CellTSSValue) + CoeffB_Turb) C1(K,SegNumI,nturb) = exp(CoeffA_Turb*log(TOTSS(K,SegNumI)) + CoeffB_Turb) ! cb 2/18/13 C2(K,SegNumI,nturb) = C1(K,SegNumI,nturb) !exp(CoeffA_Turb*log(CellTSSValue) + CoeffB_Turb) </pre>	11/15/2019
28	W2	Examples and User Manual	The kinetic flux output order for DOEP and DOAR was reversed in a couple of the input files in the model example files (in w2_con.npt) and in one place in the User Manual. These were corrected.	11/15/2019

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
29	W2	Branch Active	In the process of setting inflows to upstream inactive branches to branches that were still active, the following code was updated for complex, multi-waterbody systems: [Chris Berger] <pre> IF (BR_INACTIVE(JB)) THEN ! CONVERT INFLOWS TO TRIBS SET TO THE CUS(1) LOCATION JTT=JTT+1 ! ITR(JTT)=CUS(1) ! HARDWIRED TO FIRST BRANCH ITR(JTT)=CUS(jbdn(jw)) ! changed HARDWIRE TO JBDN BRANCH ! cb 11/20/19 QTR(JTT) = QIN(JB) ... ! including tributary flows for inactive branches DO JW=1,NWB ! cb 11/20/19 KT = KTWB(JW) DO JB=BS(JW),BE(JW) IF (BR_INACTIVE(JB)) then IU = CUS(JB) ID = DS(JB) IF (TRIBUTARIES) THEN DO JT=1,JTT !***** Inflow fractions IF (JB == JBTR(JT)) THEN I = cus(jbdn(jw)) ! plac- ing tributary flows in upstream end of main branch QTRF(KT:KB(I),JT) = 0.0 KTTR(JT) = KT KBTR(JT) = KB(I) KTTR(JT) = MAX(KT,KTTR(JT)) KBTR(JT) = MIN(KB(I),KBTR(JT)) IF (KBTR(JT) < KTTR(JT)) KBTR(JT) = KTTR(JT) BHSUM = 0.0 DO K=KTTR(JT),KBTR(JT) BHSUM = BHSUM+BH2(K,I) END DO DO K=KTTR(JT),KBTR(JT) QTRF(K,JT) = BH2(K,I)/BHSUM END DO DO K=KTTR(JT),KBTR(JT) QSS(K,I) = QSS(K,I)+QTR(JT)*QTRF(K,JT) END DO END IF END DO END IF end if end do end do </pre>	11/22/2019

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
30	W2	WRN output	The warning output format was improved when there were volume balance errors – thanks to Stewart Rounds, who defies retirement to provide model improvements!! In the subroutine, balances.f90, the new code is shown below: <pre> IF (VOLUME_WARNING) THEN WRITE (WRN, ' (A,F0.4,/A,I0,A,I0,A,I0,3 (:/A,E15.8, A)) ') & 'COMPUTATIONAL WARNING AT JULIAN DAY = ',JDAY, & 'WATERBODY=', JW, ' ', BRANCH=' ', JB, ' ', KT=' ', KT, & 'SPATIAL CHANGE =', VOLSBR(JB), ' M^3', & 'TEMPORAL CHANGE =', VOLTBR(JB), ' M^3', & 'VOLUME ERROR =', VOLSBR(JB)-VOLTBR(JB), ' M^3' !SR 11/16/19 WRITE (WRN,*) 'LAYER CHANGE:', LAYERCHANGE(JW) WRITE (WRN,*) 'SZ', SZ(CUS(JB):DS(JB)), 'Z', Z(CUS(JB):DS(JB)), 'H2KT', H2(KT,CUS(JB):DS(JB)), & 'H1KT', H1(KT,CUS(JB):DS(JB)), 'WSE', ELWS(CUS(JB):DS(JB)), 'Q', Q(CUS(JB):DS(JB)), & 'QC', QC(CUS(JB):DS(JB)), 'T1', T1(KT,CUS(JB):DS(JB)), 'T2', T2(KT,CUS(JB):DS(JB)), & 'SUKT', SU(KT,CUS(JB):DS(JB)), 'UKT', U(KT,CUS(JB):DS(JB)), 'QIN', QINSUM(JB), & 'QTR', QTR, 'QWD', QWD !SR 11/16/19 WARNING_OPEN = .TRUE. </pre>	12/2/2019

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
31	W2	W2 Error Dump	The W2ErrorDump.opt file output format was changed to be easier to analyze. The file was changed to W2ErrorDump.csv to facilitate opening in Excel and for general plotting. The new code is from Stewart Rounds and is found in endsimulation.f90. Old code: <pre>!OPEN(W2ERR,FILE='W2Errordump.opt',status='unknown') !WRITE(w2err,*) 'JDAY',jday,'SZ',sz,'Z',z,'H2KT',h2(kt,1:imx),'H1KT',h1(kt,1:imx),'BHR1',bhr1(kt,1:imx),'BHR2',bhr2(kt,1:imx),'WSE',elws,'Q',q,'QC',qc,'QERR',qerr,'T1',t1(kt,1:imx),'T2',t2(kt,1:imx),'SUKT',su(kt,1:imx),& ! 'UKT',u(kt,1:imx),'QIN',qin,'QTR',qtr,'QWD',qwd</pre> New Code: <pre>OPEN (W2ERR,FILE='W2Errordump.csv',status='unknown') WRITE (W2ERR,*) 'JDAY = ', JDAY WRITE (W2ERR,'(A,1000(" ",F0.6))') 'QIN:', (QIN(J),J=1,NBR) WRITE (W2ERR,'(A,1000(" ",F0.6))') 'QTR:', (QTR(J),J=1,NTRT) WRITE (W2ERR,'(A,1000(" ",F0.6))') 'QDT:', (QDTR(J),J=1,NBR) WRITE (W2ERR,'(A,1000(" ",F0.6))') 'QWD:', (QWD(J),J=1,NWDT) WRITE (W2ERR,'(/A)') 'SEG,BRANCH,KT,WSE,SZ,Z,Q,QC,QERR,H2KT,H1KT,BHR1,BHR2,T1,T2,SUKT,UKT' DO JW=1,NWB KT = KTWB(JW) DO JB=BS(JW),BE(JW) DO I=US(JB)-1,DS(JB)+1 WRITE (W2ERR,'(I0," ",I0," ",I0,14(" ",F0.6))') I, JB, KT, ELWS(I), SZ(I), Z(I), Q(I), QC(I), QERR(I), H2(KT,I), H1(KT,I), BHR1(KT,I), BHR2(KT,I), T1(KT,I), T2(KT,I), SU(KT,I), U(KT,I) END DO END DO END DO</pre>	12/30/2019
32	W2	Multiple Waterbody	Stewart Rounds improved on the code presented in Version 4.2. Instead of writing and reading restart files, the code waited for the updates before proceeding. This led to code changes in many subroutines. Also, the downstream model can now wait for input from multiple upstream files rather than just one. This update is described in the User Manual and is part of the Version 4.2.1 model update because of a new input file format for the multiple waterbody input file.	1/2/2020

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
33	PREW2	Preprocessor	A new feature was added to the preprocessor – a new windows pops up with errors if they are present. This avoids having to open the pre.err file in a word processor in case there were errors. 	3/15/2020
34	W2	Pump	Added features were added to the pump algorithm. This allows for the flow through a pump to be controlled by the downstream water level, rather than just the upstream level. If this is the case, then a negative downstream segment number is used. This also changes the logic of ELON and ELOFF. This has been updated in the User Manual also.	3/15/2020
35	W2/PREW2	Control file	A new control file option was created in Excel using a comma delimited control file, w2_con.csv. This control file is a lot simpler to use than the text file w2_con.npt because (1) strict formatting by spaces is not required, (2) easier to cut and paste large sections withing Excel, (3) variable explanations are available in Column A, (4) the graph.npt file is no longer required and is incorporated into the Excel based input file, and (5) there is a one-button function to write the input file into csv-format for reading into the W2 code. This file also has many other advantages in setting up output and more intuitive variable placement (i.e., the order of the old w2_con.npt is not strictly followed.) The only negative is that it breaks the GUI W2_Control which is not necessary other than the nice visualization bathymetry tools. But many of those are in the w2Tools post-processor. There is currently not a converter from the old format to the new one. But the model executable and preprocessor work with either control file.	4/16/2020
36	W2	TSR	TSR model output headers were improved. The P, N, and Light limitation for algae headers (APLIM, ANLIM, ALLIM) now show the algae group number rather than a generic header, such as PLIM_ALG1.	5/3/2020
37	Converter	Control file	A control file converter utility and description was added to the download package. This utility converts the control file (w2_con.npt) to a format using Excel.	5/22/2020
39	PreW2	More checks	Several additional model checks were added to the preprocessor to improve its skill set.	5/22/2020 6/15/2020

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
40	W2	Tecplot output	A new input file Tecplotbr.csv is now used to specify which branches are output to the Tecplot contour map. Before, all model branch information was output to the Tecplot file. Now the user can control this more precisely to eliminate a lot of post-processing of the data file. This is described in the User Manual.	5/22/2020
41	W2	CO2 in atmosphere	As the basis for Version 4.2.2, the CO2 gas saturation was redone. See User Manual Part 2 under TIC.	8/10/2020
42	Converter		Fixed a bug in writing out the csv format for the control file in case there were more than 1 withdrawals.	8/27/2020
43	W2	SPR ONV	Whenever the SPR was set equal to ONV for a w2_con.csv file, this output was not implemented correctly. For the input file w2_con.npt this worked as expected.	10/8/2020
44	Converter		Fixed a bug when NTR was exactly 9 (this bug also affected cases where NWB=9 or NBR=9). Also fixed some of the output formatting.	11/27/2020
45	W2	Pipe	A dimensioning bug was found when there were no pipes. This did not affect model results with the executable but did show up using the debugger in the Intel environment. This has been fixed. Many thanks to Jun Ma for noticing that!!	12/20/2020
46	W2/Pre/Converter	Csv input	When converting to the Excel macro utility for the control file and when modeling a system with no tributaries, the columns read for distributed inflows and precipitation may have been read incorrectly. This was adjusted by fixing the converter utility, the preprocessor and W2 code.	2/17/2021
47	W2	Kinetic Flux	When there are layer subtractions, the Kinetic flux in the surface layer terms can not account for the mass. This did not affect the mass balance of the constituents, just the flux variables used for output. Many thanks for Taylor Adams of Hydros for finding that error. In layeraddsubtract.f90: <pre> KF(KT,I,KFCN(1:NAF(JW),JW)) = (KF(KT-1,I,KFCN(1:NAF(JW),JW))*VOL(KT-1,I)+ KF(KT,I,KFCN(1:NAF(JW),JW))*VOL(KT,I))/(VOL(K T-1,I)+VOL(KT,I)) ! SW Fix suggested by Taylor Adams Hydros 25Feb2021 ! KF is in units of g/m3/s KFS(KT,I,KFCN(1:NAF(JW),JW)) = KFS(KT- 1,I,KFCN(1:NAF(JW),JW))+KFS(KT,I,KFCN(1:NAF(J W),JW)) ! SW Fix suggested by Taylor Adams Hydros 25Feb2021 ! KFS is in units of g KFS=KF*VOL*DT C1(KT-1,I,CN(1:NAC)) = 0.0 </pre>	2/25/2021

W2 V4.1 Bug Fixes, Enhancements, and User Manual Changes

Table 5. Bug fixes and enhancements for Version 4.1.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
1	PREW2	Additional checks	Additional checks were added to the preprocessor for sediment decay temperature coefficients and stoichiometric coefficients. Concentration summaries in downstream and upstream head boundary conditions were added to the pre.opt file.	5/19/2017
2	W2	DO Saturation	The equation for computing dissolved oxygen saturation was a function of elevation and temperature. If the user set the water body type to SALT, the TDS or salinity was used to correct the dissolved oxygen saturation. The TDS correction for dissolved oxygen saturation was added to the fresh water computation also. The new code is highlighted below: <pre> SATO = EXP (7.7117- 1.31403*(LOG (T+45.93))) *P IF (SALT_WATER) THEN SATO = EXP (LOG (SATO) - SAL*(1.7674E-2- 1.0754E1/(T+273.15)+2.1407E3/(T+273.15)** 2)) ! SAL is in ppt ELSEIF (SAL > 100.) THEN SATO = EXP (LOG (SATO) - (SAL/1000.)* (1.7674E-2- 1.0754E1/(T+273.15)+2.1407E3/(T+273.15)** 2)) ! SAL is in mg/l ENDIF </pre>	5/21/2017
3	W2	Sediment diagenesis output	Another line was added to the sediment diagenesis input file for the frequency of output. Prior to this it used the TSR output frequency and wrote out duplicate results if there was more than 1 waterbody.	5/25/2017
4	W2	Initialize variables	DLVOL, VOLTBR, EVBR, and QSUM were added to the initialized variables in INIT.F90. This only affects the Fortran compiler when it is in debug model. In the release executable all variables are initialized to zero even if not explicitly set to zero.	7/24/2017

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
5	W2	Sediment diagenesis code updates	We have deleted unused variables and array initializations. This has improved the speed of running the model with sediment diagenesis. These code areas are: <pre> !SP CEMA if(sediment_diagenesis)then If(CEMARElatedCode .and. IncludeBedConsolidation)Call ComputeCEMARElatedSourceSinks ! If(CEMARElatedCode .and. IncludeCEMASedDiagenesis)Call ComputeCEMADIagenesisSourceSinks SW 6/27/2017 end if !End SP CEMA !SP CEMA !if(sediment_diagenesis)then ! If(CEMARElatedCode .and. IncludeBedConsolidation)TSS = 0.0 ! SW 7/27/2017 !end if !End SP CEMA !SP CEMA !if(sediment_diagenesis)then ! CEMATSSCopy = TSS !end if !End SP CEMA </pre>	7/24/2017
6	PREW2	Sediment diagenesis	Additional error checking for the sediment diagenesis model was added to the preprocessor. In this case, whenever SOD was not set to zero, an error is displayed.	7/24/2017
7	W2	Assorted code improvements	Stewart Rounds of the USGS suggested a few minor updates: eliminated extra right-parentheses in a format description for time series output (the Intel compiler allowed them!), added WARNING_OPEN and ERROR_OPEN = .TRUE. in several cases where output is written to these files, and eliminated a situation where the derived output file at a withdrawal point was not written out if the file is empty. Also, for water age, evaporation should not concentrate the 'age'. Hence code was added to recognize water age and to eliminate the effect of evaporation on water age.	7/27/2017
8	W2	Branch active or inactive	In the W2 model, if a model branch became dehydrated, the model would not continue running. In order to allow for wide varieties of water levels, users would often have to add numerous deep fictitious layers to keep a branch hydrated. Now the model can handle branches becoming active or inactive automatically. Code was added to allow branches to become active as they fill up or to become inactive if they lose their water. Also, any branch inflows or tributaries entering inactive branches are automatically moved to the current active segment of the nearest hydrated branch.	7/27/2017

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
9	W2	RPOMN	Stewart Rounds, USGS, found this one. The if test below used NRPOP rather than NRPOP. Usually both NRPOP and NRPOP are both 'ON', so for most applications this should not affect the model user. OLD Code: <pre>IF(CAC(NRPOP) == ' ON')THEN IF(RPOM(K,I).GT.0.0)THEN ORGNRP(K,I)=RPOMN(K,I)/RPOM(K,I)</pre> NEW Code: <pre>IF(CAC(NRPOP) == ' ON')THEN IF(RPOM(K,I).GT.0.0)THEN ORGNRP(K,I)=RPOMN(K,I)/RPOM(K,I)</pre>	8/2/2017
10	W2	Screen Dialog Box	Under some unique conditions, exiting the W2 dialog box reinitializes some of the output files. Added code was inserted to STOP program execution after closing the dialog box.	8/23/2017
11	W2	Sediment Diagenesis	Flux rates for P, NH3, and NO3 were added to the MASSBAL output file from sediment diagenesis so that a complete N and P balance can be evaluated for a waterbody.	8/31/2017
12	W2	Particle Tracking	Particle tracking algorithm has been added and documentation in a separate report added to the model release	8/31/2017
13	W2	Opt to csv file	Changed flowbal.opt and massbal.opt to flowbal.csv and massbal.csv in order to facilitate opening in Excel.	8/31/2017
14	W2	Gate file	The gate file was inadvertently not converted over to a csv format in the earlier Version 4.0 code. The 4.1 code was updated to include csv gate files.	9/26/2017

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
15	W2	Shading	Added code to allow for a canopy shading in addition to dynamic shading. The DYNOSH in the shade input file can now be a negative number between -1 and 0. This will activate dynamic shading and taking the absolute value of this number will reduce the short wave solar radiation by a fixed fraction as if some of the channel has a canopy. Of course canopy cover is more complex than this small correction since it also affects long-wave radiation transfer. New code is highlighted below: <pre> SN = MIN (HT*ABS (SIN (ABS (PHI0(I)- AZ00))))/TAN (A0)-EDGE,BI(KT,I)) SFACT = SRED*SN/BI(KT,I) 100 CONTINUE SHADE(I) = MAX (0.0,1-SFACT) SHADE(I) = MIN(ABS(SHADEI(I)),SHADE(I)) ! SW 10/2/2017 Allows for fixed canopy cover over top of channel - only used if shade is less than shadei only valid for -0.99 and 0.0 </pre> Hence if DYNOSH (or SHADEI) in the code were -0.9 and the dynamic shading algorithm computed the shade factor as 0.95 (which is a 5% reduction in short wave solar), the code would use 0.9 or a 10% reduction in short-wave solar. If the dynamic shade algorithm computes a shade greater than the fixed rate, the minimum of these is used.	10/3/2017
16	W2	Sediment Diagenesis	Changed back to the original segment width at the bottom for sediment diagenesis so that this algorithm replicates the original CEMA sediment diagenesis algorithm. Pulled out the CellArea as a dimensioned variable computed only once rather than for each cell at each time step. Also, added a control variable to turn ON/OFF Bubbles calculation. This saves much computational time and until the Bubbles subroutine is vetted we do not recommend its use. Also several code fixes were made in the sediment diagenesis module for mistakes in the original algorithm.	10/3/2017, 10/22/2017
17	W2	User Manual	The User Manual was updated fixing minor errors and typos and adding discussion of new features of Version 4.1. This is Revision 1 of the 4.1 Manual but includes updates and fixed typos from the Version 4.0 Manual and explanations of new features.	10/3/2017
18	W2	Kinetic fluxes for ADD Layer	A bug was corrected in the Kinetic Flux layer addition code as shown below: <pre> KFS(KT,I,KFCN(1:NAF(JW),JW)) = KFS(KT+1,I,KFCN(1:NAF(JW),JW)) !KF(KT+1,I,KFCN(1:NAF(JW),JW)) CODE ERROR FIX SW 10/24/2017 </pre> This does not affect fluxes in the TSR file, only the cumulative fluxes (KFS) during an add layer event for the surface layer only.	10/24/2017

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
19	W2	Header	Changed header for flux for DO reaeration from just source to source/sink which can occur during super-saturation. KFNAME(64) = 'DO reaeration - source/sink, kg/day'	10/24/2017
20	W2	Fluxes	In the file, kfl_wbX.opt where X is the waterbody number, the fluxes are presented in the same format as a snapshot file. The headers showed fluxes in kg/d but they were in kg. This has been corrected. The fluxes in kg/d in the file kflux_wbX.opt were already in the correct units of kg/d. Also, added the following code since KT would have been from the prior waterbody rather than the current waterbody: <pre> DO JW=1,NWB KT = KTWB(JW) ! SW 10/25/2017 IF (FLUX(JW)) CALL KINETIC_FLUXES END DO </pre>	10/24/2017
21	W2	Pumps	Changed some of the logic for pumps to avoid settings for older values influencing the current settings. The following code was added to hydroinout.f90: <pre> ILAT = 0 JWW = NWD withdrawals = jww > 0 if(nwdt>nwd)qwd(nwd+1:nwdt)=0.0 ! SW 10/30/2017 JTT = NTR tributaries = jtt > 0 if(ntrt>ntr)qtr(ntr+1:ntrt)=0.0 ! SW 10/30/2017 JSS = NSTR </pre>	10/30/2017

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
22	W2	TDG at spillway	Since the new implementation of the TDG algorithm, the withdrawal.f90 algorithm at spillways had not been updated properly. This fix applies to the computation of TDG at spillways and gates. The old and new code are shown below: <pre> !if(tdgon)then ! cb 11/6/17 !cdavg(js,jb,16) = (cavg(js,jb,ndo)/exp(7.7117- 1.31403*(log(tavg(js,jb)+45.93))))*palt(id))*1 00.0 dosat=exp(7.7117- 1.31403*(log(tavg(js,jb)+45.93))))*palt(id) cdavg(js,jb,16)=(cavg(js,jb,ndo)/dosat)*100.0 IF(ngctdg /= 0)THEN EA = DEXP(2.3026D0*(7.5D0*TDEW(JW)/(TDEW(JW)+237.3 D0)+0.6609D0))*0.001316 ! in mm Hg 0.0098692atm=7.5006151mmHg !cdavg(js,jb,NDC) = (cavg(js,jb,NGN2)/(1.5568D06*0.79*(PALT(ID)- EA)*(1.8816D-5 - 4.116D-7 * Tavg(js,jb) + 4.6D-9 * Tavg(js,jb)**2)))*100.0 ! SW 10/27/15 n2sat=1.5568D06*0.79*(PALT(ID)- EA)*(1.8816D-5 - 4.116D-7 * Tavg(js,jb) + 4.6D-9 * Tavg(js,jb)**2) cdavg(js,jb,NDC) = 100.*(0.79*(cavg(js,jb,NGN2)/n2sat) + 0.21*(cavg(js,jb,ndo)/dosat)) ENDIF !end if </pre>	11/8/2017
23	W2 Control	Updated GUI	A new version of W2Control has been made to account for a large number of small refinements in the control file to bring it up to Version 4.1. Otherwise, some of these changes had to be implemented by editing the text file, w2_con.npt. Also, guidance was added to the release notes how to use the GUI with a touch-screen laptop or desktop in cases where the touch screen does not work with the GUI.	3/10/2018

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
24	W2	Csv file input for shade file	Updated reading in csv file for shade file to be compatible with preprocessor. Sometimes one needs to add an extra column of commas in a csv input file. New code: <pre> IF (INFORMAT=='\$') THEN READ (SHD, '(/') DO I=1, IMX READ (SHD, *) J, SHADEI(I) ! SW 3/14/2018 ADDED TO BE COMPATIBLE WITH PREPROCESSOR IF (SHADEI(I) < 0.0) THEN BACKSPACE (SHD) READ (SHD, *) J, SHADEI(I), TTLB(I), TTRB(I), CLLB(I), CLRB(I), SRLB1(I), SRLB2(I), SRRB1(I), SRRB2(I), (TOPO(I, J), J=1, IANG), SRFJD1(I), SRFJD2(I) ENDIF ENDDO </pre>	3/14/2018
25	W2	TSR file output	When the model user sets the elevation as a negative value, the model outputs variables at that layer only. When the water level went below that layer, the output was fixed at the old value of the variable until the water level rose into the layer. To eliminate issues with misinterpreting or having to edit out constant values, whenever the water level is below the layer, now a -99 is written out showing that there is no water in the layer specified.	4/5/2018
26	PREW2	More checks	Added checks for NaN in input files for meteorological files and flow, temperature and concentration files for inflows, distributed tributaries, precipitation, and tributaries. Previously, the preprocessor read input files even with NaN without reporting an error since this is a proper numerical value.	4/10/2018
27	W2	WDO output	The Withdrawal files are often used for downstream models. The withdrawal output frequency is in days. In order to make this more precise numerically, the output frequency can also be entered in hours and seconds. The problem was that 1 hour is 0.04167 days and due to round off error for long term runs of many years, the hourly frequency output would not be at the same hour. The variable WDOC now can be ON/OFF/ONS/ONH where ONS means the output frequency is in sec and ONH means the output frequency is in hours. The User Manual, GUI, and Preprocessor have been updated.	4/10/2018
28	Water-balance	Water-balance	A new waterbalance utility was released that is a console application and can be used in batch file processing. It also has more features than the old water balance utility allowing the use of multiple waterbodies in the calculation of flows. There is a new file directory for this application in the download section as well as executables and a User Manual for this utility.	4/10/2018

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
29	PREW2	More checks	Several more preprocessor enhancements were added to the preprocessor to check for Unix file type where an EOL is a <CR> and not the windows <CR><LF>. Also a few more descriptions were added to the error output to clarify where a problem may be.	4/18/2018
30	Water-balance	Water-balance	Some refinements to the water balance utility console application were made.	6/5/2018
31	W2	Sediment Resuspension	The sediment resuspension algorithm computed sediment resuspension in all model layers except the surface layer. The fix below adds resuspension for the surface layer (highlighted code is new). <pre> IF (SEDIMENT_RESUSPENSION(J)) THEN FETCH = FETCHD(I,JB) IF (COS(PHI(JW)-PHI0(I)) < 0.0) FETCH = FETCHU(I,JB) FETCH = MAX(FETCH,BI(KT,I),DLX(I)) U2 = WIND(JW)*WSC(I)*WIND(JW)*WSC(I)+NONZERO COEF1 = 0.53 *(G*DEPTHB(KT,I)/U2)**0.75 COEF2 = 0.0125*(G*FETCH/U2)**0.42 COEF3 = 0.833* (G*DEPTHB(KT,I)/U2)**0.375 COEF4 = 0.077* (G*FETCH/U2)**0.25 HS = 0.283 *U2/G*0.283*TANH(COEF1)*TANH(COEF2/TANH(COEF1)) ITS = 2.0*PI*U2/G*1.2* TANH(COEF3)*TANH(COEF4/TANH(COEF3)) TS = 2.0*PI*sqrt(U2)/G*1.2* TANH(COEF3)*TANH(COEF4/TANH(COEF3)) ! cb 7/15/14 L0 = G*TS*TS/(2.0*PI) L1 = L0 ! SW 6/28/2018 Allow for resuspension of surface layer L = L0*TANH(2.0*PI*DEPTHB(KT,I)/L1) DO WHILE (ABS(L-L1) > 0.001) L1 = L L = L0*TANH(2.0*PI*DEPTHB(KT,I)/L1) END DO COEF = MIN(710.0,2.0*PI*DEPTHB(KT,I)/L) UORB = PI*HS/TS*100.0/SINH(COEF) TAU = 0.003*UORB*UORB IF (TAU-TAUCR(J) > 0.0) EPSILON = MAX(0.0,0.008/49.0*(TAU-TAUCR(J))*3*10000.0/DLT) SSR = EPSILON*DLX(I)*(BI(KT,I)- BI(KT+1,I))/VOL(KT,I) END IF SSSS(KT,I,J) = - SSS(J)*SS(KT,I,J)*BI(KT,I)/BH2(KT,I)+SSR </pre>	6/28/2018
32	W2	Sediment Flux	Added 'RECURSIVE' to the subroutine definition as it calls itself and recent updates to the Intel compiler flagged this as an error. <pre> RECURSIVE Subroutine CEMASedimentDiagenesis Use MAIN Use GLOBAL Use GEOMC Use SCREENC </pre>	6/28/2018
33	PREW2	Added checks	Assorted minor updates to error checks	7/1/2018

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
34	PREW2	Consolidation	Update to checking code in sediment diagenesis for consolidation. New code: <pre> Read(5000,*)MessageTemp, LayerAddThkFrac Read(5000,*)MessageTemp, NumConsolidRegns if(IncludeBedConsolidation .and. NumConsolidRegns == 0)then ! CB 7/9/2018 changed .or. to .and. CALL ERRORS </pre>	7/9/2018
35	W2	Sediment	The sediment model allows for first order sediment decay in the layers above the bottom layer. If a modeler set as an initial condition a finite value for the initial concentration of sediment, then the amount in layer kb would be there and not transfer eventually to the sediment diagenesis layer as downward sloughing occurs. Hence, if the model user sets an initial concentration of the first order sediment model and has turned on sediment diagenesis, then the code sets the value of sediment in layer kb to zero. The code fix for SED is shown below: <pre> SED(KT,I) = SED(KT,I)/H2(KT,I) If(CEMARelatedCode .and. IncludeCEMASedDiagenesis)then ! cb 07/23/18 SED(KT+1:KB(I)-1,I) = SED(KT+1:KB(I)-1,I)/H2(KT+1:KB(I)-1,I) sed(kb(i),i)=0.0 else SED(KT+1:KB(I),I) = SED(KT+1:KB(I),I)/H2(KT+1:KB(I),I) end if END DO </pre> Similar code fixes were applied to SEDP, SEDC and SEDN.	7/23/2018
36	W2	TSR output	When a branch goes inactive, the TSR output file continued to write output information even though the segment was not hydrated. Logic was added to skip output for a TSR file if the segment was inactive or if the fixed layer was above the surface layer. <pre> DO J=1,NIKTSR I = ITSR(J) ! find out if segment is inactive OR cell is inactive for fixed layer - do not write out tsr file ! SW 7/24/2018 IF(BR_INACTIVE(JBTSR(J)))CYCLE IF(CUS(JBTSR(J)) > I)CYCLE DO JW=1,NWB IF (I >= US(BS(JW))-1 .AND. I <= DS(BE(JW))+1) EXIT END DO IF(ETSR(J) < 0)THEN ! SW 7/24/2018 IF(INT(ABS(ETSR(J))) < KTWB(JW))CYCLE ENDIF </pre>	7/24/2018

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
37	W2	WRN	When a branch becomes ACTIVE, it is now written out to the WRN file (this was previously written out to the SNP file and only when INACTIVE to the WRN file).	7/24/2018
38	W2	SO4 in Sediment	The code was updated to compute SO4 in both the aerobic and anaerobic layers and to allow diffusion from the water column to the sediment. This was previously not implemented. Also, updates were made to the bed consolidation routine in the sediment diagenesis model.	8/1/2018
39	W2, PREW2	Constriction	This is a model enhancement: The model user can specify a maximum width between segments by specifying that in a new input file called 'constriction.csv'. This does not affect the volume of the segments – it only affects the right hand side face width. Whereas an internal weir blocks all the flow, this allows for a reduced area and eliminates the need to insert a short segment of small width that can cause stability and lowered time step issues. The new input file is 'constiction.csv' and has the following format: Line 1: Comment Line 2: # of constrictions Line 3: Comment Line 4: [Repeated by # of constrictions] Segment # of constriction, Maximum width in m of constriction # of Constrictions 1 Seg # Max width, m 10 115 The W2 code looks for 'constriction.csv' – if it is found, then it reads the file and applies this to the right hand side width between 2 segments. Updates were made to the User Manual, Preprocessor and W2 Executables	8/3/2018
40	W2	Restart	If there were WDO output files, the RESTART option may not have worked properly. A code fix was implemented for a RESTART with WDO output files.	8/3/2018
41	W2	Envir Perf	The environmental performance output was updated to include the descriptive terms: 'Sum of fractions' and 'Average'. The User Manual was updated.	8/6/2018
42	W2	Particle Transport	Tracking the time history of temperature, velocity and depth of particles was added to the particle transport algorithm. The file 'particle.csv' now includes information on histogram output for each particle released into the waterbody. The User Manual and Preprocessor were also updated for this new feature.	8/6/2018

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
43	W2	Layer subtract	When the W2 model adds a deep slot in shallow rivers, the variable AVHR in layeraddsub.f90 was defined for an inactive segment incorrectly. This code was cleaned up. New code is shown below: <pre> DEPTHM(KTWB(JW),I)=(H1(KTWB(JW),I)- (EL(KBI(I)+1,I)-EL(KB(I)+1,I)))*0.5 ! SW 1/23/06 IF(I<=DS(JB))THEN ! SW 8/6/2018 AVHR(KT,I)=(H1(KT,I)- (EL(KBI(I)+1,I)-EL(KB(I)+1,I)))+(H1(KT,I+1)- (EL(KBI(I)+1,I+1)-EL(KB(I)+1,I+1))& -H1(KT,I)+(EL(KBI(I)+1,I)- EL(KB(I)+1,I)))/(0.5*(DLX(I)+DLX(I+1)))*0.5*DLX(I) ELSE AVHR(KT,I)=AVHR(KT,I-1) ENDIF </pre>	8/6/2018
44	W2	Input bathymetry	If the model user in the bathymetry input file gives the water surface (or FRIC, or PHI0, or DLX) of inactive segments a '0', the code in the first time step may set the KT layer incorrectly because of using ELWS of an inactive segment. The code though fixes it on the 2nd time step. This does not really affect the model run as it gets started other than an unnecessary change in the surface layer designation. If the model user gave the inactive segments the same values as the active ones, the code ran normally from the first time step. In order to standardize this and allow the model user to keep inactive segment ELWS, DLX, FRIC, and PHI0 set to zero, these values are initialized with the neighboring segment value. <pre> READ (BTH(JW),'/(10F8.0)') (DLX(I), I=US(BS(JW))-1,DS(BE(JW))+1) READ (BTH(JW),'/(10F8.0)') (ELWS(I), I=US(BS(JW))-1,DS(BE(JW))+1) READ (BTH(JW),'/(10F8.0)') (PHI0(I), I=US(BS(JW))-1,DS(BE(JW))+1) READ (BTH(JW),'/(10F8.0)') (FRIC(I), I=US(BS(JW))-1,DS(BE(JW))+1) READ (BTH(JW),'/(10F8.0)') (H(K,JW), K=1,KMX) DO I=US(BS(JW))-1,DS(BE(JW))+1 READ (BTH(JW),'/(10F8.0)') (B(K,I), K=1,KMX) H2(:,I) = H(:,JW) END DO endif ! Set water surface of inactive segments to those active cells next to them SW 8/6/2018 DO JB=BS(JW),BE(JW) ELWS(US(JB)-1)=ELWS(US(JB)) ELWS(DS(JB)+1)=ELWS(DS(JB)) DLX(US(JB)-1)=DLX(US(JB)) DLX(DS(JB)+1)=DLX(DS(JB)) PHI0(US(JB)-1)=PHI0(US(JB)) PHI0(DS(JB)+1)=PHI0(DS(JB)) FRIC(US(JB)-1)=FRIC(US(JB)) FRIC(DS(JB)+1)=FRIC(DS(JB)) END DO </pre>	8/6/2018
45	W2	Particle	Added the ability of particles to be passed to other branches by connections of gates, pumps or spillways between upstream and downstream branches aligned linearly along a river.	8/7/2018

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
46	W2	Flow Balance Output	The flow balance output file, flowbalance.csv, output was affected by Branches becoming active or inactive. Code was revised in balances.f90 to allow for the flow balances to properly account for cumulative flow sources/sinks. <pre> IF (VOLUME_BALANCE(JW)) THEN DO JB=BS(JW),BE(JW) IF(.NOT.BR_INACTIVE(JB))THEN ! SW 8/8/2018 VOLSBR(JB) = VOLSBR(JB)+DLVOL(JB) VOLTBR(JB) = VOLEV(JB)+VOLPR(JB)+VOLTRB(JB)+VOLDT(JB)+VOLWD(JB)+V OLUH(JB)+VOLDH(JB)+VOLIN(JB)+VOLOUT(JB)+VOLICE(JB) if(sediment_diagenesis)then If(CEMARElatedCode .and. IncludeBedCon- solidation)Then VOLTBR(JB) = VOLTBR(JB)+ VOLCEMA(JB) End If ENDIF IF (ENERGY_BALANCE(JW)) THEN ESR(JW) = 0.0 ETR(JW) = 0.0 DO JB=BS(JW),BE(JW) IF(BR_INACTIVE(JB))CYCLE ! SW 8/8/2018 IF (MASS_BALANCE(JW)) THEN DO JB=BS(JW),BE(JW) IF(BR_INACTIVE(JB))CYCLE ! SW 8/8/2018 DO JC=1,NAC </pre>	8/8/2018
47	W2	Particle Transport	In the particle tracking algorithm, the depth statistics of are now based on the particle depth. Before this, the depth statistics of a particle were based on water depth not particle depth.	8/24/2018

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
48	W2	Long Profile Output	The volume weighted temperature was added to the longitudinal profile output. The User Manual was also updated. <pre> WRITE(NUNIT, '(1000(A, ", "))')'Seg#', 'ElevWaterSurf(m) ', 'Q(m3/s)', 'SurfaceT- emp(oC)', 'Depth(m)', 'Width(m)', 'VolWeighTemp(oC)' , (CNAME2(CN(JC)), JC=1, NAC), (CDNAME2(CDN(JD, JW)), JD=1 , NACD(JW)) ELSE WRITE(NUNIT, '(1000(A, ", "))')'Seg#', 'ElevWaterSurf(m) ', 'Q(m3/s)', 'SurfaceT- emp(oC)', 'Depth(m)', 'Width(m)', 'VolWeighTemp(oC)' ENDIF DO JJ=1, NWB K=KTWB(JJ) DO JB=BS(JJ), BE(JJ) DO I=CUS(JB), DS(JB) ! TEMP VOL WEIGHTED AVERAGE SW 8/30/2018 TVOLAVG=0.0 ! SW 8/30/2018 VOLTOT=0.0 DO KK=KTWB(JW), KB(I) VOLTOT=VOLTOT+VOL(KK, I) TVOLAVG=TVOLAVG+T1(KK, I)*VOL(KK, I) ENDDO IF(KB(I)>=KTWB(JW))tvolavg=tvolavg/voltot IF(CONSTITUENTS)THEN WRITE(NUNIT, '(I5, ", ", 100(F12.3, ", "))')I, ELWS(I), QC(I), T2(K, I), DEPTHB(KB(I), I), B(KTI(I), I), TVOLAVG, (C2(K, I, CN(JAC))*CMULT(CN(JAC)), JAC=1, NAC), (CD(K, I, CDN(JD, JW))*CDMULT(CDN(JD, JW)), JD=1, NACD(JW)) ELSE WRITE(NUNIT, '(I5, ", ", 100(F12.3, ", "))')I, ELWS(I), QC(I), T2(K, I), DEPTHB(KB(I), I), B(KTI(I), I), TVOLAVG </pre>	8/30/2018
49	W2	SPR output	The spreadsheet output file now has an option to output volume weighted values at the locations and times specified. If the user enters 'ONV' in the control file in the following line: <pre> SPR PLOT SPRC NSPR NISPR WB 1 ONV 1 2 </pre> the output will include the regular SPR output and a separate file with the suffix '_volw.csv' for the volume weighted output. The User Manual has also been updated with the new feature.	9/28/2018
50	W2	User Manual	Updated the LPOM, LPOM-P, and LPOM-N source sink equations and the LPOM source sink stencil to be in agreement with the model code. These were old and did not reflect the addition of zooplankton ingestion and excretion/mortality.	10/9/2018
51	W2	Particle Tracking	Added an enhancement to add monitoring segments that record when the particle goes by the monitor or sensor. This would be useful for comparing to a dye study in a river for example.	11/14/2018
52	W2	SPR output	A volume weighted output bug for item 49 was fixed.	11/14/2018
53	W2	TSR flux output units	If flux output was ON, the instantaneous flux values for TSR output were in the incorrect units. These units were corrected to kg/day. Thanks to Binglei Gong at Anchor! All other flux files units were not affected.	12/10/2018

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
54	W2	Volume balance	The volume balance when segments were added and subtracted because of a shallow location in the grid, caused the code to incorrectly compute the volume balance (if VBC is ON). This was corrected.	12/18/2018
55	W2	Internal solar radiation and angle of sun for shading	The internal solar radiation algorithm had a bug that if one used reference years many years from your simulation or made a simulation over 20 years, the solar maximum each year drifts. If you read in solar radiation, there are no issues or if one makes runs from 1-10 years – the drift is probably minimal. This mainly affects those who use a reference year far from the start date or run simulations over 20 years or longer while using the internal short wave solar algorithm. Fix from Dan Turner at USACE and Stewart Rounds USGS – many kudos!!	12/18/2018
56	W2	Energy and mass balance	For a simulation with an initial BRANCH INACTIVE, the energy and mass balance if ON (EBC and MBC set to ON) starts off incorrectly. The layer subtraction algorithm when segments were subtracted in some cases where there were deeper holes followed by shallows incorrectly computed the energy and constituent mass balance. Also, a couple changes were made for the heat balance and mass balance for constituents when branches were changed from active to inactive.	12/18/2018
57	W2	Particle tracking	Fixed a bug when a particle was in the “air” above the water surface when in a river a ‘deep slot’ was added to the river bathymetry to keep it hydrated. Also, fixed a reflection error off the surface for vertical movement of the particle.	12/26/2018
58	W2	Particle tracking	For file output for finalparticle.csv, adjusted header so that it would be general for any number of monitoring stations. Previously the header was hard-wired to write only 3 monitoring locations in the title. This does not affect prior model output information only the title header.	1/3/2019
59	PREW2	Pipe invert	The preprocessor gave a ‘false’ error for an invert elevation being below the grid in some rare cases. This has been fixed. The model was not affected.	1/21/2019
60	W2	TN	There was a bug in the calculation of TN when large concentrations of zooplankton were present. If zooplankton were not simulated or they were a small population, there would be no impact of this TN bug. Bug fixed.	1/29/2019

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
61	W2	Epiphyton periphyton burial and conversion to LPOM and sediments	The User Manual showed that the epiphyton parameter EB, burial rate, was in units of m/day. The model implemented this and also computed the cell layer burial rate as $EB \cdot EC$ (concentration of biomass per cell volume) * Surface area/Volume of a cell or $EB \cdot EC/H$ where H is the layer thickness. In reality though it should be the thickness of the periphyton layer. But that layer thickness is not predicted by the model. Hence, the decision was to change the burial rate to units of 1/day such that the loss of epiphyton by burial would follow a first order decay process (similar to the first order sediment model) as $EB \cdot EC$. Hence, this burial rate is no longer a function of the grid. Also, there was the conversion of organic matter and nutrients at death to the sediment P, N and C compartments whereas they should go directly to LPOM first. Hence only burial goes directly into the sediment compartments.	3/5/2019
62	W2	Output files	Code was written to print the initial condition in output files at time $t=0$ at the start of the simulation.	3/5/2019
63	W2	Restart with Sediment Diagenesis	Prior to this date, the restart option did not work with sediment diagenesis turned ON. This enhancement has been coded and debugged and is now working.	3/5/2019
64	W2	Dynamic loading	This is a new feature in CE-QUAL-W2 and applies to linear waterbodies in series. For example, if one has a river system with 9 waterbodies, the model user can break the model into 9 separate models and run them all in parallel where the downstream model dynamically updates its boundary conditions as the upstream model moves forward in time. This has resulted in significant time savings for model runs (90%). Documentation for this will be provided shortly.	3/5/2019
65	W2	User Manual	The existing User Manual will be frozen in its current version until a newly updated user manual will be released in the next couple months. Hence, the dynamic loading described above will be described in the new User Manual.	3/5/2019

W2 V4.0 Bug Fixes, Enhancements, and User Manual Changes

Table 6. Bug fixes and enhancements for Version 4.0.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
1	PREW2	Additional model checks	Additional model checks were added for Profile and Spreadsheet output model segments	6/7/16
2	User Manual	Updated	User Manual Rev 6 was released with many minor updates and better explanatory text	6/7/16
3	W2	Restart	Fixed restart to work for epiphyton and macrophytes. This was broken in case a model user used RESTART. Fixed restart for mass balance for nutrients output in the file mass-bal.opt.	6/7/16
4	W2	Location of compiler info file	Fixed location of W2 compiler information in case of using command line aware directory. File was written to the location of the model executable rather than the command line aware directory.	6/7/16
5	Waterbalance	Update for Version 4	The waterbalance utility uses a model tsr file for reading in water level over time. Since the Version 4 file format was updated with comma delimited output files, the waterbalance utility has been updated. This utility is not compatible with earlier versions.	6/10/16
6	W2	Sediment Diagenesis	Initialized the sediment width (sedcellwidth) in subroutine CEMASedimentDiagenesis.	6/11/16
7	W2	Screen output	The text fields in the Windows dialog box may 'overflow' if you have more than 160 tributaries. The field size was increased to avoid this possibility. Old code: CHARACTER(1000) :: TEXT1 New code: CHARACTER(1700) :: TEXT1	6/24/16
8	W2	Profile output	The longitudinal profile output added depth at a segment as part of the longitudinal output. User Manual updated also.	7/11/2016
9	W2	Profile output	Changed file name of longitudinal file output from integer of the Julian day to Julian day in F8.2 format in case of multiple outputs on one day	7/16/2016

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added																																																																						
10	W2	TSR output	Changed TSR file so that the first 11 lines of header are eliminated to facilitate graphing. Also, the name of the filetype in the control file is now read and used for the output file. Hence, using the TSR FILENAME of 'tsr.csv' will produce csv files that are immediately opened in Excel for viewing again making it easier for post-processing.	8/1/2016																																																																						
11	PRE	Met file checks	The preprocessor has been enhanced with more model file checks. This program now has summaries of meteorological data (min, max, average) for each waterbody in the pre.opt file as well as further logical checks on values of these averages. These summaries are another check on the correctness of the input met data file. A typical result in pre.opt is shown below: <table> <tr> <th colspan="5">Meteorological Data Input Summary</th></tr> <tr> <th>Parameter</th><th>Waterbody</th><th>Average Value</th><th>Maximum</th><th>Minimum</th></tr> <tr> <td>TAIR (C)</td><td>1</td><td>10.553</td><td>37.780</td><td>-11.940</td></tr> <tr> <td>TDEW (C)</td><td>1</td><td>6.935</td><td>19.500</td><td>-17.670</td></tr> <tr> <td>WIND (m/s)</td><td>1</td><td>1.337</td><td>12.440</td><td>0.000</td></tr> <tr> <td>PHI (rad)</td><td>1</td><td>3.426</td><td>6.280</td><td>0.000</td></tr> <tr> <td>CLOUD (0-10)</td><td>1</td><td>7.367</td><td>9.720</td><td>0.000</td></tr> <tr> <td>SRO (W/m2)</td><td>1</td><td>0.000</td><td>0.000</td><td>0.000</td></tr> <tr> <td>TAIR (C)</td><td>2</td><td>10.553</td><td>37.780</td><td>-11.940</td></tr> <tr> <td>TDEW (C)</td><td>2</td><td>6.935</td><td>19.500</td><td>-17.670</td></tr> <tr> <td>WIND (m/s)</td><td>2</td><td>1.337</td><td>12.440</td><td>0.000</td></tr> <tr> <td>PHI (rad)</td><td>2</td><td>3.426</td><td>6.280</td><td>0.000</td></tr> <tr> <td>CLOUD (0-10)</td><td>2</td><td>7.367</td><td>9.720</td><td>0.000</td></tr> <tr> <td>SRO (W/m2)</td><td>2</td><td>0.000</td><td>0.000</td><td>0.000</td></tr> </table>	Meteorological Data Input Summary					Parameter	Waterbody	Average Value	Maximum	Minimum	TAIR (C)	1	10.553	37.780	-11.940	TDEW (C)	1	6.935	19.500	-17.670	WIND (m/s)	1	1.337	12.440	0.000	PHI (rad)	1	3.426	6.280	0.000	CLOUD (0-10)	1	7.367	9.720	0.000	SRO (W/m2)	1	0.000	0.000	0.000	TAIR (C)	2	10.553	37.780	-11.940	TDEW (C)	2	6.935	19.500	-17.670	WIND (m/s)	2	1.337	12.440	0.000	PHI (rad)	2	3.426	6.280	0.000	CLOUD (0-10)	2	7.367	9.720	0.000	SRO (W/m2)	2	0.000	0.000	0.000	10/30/16
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12	PRE	Distributed concentration checks	Added checks for average, min, and max inflow concentrations for all distributed tributaries. These are written out to the pre.opt file	11/1/16																																																																						
13	PRE	LPR input	For LPR file inputs for temperature, the preprocessor reports an error when using LPR input. The code incorrectly used KT rather than KTWB(JW). [This also affects V3.7 preprocessor.]	11/9/2016																																																																						
14	W2	Model update	The model executables were updated from Intel Fortran Compiler # 14 to Intel Fortran compiler # 17. Also, the flag to initialize all variables to zero was enforced. There are many variables in the new sediment diagenesis model that need to be explicitly set to zero. These initializations will be made in the code in the future so that setting this flag will be unnecessary.	11/17/2016																																																																						
15	W2 and PRE	Code updates	A couple code updates were made as a result of using the Intel Fortran Version 17 compiler. The new compiler did not like some of the older implementations. These were minor updates.	11/22/2016																																																																						
16	W2	Output	Improved clarity of output headers for flux outputs, including units of kg/d in all header titles	11/28/2016																																																																						

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
17	W2	Output header	The order of the flux headings in the file 'kflux_jwX.opt' were switched. The header showed DOAP, DOAR, DOEP, DOER but it should have been DOAP, DOEP, DOAR, DOER. This is determined from the order in the example problem control files and the User Manual. The example problems and User Manual have all been updated.	11/28/2016
18	W2	Example problems	Updated example problems using FIX #10 above where tsr.opt filename was changed to tsr.csv allowing tsr files to open directly in Excel.	11/28/2016
19	W2	Algae-Si	The flux of Si from dying algae was incorrectly computed. This bug has existed since Version 3.0 when the algorithm was first added to W2. Below is the code fix: <pre> ENTRY PARTICULATE_SILICA PSIAM(:,IU:ID) = 0.0 DO I=IU,ID DO K=KT,KB(I) DO JA=1,NAL IF (ALG_CALC(JA)) THEN PSIAM(K,I) = PSIAM(K,I)+AMR(K,I,JA)*ALG(K,I,JA)*ASI(JA) ! PSI(K,I) HA-Z 12/2016 ENDIF </pre>	12/5/2016
20	W2	WDO output	Enhancement: The Withdrawal output file name WDOFN was unused in the main program. Now the model reads this file and uses the file type for all WDO output files. Previously this was hard-wired to 'opt' output. Now if the user sets WDOFN to 'wdo.csv' all the files will be written with the 'csv' file type facilitating opening in Excel. The files are already in comma delimited format.	12/8/2016
21	W2	DLT INTER	There was a problem computing the interpolated value of DLTMAX and DLTF when the first value of DLTD was earlier than the start date of the model. This bug was fixed.	12/9/2016
22	User Manual	Updates	Assorted typos fixed, better explanatory text added, and added definitions and units of model parameters. This is REV8.	1/6/2017, 2/10/2017
23	W2	Output format	Output format changed for Bioenergetics output file	1/6/2017
24	W2	TECPLOT output	Added derived variables to TECPLOT output files (See Contour Plot in User Manual). User Manual updated.	1/17/2017
25	PREW2	ENVIRPC	Checks were added for the ENVIRPC input file in the preprocessor.	2/16/2017
25	W2	ENVIRPC	Fixed several minor bugs in the ENVIRPC subroutine and added an enhancement to perform a histogram analysis of water depth. The User Manual was updated to reflect this new enhancement as well as the new csv output format and file names. The example problems were updated with new w2_envirprf.npt files.	2/16/2017

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Fixed or Enhancement added
26	W2	Head BC input files	For head boundary condition input files (both upstream and downstream), the W2 code was updated to include new file formats for these boundary conditions (BCs). They include the older format, a new csv format and a new csv format in case conditions are not stratified at the BC. The User Manual was updated to show these new file formats.	3/3/2017
27	PREW2	Head BC checks	With the new file format for head BCs in #26 above, the preprocessor was updated to check these new input file formats. Also, additional checks were added to the head BCs.	3/3/2017
28	Water-balance	Bug fixes/updates	The water balance utility was updated because of the new input format of TSR output files. See fix #10. Also, a bug was fixed in this code that affected cases when the water level was above the top of the grid.	3/17/2017
29	W2	CPL Tecplot	The CPL Tecplot output sometimes did not update the month in the contour plot text files. This has been fixed – thanks to Jung Ma, Hubei University of Technology in Wuhan, for finding it!	4/4/2017
30	Water-balance	Waterbalance manual	The waterbalance manual was updated for Version 4.	4/14/2017
31	W2	TSR output	Refined the TSR output so that flux terms that were not specified are no longer written out. This cleans up the TSR output and reduces the active number of flux variables when sediment diagenesis is not on.	4/15/2017
32	W2	Derived variables	Fixed a code regression for derived variable TDG when the user stopped the code and pressed restart	4/15/2017
33	W2	Withdrawal output	When a user pressed restart and he/she specified withdrawal output files, the restarted files ignored the filetype of the WDO specification in the control file and used 'opt'.	4/15/2017

W2 V3.7 Bug Fixes, Enhancements and User Manual Changes

Table 7. Bug fixes and enhancements for Version 3.7.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
1	W2	Fish habitat limits	Changed temperature and DO criteria from <code>t2(k,i)<fisht-emph(ii).and.t2(k,i)>fishtempl(ii).and.o2(k,i)>fishdo(ii)</code> to <code>t2(k,i)<=fisht-emph(ii).and.t2(k,i)>fishtempl(ii).and.o2(k,i)>=fishdo(ii)</code> This update is reflected in the manual. Hence the high temperature limit and the dissolved oxygen minimum is less than or equal to given value rather than less than.	8/7/2012
2	W2	Structure, gate, pump, pipe, withdrawal output files	Added code to ensure that if flow is '0' in an outlet structure, that the corresponding temperature and concentration in the outlet file is written as '-99.0'. Previously this was not fully implemented in the code. Code such as this was inserted in several places in the subroutine outputa2.f90: <pre> IF (QGT(JS)==0.0) THEN TAVGW(JWD)=-99.0 CAVGW(JWD, :)=-99.0 CDAVGW(JWD, :)=- 99.0 ENDIF </pre>	8/13/2012
3	PREW2	Format updates	Several output updates were made for warnings and errors	8/16/2012
4	Resource files for W2	Compiling files	Updated some corrupted resource files that were used to compile the source code. Also, zipped up source code and compiler settings together so that file locations are correct for using the Intel compiler.	9/12/2012

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
5	W2 and PREW2	Read csv files	By inserting the character '\$' as the first character of the first line, the following files can now be read in free-format or csv format: met, lpr, vpr, wsc, met, cin, ctr, cdtr, cpre, qot, and qwd. This is described in a Word document that accompanies the download package. The preprocessor has also been updated for file checks. This is part of the Version 3.71 update.	9/12/2012
6	W2	Read input file	An input format bug was fixed for a system with more than 9 waterbodies. <pre> DO JD=1,NDC !READ (CON, '(A8, (:9A8)) ') CDNAME2 (JD), (CDWBC (JD, JW), JW=1,NWB) READ (CON, '(A8, (:9A8) :/ (8X, (:9A8))) ') CDNAME2 (JD), (CDWBC (JD, JW), JW=1,NWB) !cb 9/13/12 END DO READ (CON, ' (/) ') ! DO JF=1,NFL do jf=1,73 ! Fix this later !READ (CON, '(A8, (:9A8)) ') KFNAME2 (JF), (KFWBC (JF, JW), JW=1,NWB) READ (CON, '(A8, (:9A8) :/ (8X, (:9A8))) ') KFNAME2 (JF), (KFWBC (JF, JW), JW=1,NWB) !cb 9/13/12 END DO </pre> This had the effect of turning OFF output for derived constituents for waterbody 10.	9/13/2012
7	GUI	Time series elevation	The GUI read in values of ETSR as integers rather than real numbers. This was fixed.	10/30/12
8	W2	Spillways Lateral	Lateral spillways when connected to other model segments were sometimes not connecting as a tributary to the downstream segment. This has been fixed.	10/30/12
9	W2	W2Tools output	In place of the Vector Plot Output (VPL), a new output was added that allows use of the W2Tools post-processing package. This is part of the Version 3.71 update.	10/30/12
10	W2	User Manual	The User Manual has been updated with the new model features as shown in 5 and 9 above. In addition a separate user manual file shows how to use the w2tools post-processor. This is in the directory for W2tools. This is the version 3.71 update.	10/30/12

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
11	W2	Water quality and temperature	A new calculation technique was added that eliminates calling the Tri-diagonal subroutine. These were built into the temperature and water quality subroutines. This change results in improvements in computational speed of from less than 5% to over 20% for water quality models with lots of water quality state variables.	10/30/2012
12	PREW2	More checks	Added more error trapping for input files. This is an effort for the error trapping to occur before the code bombs. Fixed a couple of regression errors as a result of this fix.	11/2/2012, 11/5/2012
13	Excel macro utility		Added an Excel macro utility to aid in writing out input files to CE-QUAL-W2	11/5/2012

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
14	W2	Withdrawal subroutine	Fixed an IF test that used the wrong variable in the dynamic port allocation algorithm. Also added code to allow the code to test for temperatures at the outlet levels specified. Deleted line of code is underlined followed by the fix. <pre> DO J=1,NUMTSPLT !REORDERING OUTLETS SO THAT HIGHEST ELEVATION STRUCTURE ON TOP (ASSUMING 2 SPLIT OUTLETS) ! <u>IF (TCNTR(J) == 'ST') THEN</u> IF (TSPLTCNTR(J) == 'ST') THEN ! cb 11/11/12 IF (ESTR(JSTSPLTT(J,1),TSPLTJB(J)) < ESTR(JSTSPLTT(J,2),TSPLTJB(J))) THEN JSTSPLT(J,1)=JSTSPLTT(J,2) JSTSPLT(J,2)=JSTSPLTT(J,1) END IF ! <u>ELSE IF (TCNTR(J) == 'WD') THEN</u> ELSE IF (TSPLTCNTR(J) == 'WD') THEN ! cb 11/11/12 IF (EWD(JSTSPLTT(J,1)) < EWD(JSTSPLTT(J,2))) THEN ... IF (TSPLTJB(J) == JB .AND. TSPLTCNTR(J) == ' ST') THEN QALL=0.0 DO JJ=1,NOUTS(J) QALL=QALL+QSTR(JSTSPLT(J,JJ),TSPLTJB(J)) ! SUM UP ALL THE FLOWS ELR = SINA(JB)*DLX(DS(JB))*0.5 DO K=KTWB(JW),KB(DS(JB)) IF (EL(K,DS(JB))-ELR < ESTR(JSTSPLT(J,JJ),TSPLTJB(J))) EXIT !SW 10/17/01 END DO KSTR = K-1 KSTRSPLT(JJ) = MIN(KSTR,KB(DS(JB))) ENDDO DO JJ=1,NOUTS(J) ! cb 11/11/12 dividing total flow between outlets for temperature test QSTR(JSTSPLT(J,JJ),TSPLTJB(J)) = qall/real(nouts(j)) ENDDO </pre>	11/13/12

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
15	W2	Reading in names of WQ variables	In case a user does not enter the units in graph.npt, the code improperly parses the WQ variable name. In this case the output name is a blank. To avoid this issue, extra code was added to preserve the variable name even if no units were added to the graph.npt list. <pre> L1 = SCAN (CNAME(JC),',')+2 IF(L1 == 2)L1=43 ! SW 12/3/2012 Implies no comma found L2 = SCAN (CNAME(JC)(L1:43),' ')+L1 IF(L2 > 43)L2=43 ! SW 12/3/2012 CUNIT(JC) = CNAME(JC)(L1:L2) CNAME1(JC) = CNAME(JC)(1:L1-3) CNAME3(JC) = CNAME1(JC) DO WHILE (L3 < L1-3) </pre>	12/3/2012
16	PREW2	SEDS and SEDK	The variable names were switched in reading the control file in the preprocessor perhaps leading to incorrect warnings/errors being tagged. The proper order was restored: <pre> !READ (CON, '(/A8/(8X,2A8,6F8.0,A8))', ERR=400) AID, (SEDC(JW), PRNSC(JW), SEDCI(JW), seds(jw), SEDDK(JW), FSOD(JW), ! FSOD(JW), sedbr(jw), DYNSEDK(JW), JW=1,NWB) ! SW 6/1/07 READ (CON, '(/A8/(8X,2A8,6F8.0,A8))', ERR=400) AID, (SEDC(JW), PRNSC(JW), SEDCI(JW), SEDDK(JW), seds(jw), FSOD(JW), FSOD(JW), sedbr(jw), DYNSEDK(JW), JW=1,NWB) ! cb 12/30/12 </pre>	12/30/12
17	Excel macro utility w2tool	Integer/Long variables	Some loose ends were corrected in the Visual Basic code built into the Excel macros.	1/2/2013
18	W2	TDG output	A series of code changes were made to fix some issues that arose for computing the impact of a structure on downstream TDG. These fixes were made in subroutines Withdrawal, outputa2w2tools, w2modules, and hydroinout. These affected calculation of output of dissolved gas concentration for output files for spillways or gates that had dissolved gas equation.	1/23/2013

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
19	W2	Reading in dynamic extinction coefficient	For temperature only studies, the model did not update the dynamic light extinction coefficient correctly. This has been fixed by the added code below: <pre> DO JW=1,NWB IF (READ_EXTINCTION(JW)) GAMMA(: ,US(BS(JW)):D S(BE(JW))) = EXH2O(JW) ! SW 1/28/13 KT = KTWB(JW) IF (.NOT. NO_HEAT(JW)) THEN </pre>	1/28/2013
20	W2	Input format when 9 WBs	A specific input read error occurred when 9 waterbodies were present as a result of an earlier bug fix: The new read statements occur in 2 places: <pre> READ (CON,'(A8,9A8,/(:8X,9A8)))') CDNAME2(JD),(CDWBC(JD,JW), JW=1,NWB) !cb 9/13/12 sw 2/18/13 READ (CON,'(A8,9A8,/(:8X,9A8)))') KFNAME2(JF),(KFWBC(JF,JW), JW=1,NWB) !cb 9/13/12 sw2/18/13 </pre>	2/18/13
21	PREW2	More checks added	Additional checks were added to warn users of gaps in meteorological data when interpolation may be inappropriate.	2/20/2013
22	W2 User Manual	Updated	Updated User Manual – many small additions and edits – REV3.	2/20/2013
23	PREW2	Improved an error check	Updated an error check for choosing inactive segments for ISNP output	3/21/2013
24	PREW2	More checks added	Added checks for inflow temperature and tributary temperatures	3/28/2013

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
25	W2	Initial WL Calculation	Changed SLOPE to SLOPEC in init—u-elws.f90 routine since the normal depth should be based on SLOPEC. <pre> END IF FUNCVALUE=FLOW- XAREA*HRAD**0.6667*SLOPEC(JB)**0.5/FMANN ! SW 4/5/2013 RETURN END SUBROUTINE MANNINGS_EQN </pre> Also changed KB(I)-1 to KB(I)+1 for ELWS: <pre> IF (ABS(DX).LT.XACC .OR. FMID.EQ.0.) THEN ELWS(I)=RTBIS+EL(KB(I)+1,I) ! SW 4/5/13 RETURN </pre> Also changed KTTOP from REAL to an INTEGER: <pre> REAL :: XAREA, WSURF ! 4/5/13 SW INTEGER :: KTTOP ! 4/5/13 SW </pre>	4/5/2013
25	W2	Output for pumps, spillways, gates	If the LAT option was chosen, the output files index for JWD was incorrect. This may have affected output temperatures and concentrations.	5/17/2013
26	PRE-W2	Mass loading calculation	There were cases where the preprocessor bombed while calculating the mass loading for output to the pre.opt file. This error has been fixed.	6/21./2013
27	W2	Assorted code updates	Minor format errors (that were ignored by compiler), update to code comments, and faster code initializations to speed up model performance were performed in several subroutines: input_PAR.f90, temperature_PAR.f90, transport_PAR.f90, update.f90, and w2_37_win.f90. An example of an initialization code speed up from temperature_PAR.f90: New code: <pre> DO K=KT,KB(I) AT(K,I) = 0.0D0; CT(K,I) = 0.0D0; VT(K,I) = 0.0D0 ! SW CODE SPEEDUP 6/15/13 ENDDO </pre> Old code <pre> AT(:,I) = 0.0D0; CT(:,I) = 0.0D0; VT(:,I) = 0.0D0 </pre>	6/21/2013
28	W2 tools Excel macro	Update	More robust tools release	6/21/2013
29	PRE-W2	Label error	A label error for one spillway error was fixed. It mistakenly used 'gate'.	7/2/2013

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
30	W2	CPL output	A slight change in output format for the 'raw' cpl output file format was made. No change was made in the tecplot output format. <pre> DO I=CUS(JB),DS(JB) WRITE (CPL(JW),'(A38/(9(F10.3,2X)))') CDNAME(CDN(JD,JW)),(CD(K,I,CDN(JD,JW))*CDMULT(C DN(JD,JW)),K=KTWB(JW),KB(I)) ! cb 6/28/13 end do !WRITE (CPL(JW),'(A38/(9(F10.3,2X)))') CDNAME(CDN(JD,JW)),(CD(K,I,CDN(JD,JW))*CDMULT(C DN(JD,JW)),& ! SW 8/12/06 !K=KTWB(JW),KB(I)),I=CUS(JB),DS(JB)) ! CB 1/03/05 </pre>	7/31/13
31	W2	Read input file	A regression error that cropped up when there were 9 or greater than 10 waterbodies has been fixed. This had to do with reading in derived and flux variables in the control file. <pre> DO JD=1,NDC If (nwb < 10) READ (CON,'(A8,(:9A8))') CDNAME2(JD),(CDWBC(JD,JW), JW=1,NWB) If (nwb >= 10) READ (CON,'(A8,9A8,/(:8X,9A8))') CDNAME2(JD),(CDWBC(JD,JW), JW=1,NWB) !cb 9/13/12 sw 2/18/13 6/16/13 END DO READ (CON,'(/)') ! DO JF=1,NFL do jf=1,73 ! Fix this later If (nwb < 10) READ (CON,'(A8,(:9A8))') KFNAME2(JF),(KFWBC(JF,JW), JW=1,NWB) If (nwb >= 10) READ (CON,'(A8,9A8,/(:8X,9A8))') KFNAME2(JF),(KFWBC(JF,JW), JW=1,NWB) !cb 9/13/12 sw2/18/13 6/16/13 </pre>	8/13/13
32	W2	New compiler	Upgraded to the Intel XE 13.1.3.198 compiler. New W2 executables for 32 bit and 64 bit.	8/13/13
33	W2	INIT WL	An error was fixed in the initial water level computation program for rivers. The code below should have the subscript JB instead of J. <pre> DO JJW=1,NWB DO JJB=BS(JJW),BE(JJW) IF (DHS(JB) > US(JJB) .AND. DHS(J) < DS(JJB)) THEN JBD=JJB END IF END DO </pre>	8/20/13

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
34	W2	INIT WL	There was an index error with gates in the initial water level computation. The old code is shown below: <pre> IF (ELWS (ID) < WSUP) THEN IF (ELWS (IDSP (JS)) > WSUP) WSUP = ELWS (IDSP (JS)) ! CHECKING TO SEE IF DOWNSTREAM WS ELEVATION ISN'T ALREADY 'HIGH' ELWS (ID) = WSUP </pre> The new code is <pre> IF (ELWS (IDGT (JG)) > WSUP) WSUP = ELWS (IDGT (JG)) ! CHECKING TO SEE IF DOWNSTREAM WS ELEVATION ISN'T ALREADY 'HIGH' WX 8/21/13 </pre>	8/21/2013
35	W2	GATE	Cleaning up some code in the gate algorithm. Old code: <pre> IF (A2GT (JG) /= 0.0 .AND. IDGT (JG) /= 0.0) THEN </pre> New code: <pre> IF (A2GT (JG) /= 0.0 .AND. IDGT (JG) /= 0) THEN </pre>	8/21/2013
36	W2	TSS computation	Updated the computation for the derived variable TSS to include zooplankton and the particulate form of CBOD. A formula was added to the User Manual reflecting this change. New code includes <pre> IF (CBODS (IBOD) > 0.0) TOTSS (K, I) = TOTSS (K, I) + CBOD (K, I, IBOD) / O2OM (JW) ! SW 9/5/13 Added particulate CBOD to TSS computation TOTSS (K, I) = TOTSS (K, I) + ZOO (K, I, JZ) ! SW 9/5/13 Added zooplankton to TSS computa- tion </pre>	9/6/2013
37	W2	Spillway-LAT	When a spillway was defined with IDSP=0 and LAT, a tributary was defined incorrectly. The new code is shown below: <pre> IF (IDSP (JS) /= 0) then ! cb 9/11/13 JTT = JTT + 1 QTR (JTT) = QSP (JS) ITR (JTT) = IDSP (JS) PLACE_QTR (JTT) = PDSPC (JS) == ' DENSITY' SPECIFY_QTR (JTT) = PDSPC (JS) == ' SPECIFY' IF (SPECIFY_QTR (JTT)) THEN ELTRT (JTT) = ETDSP (JS) ELTRB (JTT) = EBDSP (JS) END IF JBTR (JTT) = JBD end if ! cb 9/11/13 </pre>	9/11/2013

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
38	W2	32 bit exe on XP	Recompiled with new settings from Visual Studio 2012 to (hopefully) run on XP systems with 32 bit OS	9/11/2013
39	W2	End Simulation	Added new close open files in the end_simulation sub-routine. This is merely cleaning up the code to be consistent in closing all open files when a 'Stop' is executed. This should have no effect on the end user. Part of this new code is shown below: <pre> IF(SELECTC == ' ON')then ! SW 9/25/13 New Section on closing files ifile=1949 do jb=1,nbr if(nstr(jb) > 0)then ifile=ifile+1 close(ifile) endif enddo if(nwd > 0)then ifile=ifile+1 close(ifile) endif endif IF (DOWNSTREAM_OUTFLOW) THEN JFILE=0 DO JWD=1,NIWDO CLOSE(WDO(JWD,1)) CLOSE(WDO(JWD,2)) IF (CONSTITUENTS) THEN CLOSE (WDO(JWD,3)) END IF IF (DERIVED_CALC) THEN CLOSE(WDO(JWD,4)) END IF </pre>	9/25/13

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
40	W2	Pumps – Lateral	Fixed several sections of code in the PUMP algorithm in the hydroinout.f90 routine. Under some conditions such as specifying “Lateral”, the PUMP algorithm may not have moved the water from the upstream to the downstream segment correctly. This has been fixed and tested. Part of the code changes are shown below: <pre> IF (LATERAL_PUMP(JP)) THEN ELW = EL(KTWB(JWU),IUPU(JP))- Z(IUPU(JP))*COSA(JBU) ! JWU = JWU+1 ! SW 9/25/13 ! JBD(JWU) = JBU ! IWD(JWU) = IUPU(JP) ELSE ELW = EL(KTWB(JWU),IUPU(JP))- Z(IUPU(JP))*COSA(JBU)- SINA(JBU)*DLX(IUPU(JP))*0.5 ! JSS(JBU) = JSS(JBU)+1 ! SW 9/25/13 END IF ... IF (PUMPON(JP)) THEN IF (LATERAL_PUMP(JP)) THEN JLAT = 1 JWU = JWU+1 ! SW 9/25/13 ... CALL LATERAL_WITHDRAWAL ! (JWU) DO K=KTW(JWU),KBW(JWU) QSS(K,I) = QSS(K,I)-QSW(K,JWU) END DO IF (IDPU(JP) /= 0) THEN ! MOVED CODE SW 9/25/13 JTT = JTT+1 ... ELSE JSS(JBU) = JSS(JBU)+1 ! SW 9/25/13 KTSW(JSS(JBU),JBU) = KTPU(JP) ... </pre>	9/25/13

41	W2	Clean up memory issues	A series of minor memory issues were cleaned up. This should have no impacts on current model runs. These were usually uninitialized memory. Code changes made include: <pre> READ (CON,'(/)') KFNAME2=' ' ! SW 9/27/13 INITIALIZE ENTIRE ARRAY KFWBC = ' ' ! SW 9/27/13 INITIALIZE ENTIRE ARRAY READ (CON,'(/(:8X,9I8))') (KBWD(JW), JW=1,NWD); TRC= ' ' ! SW 9/27/13 INITIALIZATION SINCE ALLOCATION IS TO NTRT READ (CON,'(/(:8X,9A8))') (TRC(JT), JT=1,NTR) EHSN(JE), EHSSI(JE), JE=1,NEPT) !JE=1,NEP) SW 9/27/13 READ (CON,'(/(8X,2F8.0,I8,F8.0))') (ESAT(JE), EHS(JE), ENEQN(JE), ENPR(JE), JE=1,NEPT) !JE=1,NEP) SW 9/27/13 READ (CON,'(/(8X,8F8.0))') (ET1(JE), ET2(JE), ET3(JE), ET4(JE), EK1(JE), EK2(JE), & EK3(JE), EK4(JE), JE=1,NEPT) !JE=1,NEP) SW 9/27/13 READ (CON,'(/(8X,6F8.0))') (EP(JE), EN(JE), EC(JE), ESI(JE), ECHLA(JE), EPOM(JE), JE=1,NEPT) !JE=1,NEP) SW 9/27/13 READ (CON,'(/8X,A8,I8,A8)') RSOC, NRSO, RSIC; RSOD=0.0 ! SW 9/27/13 INITIALIZE SINCE ALLOCATED AS NOD BUT ONLY NRSO USED READ (CON,'(/(:8X,9F8.0))') (RSOD(J), J=1,NRSO) READ (CON,'(/8X,I8,F8.0,a8)') NDLT, DLTMIN, DLTINTER; DLTD=0.0 ! SW 9/28/13 INITIALIZE ARRAY TO NOD SINCE ONLY NDLT ASSIGNED READ (CON,'(/(:8X,9F8.0))') (DLTD(J), J =1,NDLT) SINKC(1:NSTR(JB),JB) = SINKCT(1:NSTR(JB),JB) POINT_SINK(1:NSTR(JB),JB) = SINKC(1:NSTR(JB),JB) == ' POINT' ! SW 9/27/13 END DO ! POINT_SINK = SINKC == ' POINT' COLDEP=ELWS(I)-COLB ! MACT(J,KT,I)=MACT(J,KT+1,I) IF(MACROPHYTE_ON)MACT(J,KT,I)=MACT(J,KT+1,I) ! SW 9/28/13 ! SDKV(:,US(JB):DS(JB))=SDK(JW) SDKV(:,US(JB)-1:DS(JB)+1)=SDK(JW) ! SW 9/28/13 </pre>	9/27/13
42	W2	CPL output	Code was added to eliminate writing out the habitat index to the CPL file for Tecplot when HABITATC is OFF. <pre> IF (I /= DS(JB)+1) THEN IF (HABTATC == ' ON') THEN </pre>	9/28/13

#	Code: W2 or PREW2 or GUI	Fix or En- hancement Type	Description of Bug/Enhancement	Date Bug Fixed or En- hancement added
			<pre> WRITE (CPL(JW),9999) X1(I),ELWS(I),U(K,I),- W(K,I),T1(K,I),RHO(K,I),HAB(K,I),(C2(K,I,CN(JC)) ,JC=1,NAC) ELSE WRITE (CPL(JW),9999) X1(I),ELWS(I),U(K,I),- W(K,I),T1(K,I),RHO(K,I),(C2(K,I,CN(JC)),JC=1,NAC) ENDIF ELSE XDUM=-99.0 WRITE (CPL(JW),9999) X1(I),ELWS(I),XDUM,XDUM,XDUM,XDUM,XDUM,(XDUM, JJ=1,NAC) ENDIF DO K=KTWB(JW),KMX-1 IF(I /= DS(JB)+1 .AND. K <= KB(I))THEN IF(HABTATC == ' ON')THEN WRITE (CPL(JW),9999) X1(I),ELWS(I)-DEPTHM(K,I),U(K,I),- W(K,I),T1(K,I),RHO(K,I),HAB(K,I),(C2(K,I,CN(JC)) ,JC=1,NAC) ELSE WRITE (CPL(JW),9999) X1(I),ELWS(I)-DEPTHM(K,I),U(K,I),- W(K,I),T1(K,I),RHO(K,I),(C2(K,I,CN(JC)),JC=1,NAC) ENDIF IF(K == KB(I))THEN IF(HABTATC == ' ON')THEN WRITE (CPL(JW),9999) X1(I),ELWS(I)-DEPTHB(K,I),U(K,I),- W(K,I),T1(K,I),RHO(K,I),HAB(K,I),(C2(K,I,CN(JC)) ,JC=1,NAC) ELSE WRITE (CPL(JW),9999) X1(I),ELWS(I)-DEPTHB(K,I),U(K,I),- W(K,I),T1(K,I),RHO(K,I),(C2(K,I,CN(JC)),JC=1,NAC) ENDIF WRITE (CPL(JW),*)'TITLE="CE-QUAL-W2"' IF(HABTATC == ' ON')THEN WRITE (CPL(JW),19233)(CNAME2(CN(JN)),JN=1,NAC) ELSE WRITE (CPL(JW),19234)(CNAME2(CN(JN)),JN=1,NAC) ENDIF ! sw 9/28/13 19233 FORMAT('VARIABLES="Distance, m","Elevation, m","U","W","T","RHO", "HABITAT" ',<NAC>('','',A8,'')) 19234 FORMAT('VARIABLES="Distance, m","Elevation, m","U","W","T","RHO" ',<NAC>('','',A8,'')) ! sw 9/28/13 </pre>	

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
43	W2	SPECIFY TRIB	In specifying the elevation between top and bottom for an inflow tributary, the code put the inflow 1 layer below it should have been in many cases. This has been fixed by the additional code shown below: <pre> IF (SPECIFY_QTR(JT)) THEN KTTR(JT) = 2 DO WHILE (EL(KTTR(JT),I) > ! ELTRT(JT)) DO WHILE (EL(KTTR(JT),I) > ELTRT(JT) .and. EL(KTTR(JT)+1,I) > ELTRT(JT)) ! SW 10/3/13 KTTR(JT) = KTTR(JT)+1 END DO </pre>	10/3/2013
44	W2	CWO or CWDO output	Fixed a format overflow in writing out concentrations in a withdrawal output file. <pre> IF (QWDO(J) /= 0.0) CWDO(CN(JC),J) = CWDO(CN(JC),J)/QWDO(J) WRITE (CWDOC(CN(JC)), '(F8.3)') CWDO(CN(JC),J) ! SW 9/23/13 Changed format from G8.3 to F8.3 to avoid format overflow CWDOC(CN(JC)) = ADJUSTR(CWDOC(CN(JC))) IF (QWDO(J) /= 0.0) CDWDO(CDN(JD,JW),J) = CDWDO(CDN(JD,JW),J)/QWDO(J) WRITE (CDWDOC(CDN(JD,JW)), '(F8.3)') CDWDO(CDN(JD,JW),J) ! SW 9/23/13 Changed format from G8.3 to F8.3 to avoid format overflow CDWDOC(CDN(JD,JW)) = ADJUSTR(CDWDOC(CDN(JD,JW))) </pre>	10/4/2013
45	W2 and PREW2	Inflow, Tributary, Distributary and Shade inputs	Added csv file format as a new file input format for flow and temperature files for inflows, tributaries and distributed tributaries. Also, the shade file is now in csv file format. This enhancement includes updates to the preprocessor and W2 codes. Also several minor bug fixes were made on the Preprocessor.	7/15/14
46	W2	Resuspension of inorganic solids	A resuspension formula was corrected. See the code change below: <pre> HS = 0.283 *U2/G*0.283*TANH(COEF1)*TANH(COEF2/TANH(COEF1)) !TS = 2.0*PI*U2/G*1.2* TANH(COEF3)*TANH(COEF4/TANH(COEF3)) TS = 2.0*PI*sqrt(U2)/G*1.2* TANH(COEF3)*TANH(COEF4/TANH(COEF3)) ! cb 5/9/14 </pre>	7/15/14
47	W2	Tecplot output	When the user sets CPL output for Tecplot, the output format when HABITAC=OFF was incorrect. This has been fixed.	7/15/14

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
48	PREW2	Warnings	Fixed a name inconsistency for developing warnings for input concentrations <pre>! IF (NAME /= 'Residence time' .AND. NAME /= 'Water age') THEN IF (NAME /= 'Residence time' .AND. NAME /= 'AGE') THEN ! SW 7/15/14 CALL WARNINGS</pre>	7/15/14
49	W2	TSR filename	The filename in w2_con.npt for TSR is used for the output filenames. In order to account for complex paths that include more than one '.', the following change was made with the BACK=.TRUE. command which checks from the right-hand-side rather than left-hand-side of the character string <pre>! L1 = SCAN(TSRFN, '.') L1 = SCAN(TSRFN, '.', BACK=.TRUE.)</pre>	8/22/14
50	PREW2	Hydraulic structure warnings	Added many new hydraulic structure warnings (gates, spillways, pumps, pipes, internal weirs) for cases where KBSTR was less than KB and fixed a few error messages for these structure checks.	9/10/14
51	W2	TSR output	The time series file has added the surface heat flux terms (net, short wave solar net, long wave radiation net, back radiation heat flux, evaporation heat flux, conductive heat flux) to the output. The manual was also updated.	1/15/15
52	W2	Interpolation of wind direction	In some cases, the wind direction interpolation was incorrect. Code was added to reduce the wind direction angle to less than 2*pi before the interpolation is performed and to consider another possible interpolation case. Thanks to Wenwei Xu for pointing this out. New code is shown below: <pre>! CONVERT PHIO AND PHINX TO LESS THAN 2*PI SW 2/13/15 DO WHILE (PHIO(JW) > 2.*PI) PHIO(JW) = PHIO(JW) - 2.*PI ENDDO DO WHILE (PHINX(JW) > 2.*PI) PHINX(JW) = PHINX(JW) - 2.*PI ENDDO IF (PHIO(JW) - PHINX(JW) > PI) THEN PHI(JW) = (1.0 - RATIO)*(PHINX(JW)+2.0*PI)+RATIO*PHIO(JW) ELSEIF (PHIO(JW) - PHINX(JW) < -PI) THEN ! WX 2/13/15 PHI(JW) = (1.0 - RATIO)*PHINX(JW)+RATIO*(PHIO(JW) +2.0*PI) ! WX 2/13/15 ELSE PHI(JW) = (1.0 - RATIO)*PHINX(JW)+RATIO*PHIO(JW) END IF</pre>	2/13/15

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
53	W2	Withdrawal	Stewart Rounds: Extra check to avoid divide by zero in withdrawal algorithm (this or similar code occurs in 4 subroutines in withdrawal.f90) IF ((ELSTR-HSWB) > EL(KBOT+1,ID)) THEN DLRHOB = ABS(RHO(KSTR,ID)-RHO(KBOT,ID)) ELSE IF ((EL(KBOT+1,ID)-ELR) == ELSTR) THEN !SR 03/24/13 DLRHOB = NONZERO !SR 03/24/13 ELSE DLRHOB = ABS(RHO(KSTR,ID)- RHO(KBOT,ID))*HSWB/(ELSTR-(EL(KBOT+1,ID)-ELR))	4/9/2015
54	W2	SELECTC	The USGS has developed a new automatic port selection algorithm. In the control file, w2_con.npt, one can use the new algorithm by setting SELECTC='USGS'. The old algorithm is used when this is set to SELECTC='ON'. There is new documentation in the User Manual for this new algorithm.	4/9/2015
55	W2	Restart output	Added code to write out a restart file (rso.opt) at the end of a run if restart_output is ON.	4/9/15
56	W2 Examples	Added example problems	Added new example problem for the Spokane River using new csv file inputs and 4 example problems for using the USGS auto-port algorithm	4/9/15
57	W2	Restart for file volume_wbX.opt	The file handler was not closed properly for volume_wbX.opt. Fixed it with additional code in endsimulation.f90: <pre> if(nwd > 0)then ifile=ifile+1 close(ifile) endif do jw=1,nwb ! sw 4/20/15 ifile=ifile+1 ! sw 4/20/15 close(ifile) ! sw 4/20/15 enddo ! sw 4/20/15 </pre>	4/20/15
58	W2	W2selective.npt	Changed input format for critical temperatures for the output file volume_wbX.opt from a maximum of 10 waterbodies to 100. <pre> READ(1010, '(8X,100F8.0)')(TEMPCRIT(JW, J),JW=1,NWB) ! NOTE MAX OF 100 WATERBODIES sw 4/20/15 </pre>	4/20/15
59	W2	Resuspension of SS	Changed DO loop index in suspended solids resuspension in water_quality.f90 from DO K=KT-1,KB(I)-1 to DO K=KT+1,KB(I)-1 ! cb 9/29/14	5/14/2015

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
60	W2	Wind at 2 m	The W2 model computes the wind at a 2 m height based on the used defined measurement height of the wind for evaporation computations. The formula for computing this variable was using a step function of the wind data rather than interpolation of the wind data when the user chose to interpolate meteorological data. For meteorological input data at short time intervals this is a very minor change. For meteorological data at large time intervals (like a day), this could affect the amount of evaporation. Hence, the calculation below was moved from the TVDS routine to the main routine so that the interpolated wind would be used. WIND2(I)=WIND(JW)*WSC(I)*DLOG(2.0D0/Z0(JW))/DLOG(WINDH(JW)/Z0(JW))	5/21/15
61	W2	TSR output	The TSR file output now also includes a volume weighted vertical average temperature for the segment that the TSR file is located. The manual has been updated also.	6/1/15
62	W2	Writing over output files	In some intermittent cases, when the dialog box closes, the model reinitializes some of the output files (effectively deleting the output). The following line of code was adding at the beginning of the main W2 code to prevent this: <pre> ! ** Task 1: Inputs ** ! ***** ***** ***** INTEGER(4) length, istatus character*255 dirc ! call omp_set_num_threads(4) ! set # of processors to NPROC Moved to INPUT subroutine IF(END_RUN)STOP ! SW 6/26/15 Added code to prevent a thread from reinitializing output files as dialog box is closing...intermittant error </pre>	6/26/15
63	W2	Output order for kinetic fluxes	The output columns for DOAR and DOER were switched in the output file kflux_jw*.opt. The model code was changed to fix this. <pre> ! DOAR => KF(:,56); DOEP => KF(:,57); DOER => KF(:,58); DOPOM => KF(:,59); DODOM => KF(:,60) DOEP => KF(:,56); DOAR => KF(:,57); DOER => KF(:,58); DOPOM => KF(:,59); DODOM => KF(:,60) ! cb 9/16/2015 </pre>	9/16/15

W2 V3.6 Bug Fixes, Enhancements, and User Manual Changes

Table 8. Bug fixes and enhancements for Version 3.6.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
1	W2	TKE1 model	The variable STRICK was incorrectly allocated as an INTEGER rather than REAL.	10/11/2008
2	W2	PIPE	Code was streamlined in the subroutine ZBRENT where calls were made directly to CDFUNC rather than through the dummy function FUNC	10/11/2008
3	W2 Manual	Z0	The User Manual had Z0 in an incorrect line in the control file (w2_con.npt). The write up and example control file in the User Manual were corrected.	10/28/2008
4	W2	Longitudinal profile input	The W2 program did not read initial constituent concentrations in the longitudinal profile file when CCC was 'OFF'. This has been fixed.	12/4/2008
5	W2	TECPLOT output	When using TECPLOT output for multiple waterbodies, the output format did not allow loading the information into TECPLOT. Fixed.	1/26/2009
6	W2	Epiphyton input	For entering vertical profile data for periphyton, there was an index error: OLD CODE: IF (VERT_EPIPHYTON(JW,JE)) EPD(:,I,JE) = EPIVP(K,JW,JE) NEW CODE: IF (VERT_EPIPHYTON(JW,JE)) EPD(:,I,JE) = EPIVP(:,JW,JE)	5/21/2009
7	PreW2	Constituent loads	An enhancement was added to the Preprocessor to compute loads in kg/day for all inflow, tributary and distributed tributaries. Also, these are summed up for the model application. These are shown in the file "pre.opt". These are approximate loads since the concentration data are used to set the frequency of loading update. Flow rates at the time of the concentration input data are used to compute load.	5/21/2009

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
8	W2	Gas transfer at spillways	A couple code fixes in the hydroinout.f90 subroutine: (1) CGAS needed to be initialized in some cases to CGAS=C2(K,ID,CN(JC)) prior to calling the subroutine TOTAL DISSOLVED GAS for use in the Butts and Evans (1983) equation: NEW CODE: CGAS=C2 (K, ID, CN (JC)) ! MM 5/21/2009 (2) Change logic in several lines from IF(CAC(NDO) == ' ON' to IF(CAC(NDO) == ' ON' .and. CN(JC)==NDO NEW CODE: IF (CN (JC) ==NDO .AND. CAC (NDO) == ' ON' .AND. GASSPC (JS) == ' ON' .AND. QSP (JS) > 0.0) THEN ! MM 5/21/2009	5/21/2009
9	W2	Reaeration from dams	An error was found in the formulae from Butts and Evans (1983). OLD CODE: DB = SAT-C DA = DB* (1.0+0.38*AGASGT (N) *BGASGT (N) *CGASGT (N)) * (1.0-0.11*CGASGT (N)) * (1.0+0.046*T)) C = SAT-DA NEW CODE: DA = SAT-C ! MM 5/21/2009 DA: Deficit upstream DB = DA/ (1.0+0.38*AGASSP (N) *BGASSP (N) *CGASSP (N)) * (1.0-0.11*CGASSP (N)) * (1.0+0.046*T)) ! DB: deficit downstream C = SAT-DB	5/21/2009

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
10	W2	Order of flux parameters	The order of flux parameters in the User Manual and output were incorrect. The control file has them in this order: <pre> RPOMSET CBODDK DOAP DOAR DOEP DOER DOPOM DODOM DOOM </pre> whereas the code assumed they were in this order: <pre> RPOMSET CBODDK DOAP DOEP DOAR DOER DOPOM DODOM DOOM </pre> This has been corrected. The User Manual and control file order is now reflected in the W2 code.	6/2/2009
11	Pre	False errors for inflow location	The preprocessor sometimes gave false errors in the pre.err for tributary, internal weirs, pipes, and other hydraulic features saying that the pipe or tributary was below the elevation of the bottom of the segment. The W2 model ran fine even with this error message given in the preprocessor. This has been fixed. Example of OLD CODE: <pre> IF (EBTR(JT) < EL(KB(ITR(JT)+1), ITR(JT))) THEN CALL ERRORS WRITE (ERR, FMTFI) 'Inflow placement bottom elevation [EBTR=', EBTR(JT), '] < bottom active cell elevation for tributary ', JT </pre> New CODE: <pre> IF (EBTR(JT) < EL(KB(ITR(JT))+1, ITR(JT))) THEN CALL ERRORS WRITE (ERR, FMTFI) 'Inflow placement bottom elevation [EBTR=', EBTR(JT), '] < bottom active cell elevation for tributary ', JT </pre>	6/18/09

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
12	Pre	Additional error checking	Additional error checking was added to help debug an error in the bathymetry file when the problem was in the branch connectivity specifically BS and BE. Also, a false error was given when the temperature had an isothermal initial condition, constituents were OFF, and an initial concentration was set to "-2". This was fixed.	6/22/09
13	Pre	Command line processing and working directory displayed for windows	In the windows version of the preprocessor, the user can now supply a command line argument that sets the working directory of the code. Hence, one does not need to copy the preprocessor into every directory. In a batch file, for example, one can execute the following command: <pre>preW2_ivf.exe "C:\scott\w2workshop\2009workshop\waterqual\problem3"</pre> The preprocessor now uses the supplied directory (in double quotes) as the working directory for all the files. The command line argument has one blank space between the end of the executable and the first quote. Also, the working directory is now displayed at the top of the window. Additional checks were also added for checking the grid linkage.	9/12/09
14	W2	# of processors	The model user can now control the # of physical processors the model uses. At this point, dual-processor model runs have shown an improvement of about 20% over a single processor. But, QUAD processors usually are slower. It is recommended that NPROC be set to 2 in the control file. The user can experiment on his/her own system. If this is not set by the user or is left blank, the model still runs but sets it to 2 processors. <pre> GRID NWB NBR IMX KMX NPROC CLOSEC 1 1 23 22 2 ON </pre>	9/12/09

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
15	W2	Command line processing for windows	In the windows version of the w2 model, the user can now supply a command line argument that sets the working directory of the code. Hence, one does not need to copy the model executable into every directory. In a batch file, for example, one can execute the following command: <pre>W2_ivf.exe "C:\scott\w2workshop\2009 workshop\waterqual\problem3"</pre> The w2 model now uses the supplied directory (in double quotes) as the working directory for all the files. The command line argument has one blank space between the end of the executable and the first quote. The working directory is displayed in a text box in the window.	9/12/09
16	W2	W2 window closed at end of successful execution	At the end of a windows run, the windows dialog box waits for the user to press 'close' to exit the window. This allows the user to examine the final run parameters. In the w2_con.npt file there is now an option to close this window when the run has completed. If this option is not set, then the dialog box will stay until the user clicks 'close'. This allows for efficient batch processing of the model, especially if user in conjunction with command line processing mentioned in #15. <pre>GRID NWB NBR IMX KMX NPROC CLOSEC 1 1 23 22 0 ON</pre> When CLOSEC is set to ON, then the dialog box will disappear once the run finishes. If it is set to OFF, then the dialog box will remain until the user clicks 'close'.	9/12/09
17	User Manual	Updates	Updates and changes to the control file (#13-#16) were reflected in an updated User Manual.	9/12/09

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
18	GUI	Updates	The GUI was updated with the following: (1) new control file parameters NPROC and CLOSEC were added (see #14 and 16). There is also a SELECTC that will be used in V3.7 that has been included – ignore it for now. (2) The GUI also can be controlled by command line passing of the working directory and file. In a batch program or from the command line in a DOS box you can execute the GUI as follows: <pre>"C:\scott\research\corps of engi- neers\tomcole\w2code\GUI36\w2control\w2co ntrol36.exe" C:\scott\w2workshop\2009 workshop\waterqual\problem1\w2_con.npt</pre> The first string in quotes executes the GUI. The command line argument is NOT in quotes. This program was developed in VB6 and does not take quotes around the command line. Note that this is different than the FORTRAN command line argument. So the above command will open the GUI and load the control file automatically. (3) A text box now shows the file path and name of the file that you are working on (4) In file open, earlier all *.npt files were shown. Since only "w2_con.npt" files are loaded into the GUI, only the "w2_con.npt" file was shown for opening.	9/12/09

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
19	W2	Gates, spillways, pipes	Whenever DOWN was specified for a gate, spillway or pump, the model estimated the water level at the end of the segment, rather than using the branch center water level. This is important in sloping river systems where a long segment may have a water surface elevation drop between the segment center and the edge. In the past this was computed assuming the slope of the channel. This was updated to estimate the water surface elevation using linear interpolation rather than the grid slope. Below is an example of the code fix – in this case for GATES: OLD CODE: <code>ELIU=ELWS (IUGT (JG)) - SINA (JBUGT (JG)) *DLX (IUGT (JG)) *0.5</code> NEW CODE: <code>ELIU= ELWS (IUGT (JG)) + (ELWS (IUGT (JG)) - ELWS (IUGT (JG) - 1)) / (0.5 * (DLX (IUGT (JG)) +DLX (IUGT (JG) - 1))) *DLX (IUGT (JG)) *0.5</code>	9/25/09
20	W2	New executable	A new executable was made using a new release of Intel Version 11 compiler that corrected problems with Windows 7 applications.	9/25/09

21	W2	ICE cover algorithm	There were a couple logic errors in the ice cover algorithm. These were corrected below: <pre> !***** Ice thickness ICETH(I) = ICETH(I)+ICETHU+ICETH1+ICETH2 IF (ICETH(I) < ICE_TOL) ICETH(I) = 0.0 IF (WINTER .AND. (.NOT. ICE_IN(JB))) THEN IF (.NOT. ALLOW_ICE(I)) ICETH(I) = 0.0 END IF ICE(I) = ICETH(I) > 0.0 IF (ICE(I)) THEN ! 3/27/08 SW ICESW(I) = 0.0 ELSE ICESW(I) = 1.0 ENDIF ICETHU = 0.0 ICETH1 = 0.0 ICETH2 = 0.0 IF (ICETH(I) < ICE_TOL .AND. ICETH(I) > 0.0) ICETH(I) = ICE_TOL ELSE IF (TERM_BY_TERM(JW)) CALL EQUILIBRIUM_TEMPERATURE ! SW 10/20/09 Must call this first otherwise ET and CSHE are 0 HIA = 0.2367*CSHE(I)/5.65E-8 ! JM 11/08 convert SI units of m/s to English (btu/ft2/d/F) and then back to SI W/m2/C ! ICETH(I) = MAX(0.0,ICETH(I)+DLT*((RIMT- ET(I))/(ICETH(I)/RK1+1.0/HIA)-(T2(KT,I)- RIMT))/RHOIRL1) ! OLD CODE ICETH(I) = MAX(0.0,ICETH(I)+DLT*((RIMT- ET(I))/(ICETH(I)/RK1+1.0/HIA)- HWI(JW)*(T2(KT,I)-RIMT))/RHOIRL1) ! SW 10/20/09 Revised missing HWI(JW) ICE(I) = ICETH(I) > 0.0 ICESW(I) = 1.0 IF (ICE(I)) THEN ! TFLUX = 2.392E- 7*(RIMT-T2(KT,I))*BI(KT,I)*DLX(I) ! OLD CODE TFLUX = 2.392E- 7*HWI(JW)*(RIMT-T2(KT,I))*BI(KT,I)*DLX(I) ! SW 10/20/09 Revised missing HWI(JW) TSS(KT,I) = TSS(KT,I) +TFLUX TSSICE(JB) = TSSICE(JB)+TFLUX*DLT ICESW(I) = 0.0 END IF END IF END DO END IF END IF </pre>	10/20/09
22	W2	Gates output in QWD file	The following bug was found in defining which branch a gate was located. This affected the output for the	3/24/10

#	Code: W2 or PREW2 or GUI	Fix or En- hancement Type	Description of Bug/Enhancement	Date Bug Fixed or En- hancement added
			withdrawals at a location where there were gates that were not tied to other branches. Old code: <pre> JWUGT(JG) = JW IF (IDGT(JG) > 0) THEN DO JB=1,NBR IF (IDGT(JG) >= US(JB) .AND. IDGT(JG) <= DS(JB)) EXIT END DO JBDGT(JG) = JB DO JW=1,NWB IF (JB >= BS(JW) .AND. JB <= BE(JW)) EXIT END DO JWDGT(JG) = JW else ! BUG FIX 9/27/07 jbdgt(jp)=1 jwdgt(jp)=1 END IF </pre> New code: <pre> JWUGT(JG) = JW IF (IDGT(JG) > 0) THEN DO JB=1,NBR IF (IDGT(JG) >= US(JB) .AND. IDGT(JG) <= DS(JB)) EXIT END DO JBDGT(JG) = JB DO JW=1,NWB IF (JB >= BS(JW) .AND. JB <= BE(JW)) EXIT END DO JWDGT(JG) = JW else ! BUG FIX 9/27/07 jbdgt(jg)=1 ! SW 3/24/10 jwdgt(jg)=1 ! SW 3/24/10 END IF </pre>	

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
23	PreW2	Reading of WSC	Reading in of the WSC file was limited to only 100 dates in the preprocessor. This limitation was fixed by the code shown below: <pre> ! DO J=1,100 28995 continue ! cb 3/26/10 READ (NPT, '(10F8.0:/(8X,9F8.0))', END=29000) SDAY, (WSC(I), I=1, IMX) IF (SDAY <= SDAYO) THEN CALL ERRORS WRITE (ERR, '(3(A,F0.3))') 'Julian date ', SDAY, ' <= previous date of ', SDAYO, ' in '//WSCFN END IF DO I=1, IMX IF (WSC(I) <= 0.0) THEN CALL ERRORS WRITE (ERR, '(A,F0.3,A,I4,A)') 'Julian date ', SDAY, ': WSC AT SEG(I)=', I, ' <= 0.0 in '//WSCFN ENDIF IF (WSC(I) > 2.0) THEN CALL WARNINGS WRITE (WRN, '(A,F0.3,A,I4,A)') 'Julian day ', SDAY, ': WSC(I) AT SEG(I)=', I, ' > 2.0 in '//WSCFN END IF IF (WSC(I) > 0.0 .and. wsc(i) < 0.5) THEN CALL WARNINGS WRITE (WRN, '(A,F0.3,A,I4,A)') 'Julian day ', SDAY, ': WSC(I) AT SEG(I)=', I, ' < 0.5 in '//WSCFN END IF ENDDO SDAYO=SDAY ! ENDDO go to 28995 ! cb 3/26/10 </pre>	3/26/10
24	PreW2	Check on LAT or DOWN	Added an enhancement to do a check in case a spillway, pipe, pump, or gate was specified as 'DOWN'. In all cases where 'DOWN' is specified, the segment that the hydraulic structure originates must be at the end of a branch. Additional logic was added to check for this in all the hydraulic structures.	3/26/10
25	W2 Manual	Light extinction, ice	Added more text to the section on computation of light extinction and inserted a missing reference. Revised an equation for clarity in ICE algorithm and added more explanation on how to estimate HICE.	4/13/2010
26	W2 Manual	Precipitation input file	The units of precipitation are in m/s. The example precipitation input file was changed to more realistic values.	4/14/2010

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
27	W2	ICE	Added code to account for the need to compute long wave radiation in case user chose the equilibrium temperature approach. Fixed subscript error in ice melt computation. Also, made the variable TICE double precision since it is assumed double precision in the call to Surface_terms. New code: <pre> IF (ICE(I)) THEN TICE = TAIR(JW) DEL = 2.0 J = 1 if(tair(jw).ge.5.0)then ! SW 4/19/10 RANLW(JW) = 5.31E- 13*(273.15+TAIR(JW))**6*(1.0+0.0017*CLOUD (JW)**2)*0.97 else RANLW(JW) = 5.62E- 8*(273.15+TAIR(JW))**4*(1.-0.261*exp(- 7.77E- 4*TAIR(JW)**2))*(1.0+0.0017*CLOUD(JW)**2) *0.97 endif RN1=SRON(JW)/(REFL*RHOWCP)*SHADE(I)*(1.0- ALBEDO(JW))*BETAI(JW)+RANLW(JW) ! SW 4/19/10 DO WHILE (DEL > 1.0 .AND. J < 500) CALL SURFACE_TERMS (TICE) RN(I) = RN1-RB(I)- RE(I)-RC(I) ! 4/19/10 ! RN(I) = SRON(JW)/(REFL*RHOWCP)*SHADE(I)*(1.0- ALBEDO(JW))*BETAI(JW)+RANLW(JW)-RB(I)- RE(JW)-RC(I) ! OLD CODE DEL = RN(I)+RK1*(RIMT-TICE)/ICETH(I) IF (ABS(DEL) > 1.0) TICE = TICE+DEL/500.0 J = J+1 END DO </pre>	4/19/10
28	W2	Evaporation	Units for EV in the SNP file were given in m/s but were actually m^3/s	4/21/10

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
29	W2	Ice	In the ice melt algorithm, SRON should not have been divided by RHOCP in computing RN1 and DEL in the DO WHILE loop should have been ABS(DEL) rather than DEL: <pre> RN1=SRON(JW)/REFL*SHADE(I)*(1.0- ALBEDO(JW))*BETAI(JW)+RANLW(JW) ! SW 4/19/10 eliminate spurious division of SRO by RHOCP DO WHILE (ABS(DEL) > 1.0 .AND. J < 500) ! SW 4/21/10 Should have been ABS of DEL CALL SURFACE_TERMS (TICE) </pre>	4/21/2010
30	PRE	Constituent loading	The output from the preprocessor in the pre.opt file for constituent loading was in kg rather than the output header of kg/day. The output was updated to kg/day by adding the following lines of code: <pre> cdtload(incdt(1:NACdt(Jb),Jb),jb)=cdtload(incdt(1:NACdt(Jb),Jb),jb)/(jday-tstart) ! CB 5/10/10 Change units to kg/day ctrload(trcn(1:NACtr(Jt),Jt),jt)=ctrload(trcn(1:NACtr(Jt),Jt),jt)/(JDAY-TSTART) !CB 5/11/10 convert to units of kg/day </pre>	5/10/10

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
31	W2	Gate, spillways, pipes	In the case where the user has specified that the flow is DOWN, in the case of reverse flow, the model did not assign the flow correctly if the user had no other tributaries or withdrawals specified in the control file. For this rare event, additional code was written to account for this fact. Also, a logic error was discovered in reverse flow for spillways and gates. This was corrected. New code added to hydroinout.f90: <pre> JWW = NWD withdrawals = jww > 0 ! 6/4/10 SW JTT = NTR tributaries = jtt > 0 ! 6/4/10 SW JSS = NSTR IF (SPILLWAY) THEN ... END IF tributaries = jtt > 0 ! 6/4/10 SW withdrawals = jww > 0 ! 6/4/10 SW DO JW=1,NWB KT = KTWB(JW) DO JB=BS(JW),BE(JW) </pre> New code in gate-spill-pipe.f90: For spillway: <pre> IF (ISUB == 0) THEN DLEL = ELIU-ESP(JS) IF (ELID > ESP(JS)) DLEL = ELIU-ELID ! SW 6/7/10 IF (DLEL < 0.0) THEN DLEL = -DLEL </pre> For gates: <pre> IF (A2GT(JG) == 0.0 .AND. G2GT(JG) /= 0.0) DLEL = ELIU-G2GT(JG) IF (ELID > EGT(JG)) DLEL = ELIU-ELID ! SW 6/7/10 IF (DLEL < 0.0) THEN </pre>	6/4/10

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement added
32	W2	Branch intersections with multiple waterbodies	In cases where there are branch intersections between waterbodies, it was possible that the variable KBI and KB were incorrectly set. Here is the fix: Move the statement defining KBI in the subroutine init-geom.f90 to the place shown below (delete the earlier reference): <pre> IF (B(K,ID+1) == 0.0) B(K,ID+1) = B(K-1,ID+1) IF (IEXIT == 1) EXIT END IF END IF END IF END DO END DO ! SW 1/23/06 END DO ! SW 1/23/06 bnew=b ! SW 1/23/06 KBI = KB ! SW 10/30/2010 !**** Upstream active segment and single layer ! 1/23/06 entire section moved SW DO JW=1,NWB KT = KTWB(JW) DO JB=BS(JW),BE(JW) </pre>	10/30/2010
33	W2	SS resuspension	The code index was incorrect in the loop for computing resuspension. This led in some compilers to an infinite loop. The corrected code is shown below: <pre> SSSS(KT,I,J) = - SSS(J)*SS(KT,I,J)*BI(KT,I)/BH2(KT,I)+SSR ! DO K=KT-1,KB(I)-1 DO K=KT,KB(I)-1 ! JP 2/3/12 IF (SEDIMENT_RESUSPENSION(J)) THEN </pre> Thanks to James Pasley for this bug report/fix.	2/3/2012

W2 V3.5 Bug Fixes, Enhancements, and User Manual Changes

Table 9. Bug fixes and enhancements for Version 3.5.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
1	W2	Zooplankton-algae	Sign error in the zooplankton grazing on algae term	8/23/06
2	W2	Input/output	Format for I/O was changed to allow better decimal precision of output	8/23/06
3	W2	Sediment settling rate	The sediment settling rate was accidentally used for POM settling. This was fixed. The old and new code lines are shown below: OLD: <pre>sedsum = sed- sum+seds (JW) * (LPOM (K, I) *lpomdk (jw)+RPO M (K, I) *rpomdk (jw)) *BI (K, I) /BH2 (K, I) * (1 .0-BI (K+1, I) /BI (K, I))</pre> NEW: <pre>sedsum = sed- sum+poms (JW) * (LPOM (K, I) *lpomdk (jw)+RPO M (K, I) *rpomdk (jw)) *BI (K, I) /BH2 (K, I) * (1 .0-BI (K+1, I) /BI (K, I)) ! cb 10/22/06</pre> This was an issue in the SEDIMENT, SEDIMENT C, SEDIMENT P, SEDIMENT N, and SEDIMENT DECAY RATE subroutines.	10/26/06

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
4	W2	Sediment burial	An algorithm was added for sediment burial. This is now a new parameter in the sediment part of the control file. An updated user manual description is forthcoming. The sediment burial rate SEDB (day^{-1}) can be specified in the "SEDIMENT" card section of the control file. A different burial rate can be specified for each water body. OLD/NEW line (example): <pre> ! SED (K, I) = MAX (SED (K, I) + (LPOMEP (K, I) + SEDAS (K, I) + SEDOMS (K, I) + SEDNS (K, I) - SEDD (K, I)) * DLT, 0.0) SED (K, I) = MAX (SED (K, I) + (sedem+SEDAS (K, I) + sedcb (k, i) + SEDOMS (K, I) + SEDNS (K, I) - SEDD (K, I) - sedbr (k, i)) * DLT, 0.0) ! cb 11/30/06 </pre>	11/30/06
5	Control File	Add burial rate for sediment model	This is the change in #4 above implemented in the control file. The new variable SEDBR is added in f8 format after the FSED variable. SEDBR: sediment burial rate in units of per day. <pre> SEDIMENT SEDC SEDPRC SEDCI SEDK SEDS FSOD FSED SEDBR WB 1 ON ON 0.00000 0.10000 0.1 1.00000 1.00000 1.0 </pre>	

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
6	W2	Sediment heating and sediment processes	If a model added and subtracted layers that resulted in segment addition and subtraction, there was the possibility that sediment fluxes were incorrectly computed. In the NO3 subroutine: Old code: $NO3SED(K, I) = NO3(K, I) * NO3S(JW) * NO3TRM(K, I) * (BI(K, I) - BI(K+1, I)) / BH2(K, I)$ New code: if(k == kb(i)) then $NO3SED(K, I) = NO3(K, I) * NO3S(JW) * NO3TRM(K, I) * (BI(K, I)) / BH2(K, I)$ else $NO3SED(K, I) = NO3(K, I) * NO3S(JW) * NO3TRM(K, I) * (BI(K, I) - BI(K+1, I)) / BH2(K, I)$ endif New code added in sediment routine: if(k == kb(i)) then ! SW 4/18/07 $SODD(K, I) = SOD(I) / BH2(K, I) * SODTRM(K, I) * BI(K, I)$ else $SODD(K, I) = SOD(I) / BH2(K, I) * SODTRM(K, I) * (BI(K, I) - BI(K+1, I))$ Endif New code added in suspended solids routine: if(k == kb(i)) then $SSR = EPSILON * DLX(I) * BI(K, I) / VOL(K, I)$ else $SSR = EPSILON * DLX(I) * (BI(K, I) - BI(K+1, I)) / VOL(K, I)$ Endif	4/18/07

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
6	W2	(see above)	New code added for heat flux to channel bottom: <pre> if (kt == kb(i)) then ! SW 4/18/07 SROSED = SROOUT*TSEDF(JW) else SROSED = SROOUT*(1.0- BI(KT+1,I)/BI(KT,I))*TSEDF(JW) Endif if (k==kb(i)) then ! SW 4/18/07 TFLUX = CBHE(JW)/RHOWCP*(TSED(JW)- T2(K,I))*BI(K,I)*DLX(I) else TFLUX = CBHE(JW)/RHOWCP*(TSED(JW)- T2(K,I))*(BI(K,I)-BI(K+1,I))*DLX(I) endif </pre> New code added for sediment subrou- tine: <pre> if (k == kb(i)) then ! SW 4/18/07 SEDAS(K,I) = SEDAS(K,I)+MAX(AS(JA),0.0)*ALG(K,I,JA) *BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I)) else SEDAS(K,I) = SEDAS(K,I)+MAX(AS(JA),0.0)*ALG(K,I,JA) *BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I)) endif if (k == kb(i)) then ! SW 4/18/07 SEDOMS(K,I) = POMS(JW)*(LPOM(K,I)+RPOM(K,I))*BI(K,I) /BH2(K,I) SEDSO = POMS(JW)*SED(K,I)*BI(K+1,I)/BH2(K,I) else SEDOMS(K,I) = POMS(JW)*(LPOM(K,I)+RPOM(K,I))*BI(K,I) /BH2(K,I)*(1.0-BI(K+1,I)/BI(K,I)) SEDSO = POMS(JW)*SED(K,I)*BI(K+1,I)/BH2(K,I)*(1.0-BI(K+1,I)/BI(K,I)) endif </pre>	4/18/07

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
7	W2	Zoo-plankton fixes	Several fixes in the zooplankton routine were made. Many thanks to Dr. Kellie Vache, Institute for Landscape Ecology and Resources Management (ILR) Justus-Liebig-University Giessen Heinrich-Buff-Ring 26 35392 Giessen, Germany, for finding these which are documented below: <pre> DO K=KT,KB(I) do jz = 1, nzp zgztot=0.0 !kv 5/9/2007 do jjz = 1,nzp ! zooss(k,i,jz)= (zmu(k,i,jz)*zeff(jz)-zrt(k,i,jz)- zmt(k,i,jz))*zoo(k,i,jz) - zgztot ! omnivorous zooplankton zgztot=zgztot+zgztot+zgztot*(k,i,jz,jz)*zoo(k,i,jz) !kv 5/9/2007 end do zooss(k,i,jz)= (zmu(k,i,jz)*zeff(jz)-zrt(k,i,jz)- zmt(k,i,jz))*zoo(k,i,jz) - zgztot ! kv 5/9/2007 end do do jjz = 1, nzp ! tgraze(k,i,jz) = tgraze(k,i,jz) + prefz(jz,jjz)*zoo(k,i,jjz) tgraze(k,i,jz) = tgraze(k,i,jz) + prefz(jjz,jz)*zoo(k,i,jjz) !cb 5/17/2007 end do do jjz = 1,nzp ! omnivorous zooplankton ! ZGZ(k,i,jjz,jz) = Zmu(K,I,jz)*ZOO(K,I,jz)*prefZ(jz,jjz)/ tgraze(K,I,jz) ZGZ(k,i,jjz,jz) = Zmu(K,I,jz)*ZOO(K,I,jz)*prefZ(jjz,jz)/ tgraze(K,I,jz) !kv 5/9/2007 end do </pre>	5/21/07
8	PRE	More checks	Added checks for Sediment burial rate and some further checks on grid geometry; added output on SEDS and SEDBR to the pre.opt file; fixed condition where NZP had to equal 1 to work.	6/2/2007

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
9	W2	Array deallocation	The deallocate command on line 7557 was commented out to avoid a deallocation error when the 'STOP' button is pushed during execution on a PC. ! deallocate (sedbr,sedbrp,sedbrn,sedbrc) ! SW 6/4/07 No need to deallocate pointers	6/4/2007
10	W2	Initialization of IUT	For code setting up an external head BC, the variable IUT was not initialized before it was used. This was fixed below: !**** Boundary bottom layers ! IF (UH_EXTERNAL(JB)) KB(IUT-1) = KB(IUT) IF (UH_EXTERNAL(JB)) KB(IU-1) = KB(IU) !cb 6/12/07 IF (UH_INTERNAL(JB)) THEN IF (JBUH(JB) >= BS(JW) .AND. JBUH(JB) <= BE(JW)) THEN ! KB(IUT-1) = MIN(KB(UHS(JB)),KB(IUT)) KB(IU-1) = MIN(KB(UHS(JB)),KB(IU)) !cb 6/12/07 ELSE ! IF (EL(KB(IUT),IUT) >= EL(KB(UHS(JB)),UHS(JB))) THEN IF (EL(KB(IU),IU) >= EL(KB(UHS(JB)),UHS(JB))) THEN !cb 6/12/07 ! KB(IUT-1) = KB(IUT) KB(IU-1) = KB(IU) ELSE ! DO K=KT,KB(IUT) ! IF (EL(KB(UHS(JB)),UHS(JB)) >= EL(K,IUT)) THEN ! KB(IUT-1) = K; EXIT DO K=KT,KB(IU) !cb 6/12/07 IF (EL(KB(UHS(JB)),UHS(JB)) >= EL(K,IU)) THEN !cb 6/12/07 KB(IU-1) = K; EXIT !cb 6/12/07 END IF	6/17/2007
11	W2	CBOD settling	The CBOD settling rate earlier was not converted from m/d in the control file to m/s in the code. Added code: cbods = cbods/day !cb 7/23/07	7/23/07

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
12	W2	TSR output	The surface width was not correctly being output. Changed BI(KT) to BI(KTWB(JW)). FIX: BI (KTWB (JW) , I) , SHADE (I) , ICETH (I) , (ADJUSTR (C2CH (JAC)) , JAC=1 , NAC) , & ! CB 7/26/07	7/26/07
13	PREW2	Pumps	The pump control for DOWN or LAT was not being checked properly, also a check on IUPUC was incorrect. Fixed.	8/14/07
14	W2	Algae	The logic for negative settling velocities for algae had an error. Old code: ! ASR (K, I, JA) = - AS (JA) * (ALG (K+1, I, JA) * B (K+1, I) / (B (K, I) * H2 (K, I)) - ALG (K, I, JA)) * BI (K, I) / BH2 (K, I) New code: ASR (K, I, JA) = - AS (JA) * (ALG (K+1, I, JA) * BI (K+1, I) / BH2 (K, I) - ALG (K, I, JA) * BI (K, I) / BH2 (K, I)) ! SP 8/27/07 Shwet Prakash	8/27/07
15	GUI	NZOOP	When # of zooplankton was set equal to zero, there was an array dimensioning error that caused the writing of the control file to only proceed part way. Fixed.	9/17/07

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
16	W2	Open channel flow	Variable passed between subroutines had inconsistent declaration between routines. <pre> ! REAL, ALLOCATABLE, DIMENSION(:) :: Y, D, B, V, CAREA, TOPW, BELEV, Q, VOLD, YOLD REAL, ALLOCATABLE, DIMENSION(:) :: Y, B, V, CAREA, TOPW, BELEV, Q, VOLD, YOLD ! cb 10/1/07 ! ALLOCATE (Y(NN), V(NN), CAREA(NN), TOPW(NN), BELEV(NN), Q(NN), VOLD(NN), YOLD(NN), D(NN), B(NN)) ALLOCATE (Y(NN), V(NN), CAREA(NN), TOPW(NN), BELEV(NN), Q(NN), VOLD(NN), YOLD(NN), B(NN)) ! cb 10/1/07 ! DEALLOCATE (Y, V, CAREA, TOPW, BELEV, Q, VOLD, YOLD, D, B, YT, VT, VPR, YPR, TAREA, TOPWT, RT, INDX, AL, DAA) DEALLOCATE (Y, V, CAREA, TOPW, BELEV, Q, VOLD, YOLD, B, YT, VT, VPR, YPR, TAREA, TOPWT, RT, INDX, AL, DAA) ! cb 10/1/07 </pre>	10/4/07
17	W2	TKE model	The TKE algorithm had several bugs that have been fixed, these included making the loop over layers go to KBMIN (rather than KB), the original code overwrote the boundary conditions when using the Thomas algorithm, the original code overwrote vertical eddy viscosity at the bed during the averaging process, Δz_k changed to $\Delta z_{k+1/2}$, TKE array was initialized to zero, TKE was implemented in add/sub layers like AZ. Many of these fixes are a result of the work of Sam Gould (Gould, 2006) who wrote an MS project report at PSU entitled "k-e Turbulence Model." Further recommendations by Gould (2006) will be incorporated into the next version of CE-QUAL-W2.	10/4/07

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
			The old code is shown below as a reference to the new code in the release version. OLD CODE <pre> ENTRY CALCULATE_TKE USTAR = SQRT(1.25*CZ(I)*WIND10(I)**2/RHO(KT,I)) IF (MANNINGS_N(JW)) THEN HRAD = BHR1(KT,I)/(BR(KTI(I),I)- BR(KT+1,I)+2.*AVH1(KT,I)) if(macrophyte_on.and.mannings_n(jw)) th en call macrophyte_friction(hrad,fric(i),effri c,kt,i) gc2=g*effric*effric/hrad**0.3333333 else if(.not.macrophyte_on.and.mannings_n(j w)) then gc2=g*fric(i)*fric(i)/hrad**0.3333333 end if ELSE GC2 = 0.0 IF (FRIC(I) /= 0.0) GC2 = G/(FRIC(I)*FRIC(I)) END IF USTARB = SQRT(GC2)*ABS(0.5*(U(KT,I)+U(KT,I-1))) TKE(KT,I,1) = 0.5*(3.33*(USTAR*USTAR+USTARB*USTARB)+ TKE(KT,I,1))*(BH2(KT,I)/BH1(KT,I)) TKE(KT,I,2) = 0.5*(USTAR*USTAR*USTAR+USTARB*USTARB*U STARB*5.0/H1(KT,I)+TKE(KT,I,2))*(BH2(K T,I)/BH1(KT,I)) DO K=KT+1,KB(I)-1 BOUK = MAX(AZ(K,I)*G*(RHO(K+1,I)- RHO(K,I))/(H(K,JW)*RHOW),0.0) PRDK = AZ(K,I)*(0.5*(U(K,I)+U(K,I- 1)-U(K+1,I)-U(K+1,I-1))/H(K,JW))**2.0 PRHE = 10.0*GC2**1.25*ABS(0.5*(U(K,I)+U(K,I- 1)))*4.0/(0.5*B(K,I))**2.0 IF (MANNINGS_N(JW)) THEN ! v3.5 start HRAD = BHR(K,I)/(BR(K,I)- BR(K+1,I)+2.0*H(K,JW)) ! GC2 = G*FRIC(I)*FRIC(I)/HRAD**0.333 if(macrophyte_on.and.mannings_n(jw)) th en call macrophyte_friction(hrad,fric(i),effri c,k,i) gc2=g*effric*effric/hrad**0.3333333 </pre>	

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
			<pre> else if (.not.macrophyte_on.and.mannings_n(jw))then gc2=g*fric(i)*fric(i)/hrad**0.33333333 end if ! v3.5 end END IF PRHK = GC2/(0.5*B(K,I))*ABS(0.5*(U(K,I)+U(K,I -1)))*3.0 UNST = PRDK-TKE(K,I,2) UNSE = 1.44*TKE(K,I,2)/TKE(K,I,1)*PRDK- 1.92*(TKE(K,I,2)/TKE(K,I,1)*TKE(K,I,2)) TKE(K,I,1) = TKE(K,I,1)+DLT*(UNST+PRHK-BOUK) TKE(K,I,2) = TKE(K,I,2)+DLT*(UNSE+PRHE) END DO USTARB = SQRT(GC2)*ABS(0.5*(U(KB(I),I)+U(KB(I), I-1))) TKE(KB(I),I,1) = 0.5*(3.33*USTARB*USTARB+TKE(KB(I),I,1)) TKE(KB(I),I,2) = 0.5*(USTARB*USTARB*USTARB*5.0/H(KB(I), JW)+TKE(KB(I),I,2)) AT = 0.0; CT = 0.0; VT = 0.0; DT = 0.0 DO J=1,2 DO K=KT,KB(I) AT(K,I) = -DLT/BH1(K,I)*BB(K- 1,I)/SIG(J)*AZ(K-1,I)/AVH1(K-1,I) CT(K,I) = - DLT/BH1(K,I)*BB(K,I)/SIG(J)*AZ(K,I)/AV H1(K,I) VT(K,I) = 1.0-AT(K,I)-CT(K,I) DT(K,I) = TKE(K,I,J) END DO CALL TRIDIAG(AT(:,I),VT(:,I),CT(:,I),DT(:,I),KT,KB(I),KMX,TKE(:,I,J)) END DO DO K=KT,KB(I) TKE(K,I,1) = MAX(TKE(K,I,1),TKEMIN1) TKE(K,I,2) = MAX(TKE(K,I,2),TKEMIN2) AZ(K,I) = 0.09*TKE(K,I,1)*TKE(K,I,1)/TKE(K,I,2) END DO ! Center at cell faces DO K=KT,KB(I)-1 AZ(K,I) = 0.5*(AZ(K,I)+AZ(K+1,I)) AZ(K,I) = MAX(AZMIN,AZ(K,I)) AZ(K,I) = MIN(AZMAX(JW),AZ(K,I)) DZ(K,I) = MAX(DZMIN,FRAZDZ*AZ(K,I)) END DO </pre>	

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
18	W2	Restart	Added TKE to restart variables written out and read in.	10/5/07
19	GUI	ET	The equilibrium temperature option in the drop down menu was 'EQT' rather than 'ET'. Fixed.	10/9/07
20	W2	Sediment	The SEDIMENT subroutine did not have any computational mistakes, just an error in assigning all array variables to the value at K,I. This resulted in excessive computational time. The fix is shown below: OLD sedbr = sedb(jw)*sed(k,i) NEW sedbr(K,I) = sedb(jw)*sed(k,i)	10/15/07

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
21	W2	TKE	Turbulence model had an improper averaging between layers. A new temporary variable was defined to temporarily store the values for AZ prior to averaging to the bottom/top of the layers and the horizontal layers. This also affected the computation of DZ. Fixed. New code defined AZT and allocated memory for it, such that $AZT(K, I) = 0.09 * TKE(K, I, 1) * TKE(K, I, 1) / TKE(K, I, 2)$ and $AZ(K, I) = 0.5 * (AZT(K, I) + AZT(K+1, I))$ Similarly for the horizontal averaging and for DZ. Also, the values of DZ were fixed to be at the bottom of a cell and AZ was fixed to be at the bottom right-hand edge of a cell as shown below:	12/17/07

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
22	W2	SS settling	The incorrect cell width was used for SSSO. BI(KT,I) was changed to BI(K,I). OLD CODE: <pre> SSSO (K, I) = (TOTSS0+FES (JW) *FPFE (K, I)) *B I (KT, I) /BH2 (K, I) *DO1 (K, I) FPSS (K, I) = FPSS (K, I) *TISS (K, I) NEW CODE: SSSO (K, I) = (TOTSS0+FES (JW) *FPFE (K, I)) *B I (K, I) /BH2 (K, I) *DO1 (K, I) FPSS (K, I) = FPSS (K, I) *TISS (K, I) </pre>	12/17/07
23	W2	Initial-ization of one-layer	The definition of KBMIN was not updated if the model started out in some segments with only one_layer. This has been fixed. Added code highlighted: <pre> DO I=IU, ID IF (KB(I)-KT < NL(JB)-1) IUT = I+1 ONE_LAYER(I) = KT == KB(I) END DO CUS(JB) = IUT ! reinitialize KBMIN DO I=IU-1, ID KBMIN(I) = MIN(KB(I), KB(I+1)) END DO KBMIN(ID+1) = KBMIN(ID) !**** Areas and bottom widths IF (.NOT. TRAPEZOIDAL (JW)) THEN </pre>	12/17/07

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
24	W2	Bottom processes	This is a couple more fixes related to bug fix #6 above. The Denitrification rate and epiphyton burial rates could be affected based on unique combinations of adding/subtracting segments that left the value of BI in an inactive layer below KB defined incorrectly. In order to prevent the possibility of problems, the following fixes were made: Old Code: <pre> sedNO3(K,I) = NO3(K,I)*NO3S(JW)*NO3TRM(K,I)*(BI(K,I) -BI(K+1,I))/BH2(K,I) EPM(K,I,J) = EPD(K,I,J)*(BI(K,I)- BI(K+1,I)+2.0*H1(K,I))*DLX(I) </pre> New code: <pre> if(k == kb(i)) then ! SW 12/16/07 sedNO3(K,I) = NO3(K,I)*NO3S(JW)*NO3TRM(K,I)*(BI(K,I))/BH2(K,I) else sedNO3(K,I) = NO3(K,I)*NO3S(JW)*NO3TRM(K,I)*(BI(K,I) -BI(K+1,I))/BH2(K,I) endif if(k == kb(i)) then ! SW 12/16/07 EPM(K,I,J) = EPD(K,I,J)*(BI(K,I)+2.0*H1(K,I))*DLX(I) else EPM(K,I,J) = EPD(K,I,J)*(BI(K,I)- BI(K+1,I)+2.0*H1(K,I))*DLX(I) endif </pre>	12/17/2007

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
25	W2	CBODS	If the user defined particulate CBOD that settles to the bottom and had SED turned ON, the conversion from oxygen to organic matter was missing in the accumulation on the channel bottom or sides. OLD <pre> do jd=1,nbod SEDcb(K,I) = SEDcb(K,I)+MAX(cbods(jd),0.0)*cbod (K,I,Jd)*BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I)) end do </pre> NEW <pre> do jd=1,nbod SEDcb(K,I) = SEDcb(K,I)+MAX(cbods(jd),0.0)*(cbo d(K,I,Jd)/O2OM(JW))*BI(K,I)/BH2(K, I)*(1.0-BI(K+1,I)/BI(K,I)) ! 1/16/08 end do </pre>	1/18/08
26	W2	SEDBR	Eliminated a redundant definition of SEDBR in the Sediment routine since it is already defined in the Kinetic rates sub-routine.	1/18/08

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
27	W2	SEDDK	The first order sediment decay rate is an average of the decay rates of all the in-fluxes of organic matter and their respective decay rates. There was an error in computing this average decay rate for CBOD treated as particulate. Code fix is shown below: OLD <pre>do jd=1,nbod sedsum = sed- sum+MAX(cbods(jd),0.0)*cbod(K,I,Jd) *BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I)) end do</pre> NEW <pre>do jd=1,nbod sedsum = sed- sum+MAX(cbods(jd),0.0)*cbod(K,I,Jd) *BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I))*RBOD(JD)*CBODD(K,I,JD)/O2OM(JW) end do</pre>	1/18/08
28	W2	SRO	There are some cases when segments were added/subtracted that the value of BI was not correctly initialized. This code is a fix to prevent such occurrences: OLD CODE: <pre>SRONET = SROIN-SROOUT SROSED = SROOUT*(1.0- BI(K+1,I)/BI(K,I))*TSEDF(JW)</pre> NEW CODE: <pre>SRONET = SROIN-SROOUT if(k /= kb(i)) then ! SW 1/18/08 SROSED = SROOUT*(1.0- BI(K+1,I)/BI(K,I))*TSEDF(JW) else SROSED = SROOUT*TSEDF(JW) endif</pre>	1/18/2008

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
29	W2	Water Qual- ity	Added several calls to prevent computation of kinetic variables if epiphyton are defined in the control file with NEP=1 or more but is not ACTIVE or turned ON. If the kinetic expressions are non-zero and the initial concentration is given, then this could add source/sink terms to the oxygen balance. This is typical of the code changes – since several of this type were made: OLD CODE: DO JE=1,NEP PO4EG(K,I) = PO4EG(K,I)+EGR(K,I,JE)*EPC(K,I,JE)*EP(JE) PO4ER(K,I) = PO4ER(K,I)+ERR(K,I,JE)*EPC(K,I,JE)*EP(JE) END DO NEW CODE: IF (EPIPHYTON_CALC(JW,JE))then ! SW 1/18/2008 PO4EG(K,I) = PO4EG(K,I)+EGR(K,I,JE)*EPC(K,I,JE)*EP(JE) PO4ER(K,I) = PO4ER(K,I)+ERR(K,I,JE)*EPC(K,I,JE)*EP(JE) endif	1/18/2008

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
30	W2	Bottom processes	Continuation of bug fix #24 in such places as New code: <pre> IF (K == KB(I)) THEN xdum=BI(K,I)/BH2(K,I) ! SW 1/18/08 ELSE xdum=BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I)) ENDIF SEDAS(K,I) = SEDAS(K,I)+MAX(AS(JA),0.0)*ALG(K,I,JA) *x dum ! SW 1/18/08 SEDOMS(K,I) = pomS(JW)*(LPOM(K,I)+RPOM(K,I))*xdum !sw 1/18/08 cb 10/22/06 IF (K==KB(I)) THEN ! SW 1/18/08 SEDSO = 0.0 ELSE SEDSO = sedS(JW)*SED(K,I)*BI(K+1,I)/BH2(K,I)*(1.0-BI(K+1,I)/BI(K,I)) Endif DO K=KT,KB(I) IF (K == KB(I)) THEN xdum=BI(K,I)/BH2(K,I) ! SW 1/18/08 ELSE xdum=BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I)) ENDIF DO JA=1,NAL SEDASp(K,I) = SEDASp(K,I)+MAX(AS(JA),0.0)*ap(ja)*ALG (K,I,JA)*xdum ! SW 1/18/08 END DO DO JE=1,NEP IF (EPIPHYTON_CALC(JW,JE)) LPOMEpP(K,I) = LPOMEpP(K,I)+EPOM(JE)*ep(je)*(EMR(K,I, JE)*EPC(K,I,JE)) END DO do jd=1,nbod </pre> This code is repeated similarly in many of the sediment routines.	1/18/2008

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
31	W2	Add segment initial-ization	The DEPTHM and DEPTHB were not initialized correctly when a segment was added – this does not affect internal computations, just output for SPR and SNP files. OLD CODE: $\begin{aligned} \text{BKT(I)} &= \text{BH1(KT,I)}/\text{H1(KT,I)} \\ \text{DEPTHB(K,I)} &= \text{H1(KT,I)} & ! \\ \text{DEPTHM(K,I)} &= \text{H1(KT,I)}*0.5 \end{aligned}$ NEW CODE: $\begin{aligned} \text{BKT(I)} &= \text{BH1(KT,I)}/\text{H1(KT,I)} \\ \text{DEPTHB(KT,I)} &= \text{H1(KT,I)} & ! \end{aligned}$ SW 1/27/08 $\text{DEPTHM(KT,I)} = \text{H1(KT,I)}*0.5$! SW 1/27/08	1/27/08

W2 V3.2 Bug Fixes, Enhancements, and User Manual Changes

Table 10. Bug fixes and enhancements for Version 3.2.

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
1	W2	Waterbody-waterbody connection	When there was negative velocities at a waterbody-waterbody connection, there was a possibility (dependent on the bathymetry of the connection at the waterbody-waterbody intersection) that there could be temperature or concentration anomalies.	8/31/04

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
2	W2	Lateral withdrawal	Added limit to the DLRHOMAX function: Old code: DLRHOMAX=MAX(DLRHOT,DLRHO B) New code: DLRHOMAX=MAX(DLRHOT,DLRHO B,1.0E-10)	1/25/05
3	W2	Branch connectivity	Logic in branch connectivity set-up was fixed Old code: IF(UHS(JB) == DS(JJB))EXIT New code: IF(abs(UHS(JB)) == DS(JJB))EXIT	1/25/05
4	W2	Pumpback	Pumpback logic was corrected – this is legacy code that will probably be removed from later versions of W2 Old code: DO JB=1,NBR IF (JB == JBP) JWBP = JW END DO New code: DO JW=1,NWB DO JB=BS(JW),BE(JW) IF(JB == JBP) JWBP = JW END DO END DO	1/25/05
5	W2	CPL write	Switched order of implied DO loop on CPL write statement for output of constituents	1/25/05
6	W2	PRF write	Changed output format for PRF output for constituents from f10.2 to e13.6	1/25/05

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
7	W2	Heat balance	Added the Idso and Jackson long wave radiation equation when air temperatures are below 5C. The Swinbank model underpredicts long wave incoming radiation at low air temperatures by as much as 10%. The computation of long wave atmospheric radiation is done using the approach of Swinbank (1963) unless air temperatures are less than 5°C, when the Idso and Jackson (1969) formula is used (Wells, et al., 1982). The Swinbank formula for clear sky long wave atmospheric radiation is $\phi_{ac} = 5.31E - 13(T_a + 273)^6$ where units are W/m², °C at 2 m height. Below 40°F (5°C) the formula of Idso and Jackson is recommended (above 10°C both equations are almost identical): $\phi_{ac} = \sigma(T_a + 273)^4(1 - 0.261\exp(-7.77E - 4T_a^2))$ where units are W/m² and T_a is in units of °C. The Stefan-Boltzmann constant = 5.62E-8 W/m²/(°K)⁴.	1/25/05

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
7	W2	Layer addition algorithm	Mistyped subscript K instead of I: Old code: <pre> IF (KB(I) > KBI(I)) THEN B(KB(K),I) = 0.0 DX(KB(I),I) = 0.0 KB(I) = KB(I)-1 IF (I /= DS(JB)+1) KBMIN(I) = MIN(KB(K),KB(I+1)) IF (I /= US(JB)-1) KBMIN(I-1) = MIN(KB(I-1),KB(I)) </pre> New Code: <pre> IF (KB(I) > KBI(I)) THEN B(KB(I),I) = 0.0 ! SW 3/2/05 DX(KB(I),I) = 0.0 KB(I) = KB(I)-1 IF (I /= DS(JB)+1) KBMIN(I) = MIN(KB(I),KB(I+1)) ! SW 3/2/05 IF (I /= US(JB)-1) KBMIN(I-1) = MIN(KB(I-1),KB(I)) </pre>	3/2/05
8	W2	Variable initialization	In some cases when there was a layer subtraction and a time step violation immediately afterward, the variable SW was not initialized properly. This caused problems in the Tomas Algorithm for the water surface computation. The following line of code was added to the SUB layer algorithm: <pre> SW(KT-1,IU-1:ID+1) = 0.0 !TC 3/9/05 </pre> Also, the variable AVHR was defined in the Update variables for DS+1. The following new code was added: <pre> AVHR(KT,DS(JB)+1)=H1(KT,DS(JB)+1) !SW 03/08/05 </pre>	3/9/05

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
9	W2	Interpolation multipliers	Possible index error if there are multiple waterbodies. Old code: <pre> RATZ (K, JW) = AVH2 (K-1, I) / AVH2 (K, I) CURZ1 (K, JW) = 2.0 * H (K, JW) ** 2 / (AVH2 (K-1, I) + AVH2 (K, I)) / AVH2 (K-1, I) CURZ2 (K, JW) = - 2.0 * H (K, JW) ** 2 / (AVH2 (K-1, I) * AVH2 (K, I)) CURZ3 (K, JW) = 2.0 * H (K, JW) ** 2 / (AVH2 (K-1, I) + AVH2 (K, I)) / AVH2 (K, I) END DO </pre> New code: <pre> RATZ (K, JW) = AVH2 (K-1, DS (BE (JW))) / AVH2 (K, DS (BE (JW))) CURZ1 (K, JW) = 2.0 * H (K, JW) ** 2 / (AVH2 (K-1, DS (BE (JW))) + AVH2 (K, DS (BE (JW)))) / AVH2 (K-1, DS (BE (JW))) CURZ2 (K, JW) = - 2.0 * H (K, JW) ** 2 / (AVH2 (K-1, DS (BE (JW))) * AVH2 (K, DS (BE (JW)))) CURZ3 (K, JW) = 2.0 * H (K, JW) ** 2 / (AVH2 (K-1, DS (BE (JW))) + AVH2 (K, DS (BE (JW)))) / AVH2 (K, DS (BE (JW))) </pre>	5/10/05
10	W2	Spillway and Gates	Older code in order to check if it was submerged or not used the elevation difference relative to the channel bed on either side of the weir, rather than the weir crest. Also removed code line: <pre>IF (ELDN > ESP (JS)) DH + ELUP - ELDN</pre>	5/10/05
11	W2	Reaeration	Corrected formula errors in Thackston and Krenkel formula: Old code: <pre> USTAR = SQRT (ADEPTH * SLOPE (JB) * 32.2) ** 0.5 REAER (I) = 24.88 * (1.0 + SQRT (0.176 * UAVG / SQRT (ADEPTH))) * USTAR </pre> New code: <pre> USTAR = SQRT (ADEPTH * SLOPE (JB) * 32.2) REAER (I) = 24.88 * (1.0 + SQRT (0.176 * UAVG / SQRT (ADEPTH))) * USTAR / ADEPTH </pre> Similar changes were made to the updated Thackston model (Eqn 10)	5/10/05
12	W2	Violations NV	The variable BI and VOL was not initialized properly during a time-step violation.	8/25/05

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
13	W2	ADD a layer	The variable BI was not initialized properly during an ADD layer.	8/25/05
14	W2	TRIDIAG sub-routine	<u>Insert Deallocate Statement in Tridiag</u> <pre> SUBROUTINE TRIDIAG (A,V,C,D,S,E,N,U) USE PREC INTEGER, IN TENT (IN) :: S, E, N REAL (R8), DIMENSION (:), INTENT (IN) :: A (E), V (E), C (E), D (E) REAL, DIMENSION (:), INTENT (OUT) :: U (N) REAL (R8), ALLOCATABLE, DIMENSION (:), :: BTA, GMA ALLOCATE (BTA (N), GMA (N)) BTA (S) = V (S) GMA (S) = D (S) DO I=S+1,E BTA (I) = V (I)-A (I) /BTA (I-1) *C (I-1) GMA (I) = D (I)-A (I) /BTA (I-1) *GMA (I-1) END DO U (E) = GMA (E) /BTA (E) DO I=E-1,S,-1 U (I) = (GMA (I)-C (I) *U (I+1)) /BTA (I) END DO Deallocate (BTA, GMA) <----- - ! SW 10/17/05 END SUBROUTINE TRIDIAG </pre>	10/17/05
15	W2	SUB layer	<u>In SUB Layer/Sub Seg - eliminate parentheses which caused a sign error</u> <pre> IF (.NOT. TRAPEZOIDAL (JW)) THEN BI (KT, IU-1) = B (KTI (IU-1), I) H1 (KT, IU-1) = H (KT, JW) - Z (IU-1) BH1 (KT, IU-1) = B (KTI (IU-1), IU-1) * (EL (KT, IU-1) - EL (KTI (IU-1) + 1, IU-1) - Z (IU-1) * COSA (JB)) / COSA (JB) -- <----- -- ! SR 10/17/05 IF (KT >= KB (IU-1)) BH1 (KT, IU-1) = B (KT, IU-1) * H1 (KT, IU-1) DO K=KTI (IU-1) + 1, KT BH1 (KT, IU-1) = BH1 (KT, IU-1) + BH1 (K, IU-1) END DO ELSE </pre>	10/17/05

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
16	W2	SUB layer for shallow systems	<u>Layer SUB - improve model running in shallow segments</u> <pre> ! ** Water surface minimum thickness DO JW=1,NWB KT = KTWB(JW) ZMIN(JW) = -1000.0 KTMAX = 2 <----- ! SR 10/17/05 DO JB=BS(JW),BE(JW) DO I=CUS(JB),DS(JB) IF (KB(I) > KTMAX) KTMAX = KB(I) <----- ! SR 10/17/05 IF (Z(I) > ZMIN(JW)) THEN IZMIN(JW) = I JBIZ = JB END IF ZMIN(JW) = MAX(ZMIN(JW),Z(I)) END DO END DO ADD_LAYER = ZMIN(JW) < -0.85*H(KT- 1,JW) .AND. KT /= 2 SUB_LAYER = ZMIN(JW) > 0.60*H(KT,JW) .AND. KT < KTMAX <-- ----- ! SR 10/17/05 !***** Upstream active segment IUT = US(JB) IF (SLOPE(JB) /= 0.0) THEN DO I=US(JB)-1,DS(JB)+1 IF (KB(I) < KT) THEN <----- ----- ! SR 10/17/05 KB(I) = KT B(KB(I),I) = 0.000001 DX(KB(I),I) = DXI(JW) ... !***** Additional layer subtractions ZMIN(JW) = -1000.0 DO JB=BS(JW),BE(JW) DO I=CUS(JB),DS(JB) ZMIN(JW) = MAX(ZMIN(JW),Z(I)) END DO END DO SUB_LAYER = ZMIN(JW) > 0.60*H(KT,JW) .AND. KT < KTMAX <----- - ! SR 10/17/05 END DO END DO <u>Also done for the initial set-up of the branch geometry:</u> !**** Upstream active segment and single layer IF (SLOPE(JB) /= 0.0) THEN DO I=US(JB)-1,DS(JB)+1 IF (KB(I) < KT) THEN <----- ! .AND. I /= IZMIN(JW) SW 10/17/05 B(KT,I) = 0.000001 </pre>	10/17/05

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
17	W2	Shade algorithm	No errors just an improvement in computational efficiency. <u>Delete this from the SHADING subroutine:</u> !*** Set the angles for which topographic shade data are available DO II=1, IANG ANG(II)=((II-1)*(360.0/FLOAT(IANG)))*PI/180.0 END DO GAMMA = (2*PI)/IANG and change the 2 occurrences of gamma to gama (only in shading subroutine): ANG2 = (TOPO(I,J+1)-TOPO(I,J))/GAMA <---- ! SW 10/17/05 TOPOANG = TOPO(I,J)+ANG2*ANG1 ENDIF END DO IF (AZ00 > ANG(IANG) .AND. AZ00 <= 2*PI) THEN ANG1 = AZ00-ANG(IANG) ANG2 = (TOPO(I,1)-TOPO(I,IANG))/GAMA <----! SW 10/17/05 ADD a line to the module SHADEC: MODULE SHADEC PARAMETER (IANG=18) REAL, PARAMETER :: GAMA=(3.1415926*2.)/REAL(IANG) <--- ! SW 10/17/05 REAL, DIMENSION(IANG):: ANG <----! SW 10/17/05 REAL, ALLOCATABLE, DIMENSION(:) :: A00, DECL, HH, TTLB, TTRB, CLLB, CLRB <----- ! SW 10/17/05 REAL, ALLOCATABLE, DIMENSION(:) :: SRLB1, SRRB1, SRLB2, SRRB2, SRFJD1, SRFJD2, SHADEI REAL, ALLOCATABLE, DIMENSION(:,:) :: TOPO LOGICAL, ALLOCATABLE, DIMENSION(:) :: DYNAMIC SHADE DATA ANG /0.00000, 0.34907, 0.69813, 1.04720, 1.39626, 1.74533, 2.09440, 2.44346,& 2.79253, 3.14159, 3.49066, 3.83972, 4.18879, 4.53786, 4.88692, 5.23599, 5.58505, 5.93412/ <----- ! SW10/17/05 END MODULE SHADEC Delete allocation statement for ang: ALLOCATE (SRLB1(IMX), SRRB1(IMX), SRLB2(IMX), SRRB2(IMX), SRFJD1(IMX), SHADEI(IMX), SRFJD2(IMX)) ALLOCATE (TOPO(IMX,IANG)) <--- - !SW10/17/05 ALLOCATE (QSW(KMX,NWDT), CTR(NCT,NTRT), HPRWBC(NHY,NWB)) Delete ang from the deallocate statement: DEALLOCATE(TTLB, TTRB, CLLB, CLRB, SRLB1, SRRB1, SRLB2, SRRB2, SRFJD1, SHADEI, SRFJD2, TOPO, QSW, CTR) <--- ---! SW 10/17/05	10/17/05

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
18	W2	Epiphyton algorithm	Several changes were made that corrected errors in shallow systems where adding and subtracting layers did not reinitialize macrophyte layers when the current KT was below KB; the epiphyton burial rate was greater than specified in the control file; epiphyton that are buried become part of the 1 st order organic sediment (as before); epiphyton mortality now becomes part of the LPOM pool (based on the EPOM fraction) and is settled and transported downstream rather than going into the organic 1 st order sediment model directly. Currently this is non-photosynthesizing – but we will change in the next version.	5/26/06
19	W2	ADD/SUB layers	There was a bug in addition and subtraction of layers that led to water quality variables not being initialized correctly during riverine shallow flow	5/26/06
20	User Manual	Typos corrected	The manual had a few typos that were corrected.	6/11/2006
21	W2	Waterbody-waterbody connection	The subroutine Upstream_velocity under specific conditions did not maintain flow continuity across a waterbody-waterbody connection	6/29/2006
22	W2	SNP output	The algal limiting nutrient SNP output had a bug under specific conditions in writing out the information.	6/30/2006

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
23	W2	Sediment heating and sediment processes	If a model added and subtracted layers that resulted in segment addition and subtraction, there was the possibility that sediment fluxes were incorrectly computed. In the NO3 subroutine: Old code: $NO3SED(K, I) = NO3(K, I) * NO3S(JW) * NO3TRM(K, I) * (BI(K, I) - BI(K+1, I)) / BH2(K, I)$ New code: <pre> if(k == kb(i)) then NO3SED(K,I) = NO3(K,I)*NO3S(JW)*NO3TRM(K,I)*(BI(K,I))/BH2(K,I) else NO3SED(K,I) = NO3(K,I)*NO3S(JW)*NO3TRM(K,I)*(BI(K,I)- BI(K+1,I))/BH2(K,I) endif </pre> New code added in sediment routine: <pre> if(k == kb(i)) then ! SW 4/18/07 SODD(K, I) = SOD(I) / BH2(K, I) * SODTRM(K, I) * BI(K, I) else SODD(K, I) = SOD(I) / BH2(K, I) * SODTRM(K, I) * (BI(K, I) - BI(K+1, I)) endif </pre> New code added in suspended solids routine: <pre> if(k == kb(i)) then SSR = EPSILON*DLX(I) * BI(K, I) / VOL(K, I) else SSR = EPSILON*DLX(I) * (BI(K, I) - BI(K+1, I)) / VOL(K, I) endif </pre>	4/18/07

#	Code: W2 or PREW2 or GUI	Fix or En- hance-ment Type	Description of Bug/Enhancement	Date Bug Fixed or Enhance- ment Added
23	W2	(see above)	New code added for heat flux to channel bottom: <pre> if (kt == kb(i)) then ! SW 4/18/07 SROSED = SROOUT*TSEDF(JW) else SROSED = SROOUT*(1.0- BI(KT+1,I)/BI(KT,I))*TSEDF(JW) Endif if (k==kb(i)) then ! SW 4/18/07 TFLUX = CBHE(JW)/RHOWCP*(TSED(JW)- T2(K,I))*BI(K,I)*DLX(I) else TFLUX = CBHE(JW)/RHOWCP*(TSED(JW)- T2(K,I))*(BI(K,I)-BI(K+1,I))*DLX(I) endif </pre> New code added for sediment subrou- tine: <pre> if (k == kb(i)) then ! SW 4/18/07 SEDAS(K,I) = SEDAS(K,I)+MAX(AS(JA),0.0)*ALG(K,I,JA) *BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I)) else SEDAS(K,I) = SEDAS(K,I)+MAX(AS(JA),0.0)*ALG(K,I,JA) *BI(K,I)/BH2(K,I)*(1.0- BI(K+1,I)/BI(K,I)) endif if (k == kb(i)) then ! SW 4/18/07 SEDOMS(K,I) = POMS(JW)*(LPOM(K,I)+RPOM(K,I))*BI(K,I) /BH2(K,I) SEDSO = POMS(JW)*SED(K,I)*BI(K+1,I)/BH2(K,I) else SEDOMS(K,I) = POMS(JW)*(LPOM(K,I)+RPOM(K,I))*BI(K,I) /BH2(K,I)*(1.0-BI(K+1,I)/BI(K,I)) SEDSO = POMS(JW)*SED(K,I)*BI(K+1,I)/BH2(K,I)*(1.0-BI(K+1,I)/BI(K,I)) endif </pre>	4/18/07

#	Code: W2 or PREW2 or GUI	Fix or Enhancement Type	Description of Bug/Enhancement	Date Bug Fixed or Enhancement Added
24	W2	Algae	The logic for negative settling velocities for algae had an error. Old code: <pre>! ASR(K,I,JA) = - AS(JA)*(ALG(K+1,I,JA)*B(K+1,I)/(B(K,I) *H2(K,I))- ALG(K,I,JA))*BI(K,I)/BH2(K,I)</pre> New code: <pre>ASR(K,I,JA) = - AS(JA)*(ALG(K+1,I,JA)*BI(K+1,I)/BH2(K, I)-ALG(K,I,JA)*BI(K,I)/BH2(K,I))</pre> !SP 8/27/07 Shwet Prakash	8/27/07